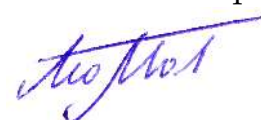


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As a manuscript



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TECHNOLOGY AND EQUIPMENT FOR THE SYNTHESIS OF
COLLOIDAL QUANTUM DOTS
CsPbX₃ (X = Cl, Br, I) AND CdSe/ZnS,
Ag/SiO₂ PLASMONIC NANOPARTICLES,
AND HYBRID STRUCTURES BASED ON THEM

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List of Symbols and Abbreviations

A — absorbance

β — integral breadth of an X-ray reflection

β_D — microstrain contribution to the X-ray reflection breadth

β_S — size contribution to the X-ray reflection breadth

C_{Ag} — molar concentration of the silver salt

C_{ext} — extinction cross section

d_c — diameter of the metal core of a core/shell structure

e — elementary charge

E-ALD — electrochemical atomic layer deposition

E_g — band gap

EDX — energy-dispersive X-ray spectroscopy

FWHM — full width at half maximum

h — Planck constant

h_{opt} — optimal dielectric shell thickness

h_s — dielectric shell thickness

$\hbar$ — reduced Planck constant

HOMO — highest occupied molecular orbital

λ_{max} — plasmon resonance wavelength

LSPR — localized surface plasmon resonance

LUMO — lowest unoccupied molecular orbital

m — reduced exciton mass

m_e — electron effective mass

m_h — hole effective mass

m_0 — electron mass

MCU — microcontroller

μ — dynamic viscosity of the liquid

N — rotation speed

n_m — refractive index of the medium

n_e — electron concentration

ω_{ex} — luminescence excitation frequency

ω_{in} — excitation frequency

ω_p — plasma frequency in the bulk metal

ω_{pl} — plasmon resonance frequency

ω_{qd} — frequency of the quantum dot luminescence maximum

PCs — photonic crystals

PL — photoluminescence

PNPs — plasmonic nanoparticles

PTFE — polytetrafluoroethylene

QD — quantum dot

r_B — exciton Bohr radius

R_{max} — distance from the bottom of the suspension to the rotation axis

R_{min} — distance from the top of the suspension to the rotation axis

ρ — liquid density

ρ_{Ag} — particle density

SERS — surface-enhanced Raman scattering

SILAR — successive ionic layer adsorption and reaction

T — transmittance

t — separation time in centrifugation

T_{G} — growth temperature

T_{N} — nucleation temperature

TEM — transmission electron microscopy

TEOS — tetraethyl orthosilicate

V_{sol} — colloidal solution volume

ε — permittivity

ε_{i} — imaginary part of the metal permittivity

ε_{m} — relative permittivity of the medium

ε_{r} — real part of the metal permittivity

XRD — X-ray diffraction

Introduction

Relevance of the research. Structures with linear dimensions of 1–100 nm substantially change how electromagnetic radiation in the optical range propagates [1]. For example, the localized surface plasmon resonance of metal nanoparticles produces a local enhancement of the electromagnetic field near their surface [2]. A quantum emitter can be placed in this field: an organic fluorophore molecule [3, 4] or a quantum dot (QD) [5]. In such a hybrid system, the photoluminescence (PL) of a QD can be either enhanced in the local field of a plasmonic nanoparticle or quenched when the two are in direct contact. The distance at which plasmonic enhancement prevails over quenching is set by the thickness of the dielectric layer (spacer) between the plasmonic and the quantum subsystems [4]. Plasmonic enhancement manifests itself differently in different hybrid structures [6, 7], and it remains relatively poorly studied experimentally.

Semiconductor and metal nanoparticles have many applications, both in combination and separately. In biomedicine, semiconductor nanoparticles are of interest as a basis for biomarkers with high contrast and chemical stability [8] that can replace organic fluorophores [9]. However, the technology of synthesizing nanostructures with a specified set of functional properties still faces open problems: spectral range, quantum efficiency [10], biocompatibility [11], radiation hardness [12], and others. Active research directions include the search for methods to create new materials [13, 14] and the use of energy transfer between nano-objects of different composition to improve the output characteristics of hybrid nanostructures [15, 16]. Such effects are important for the amplification of stimulated emission in lasers [3, 17], for higher solar cell efficiency [18], for active elements of plasmonics and photonics [19], and for the corresponding structures of flexible electronics [20].

Therefore, the topic of this thesis—the synthesis and study of colloidal quantum dots, plasmonic nanoparticles, and hybrid structures based on them—is relevant and of scientific and practical interest.

The aim of this work was to develop the physical and technological principles of the colloidal synthesis of low-dimensional semiconductor and metal nanostructures, and to fabricate and study hybrid structures based on them.

To achieve this aim, the following **objectives** were addressed:

1. To study the formation of semiconductor and metal nanoparticles of different chemical composition during colloidal synthesis.
2. To study the formation of solid layered structures based on colloidal nanoparticles.
3. To study the plasmonic enhancement of photoluminescence in hybrid structures based on quantum dots and plasmonic nanoparticles.

The practical significance of the thesis is as follows:

1. A hardware system was developed. It includes a) a flow reactor for the synthesis of semiconductor and metal nanoparticles, b) an experimental setup for real-time measurement of light-scattering indicatrices, c) an experimental setup for layer-by-layer deposition, including successive ionic layer adsorption and reaction (SILAR) and electrochemical atomic layer deposition (E-ALD) under a variable external electric field. These engineering solutions are protected by three Russian utility model patents [21–23].

2. The photoluminescence of CsPbBr₃ perovskite nanocrystals was found to be radiation-hard at high absorbed doses of γ -radiation. This makes CsPbBr₃ of practical interest as a photoluminescent material for high-radiation environments.

3. Halide perovskite CsPbX₃ (X = Cl, Br, I) nanocrystals were shown to order efficiently when deposited from colloidal solutions. This is of interest for the active layers of light-emitting structures.

4. The photoluminescence intensity of A^{II}B^{VI} and A^{IV}B^{VI} semiconductor nanocrystals was shown to increase substantially when they are in contact with monodisperse Ag/SiO₂ nanoparticles and the photoluminescence of the hybrid system is excited at the localized surface plasmon resonance (LSPR) frequency.

As a result of the research, the following **main statements for defense** were formulated:

1. The triangulation of the Cs–Pb–X system (X = Cl, Br, I) and the growth of CsPbX₃ nanocrystals are governed not only by the binary compounds on the edges of the Gibbs triangle, but also by the composition points that correspond to the [PbX₆] anionic octahedra.

2. The physicochemical features of the Cs–[PbX₆]–[PbY₆] interaction (X, Y = Cl, Br or Br, I) predetermine the efficient formation of cesium lead halide

solid-solution nanocrystals and the change of their photoluminescence spectra during anion exchange.

3. A model of photoluminescence in cesium lead halide perovskites, in which the photoluminescence is determined by the anionic octahedra as structural units, explains their radiation hardness. This effect gives CsPbBr₃ nanocrystals an advantage over CdSe/ZnS quantum dots in high-radiation environments.

4. The photoluminescence enhancement in a hybrid system of CdSe/ZnS core/shell quantum dots and Ag/SiO₂ plasmonic nanoparticles arises from a manifold increase in the extinction of the combined system. The effect is controlled by varying the thickness of the dielectric shell.

Scientific novelty. All the main statements for defense are new.

The reliability of the results is supported by the reproducibility of the experimental data on materials of different chemical composition, and by comparison of the results with previously known theoretical models and published data.

Implementation of the results. The results were implemented in the Confocal Microscopy Laboratory of the Department of Pathomorphology of the D. O. Ott Research Institute of Obstetrics, Gynecology and Reproductology, within the Russian Science Foundation grant 14-15-00324 “Colloidal quantum dots as biomarkers in exploratory research on pathological processes of the female reproductive system” (2014–2016); in the Laboratory of Nonequilibrium Processes in Semiconductors of the Ioffe Institute, within joint research in 2017–2018; and in the teaching at Saint Petersburg Electrotechnical University “LETI”, in the courses “Materials Science of Micro- and Nanosystems” and “Nanomaterials” of the master’s program “Nanotechnology and Diagnostics” (field of study 28.04.01 “Nanotechnologies and Microsystem Engineering”). Certificates of implementation were obtained for all three.

The results of the thesis were also used in the following projects: agreement 14.132.21.1703 of the Federal Target Program “Research and Scientific-Pedagogical Personnel of Innovative Russia (2009–2013)” on “Computer modeling of materials based on carbon nanotubes and lead sulfide quantum dots”; the UMNIK program project 3578GU2/2014 “Synthesis, study, and modeling of the formation of colloidal lead sulfide quantum dots”; the project part of

the state assignment of the Ministry of Education and Science of the Russian Federation No. 16.2112.2014/K “Fabrication and study of porous systems functionalized with nanomaterials for photonics, sensing, and medicine”; and the Russian Foundation for Basic Research grant 17-33-80010 “Metal coatings on porous materials by additive technologies”.

Objects and methods of research. Unless stated otherwise, all materials described in this work were prepared on purpose-built experimental setups described in the thesis. Commercial colloidal quantum dots from Sigma-Aldrich and samples from the University of Toronto (Canada) served as certified reference samples.

Microscopy studies of materials and structures used an NTEGRA Thermo scanning probe microscope (NT-MDT), a MIRA3 scanning electron microscope (TESCAN), and a JEM-2100F transmission electron microscope (JEOL). Transmission spectra were measured on PE-5400UV (Ekros) and SF-256BIK (LOMO Photonica) spectrophotometers. Photoluminescence spectra were studied with a setup based on an MDR-206 monochromator (LOMO Photonica), a silicon photodiode with a preamplifier, and a 405 nm solid-state laser with a power below 10 mW. X-ray diffraction patterns were recorded on a DRON-3 diffractometer (Cu-K $_{\alpha}$). The radiation hardness experiments used a gamma irradiator with a ^{137}Cs radionuclide source, a photon energy of 661 keV, and a dose rate of 1.7 Gy/s.

Presentation of the results. The main results of the thesis were presented and discussed at the following conferences, seminars, and schools:

- *international conferences*: Nanostructured Materials (2012), Amorphous and Microcrystalline Semiconductors (2012), Nanostructures: Physics and Technology (2013), IEEE NW Russia Young Researchers in Electrical and Electronic Engineering Conference (EIconRusNW) (2015, 2016), Saint-Petersburg OPEN (2015, 2016), Photonics North (2015, 2016), Photonic Colloidal Nanostructures: Synthesis, Properties, and Applications (2016), E-MRS Fall Meeting (2017);
- *all-Russian conferences*: Current Problems of Nano- and Microelectronics (Ufa, 2012), Diagnostics of Nanomaterials and Nanostructures (Ryazan, 2012–2015), Scientific Youth School on Solid-State Electronics (2009–2015), All-Russian Youth Conference on the Physics of Semiconductors and Nanostruc-

tures, Semiconductor Opto- and Nanoelectronics (2015), Russian Youth Conference on Physics and Astronomy PhysicA.SPb (2012–2015), All-Russian Inter-University Scientific and Technical Conference of Students and Postgraduates “Microelectronics and Informatics” (Moscow, 2013), Nanostructured Materials and Conversion Devices for Solar Energy (2014), 3rd All-Russian Conference on Photonics and Information Optics (Moscow, 2014), Congress of Young Scientists (2014), Interdisciplinary Youth Scientific Forum “New Materials. Days of Science” (2015).

– *regional and university conferences*: Scientific and Technical Conference of the Teaching Staff of Saint Petersburg Electrotechnical University “LETI” (2012–2016) and the Scientific and Technical Conference dedicated to Radio Day (2012–2016).

Interim results were presented on 20 February 2017 at the Ioffe Institute at the [Low-Dimensional Systems Seminar](#) and on 22 February 2017 at the SPM/IR/Raman seminar, and on 10 March 2017 at Saint Petersburg State University at the seminar of the research group “Photoactive Nanocomposite Materials”.

Author’s personal contribution. The author developed automated experimental setups and characterization equipment, and proposed sets of process steps for the following tasks: colloidal synthesis of nanoparticles by hot injection, synthesis of nanoparticles in a flow of immiscible liquids, layer-by-layer deposition of film nanostructures, measurement of static light-scattering indicatrices of colloidal solutions and thin films, and measurement of the optical characteristics of layered structures. The author synthesized the nanoparticles, deposited the layers, fabricated the structures, studied the materials by absorption and luminescence spectroscopy and static light scattering, and analyzed and systematized the results.

Publications. 25 works on the topic of the thesis have been published: 10 papers in leading peer-reviewed journals listed by the Higher Attestation Commission (VAK) of Russia, 20 papers in journals indexed in Web of Science and Scopus, three Russian utility model patents, 1 textbook, and 1 monograph.

Structure and scope of the thesis. The thesis comprises 121 pages. It consists of an introduction, four chapters with conclusions, a conclusion, and a list of references with 139 items. It contains 79 figures and 7 tables.

1 Basic Physical and Chemical Properties of Colloidal Quantum Dots and Plasmonic Nanoparticles

The first chapter reviews the current state of the colloidal synthesis of monodisperse semiconductor and metal nanoparticles, the concepts of their electronic structure, and the electronic and optical interaction between quantum dots and plasmonic nanoparticles in contact.

1.1 Colloidal quantum dots

1.1.1 Semiconductor materials

Control of the band gap E_g , a parameter of the electronic spectrum of solids, determines how the electronic and optical properties of a material can be changed. Throughout the second half of the 20th century, the value of E_g required for a given application was achieved by varying and extending the composition of semiconductor materials: from binary $A^{III}B^V$, $A^{II}B^{VI}$, $A^{IV}B^{VI}$ and other compounds to ternary and quaternary compounds and solid solutions.

At the end of the 20th century, this paradigm gave way to low-dimensional materials [24], in which the energy gap is determined by the spatial confinement of charge carriers. Waves of any physical nature excited in a finite volume have strictly defined frequencies. In the same way, such confinement makes the energy spectrum discrete: it becomes a set of discrete levels of allowed energy states. A material confined in all three spatial directions has a density-of-states spectrum similar to that of a single atom (a set of δ -functions). It is therefore called an artificial atom or, following M. Reed [25], a quantum dot. From the geometric point of view the term is not correct, since the particle has a finite size and a three-dimensional structure. In physics, however, where the model concept of a point mass (“material point”) is common, the term is acceptable. It reflects both the *small size* of the particle and the quantum nature of the change in its energy spectrum.

Colloidal quantum dots are nanocrystals of a semiconductor material with linear dimensions in the nominal range of 1–20 nm. They are grown by colloidal methods in a liquid medium and coated with a layer of an organic stabilizer, usually surfactant molecules (Fig. 1.1) [26]. The range is nominal because size criteria and the corresponding quantum limits can differ even for the same

substance, depending on the physical property considered [27].

The most important optical properties of QDs for practical use are light absorption over a wide wavelength range with a narrow emission band, and the dependence of the emission wavelength on particle size. The luminescence bands are symmetric and narrow (full width at half maximum, FWHM, down to 30 nm), and the absorption spectra extend far into the short-wavelength region. This makes these structures attractive for applications that need “fine tuning” of the electronic spectrum of the absorbing or emitting material.

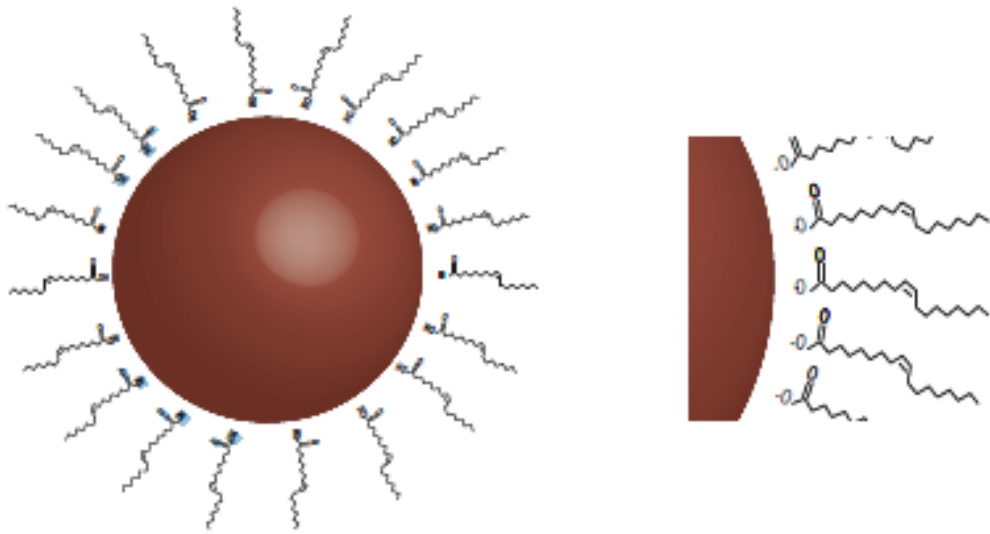


Figure 1.1. Schematic structure of a single colloidal quantum dot and arrangement of ligands at its surface

Colloidal QDs can be used for anti-counterfeiting protection of securities [28]; in biosensors for biology and medicine [29]; as the active layer of photo-conversion devices [30] (photodiodes, solar cells) and light-emitting devices [31] (LEDs, displays, lasers). Colloidal QDs can make the production of such systems much cheaper: concentrated nanoparticle solutions can serve as inks for the printing processes used in modern flexible electronics manufacturing [32, 33].

The tuning range of the PL spectrum depends on the material. The long-wavelength limit is set by the band gap of the bulk material. The short-wavelength limit is set by the smallest technologically achievable QD size at which the particles remain thermodynamically stable. Therefore, different compounds are used for different spectral regions. Fig. 1.2 shows examples of compounds and solid solutions of the $A^{II}B^{VI}$ and $A^{IV}B^{VI}$ semiconductor groups,

which in our view are the most widely represented in the literature on colloidal QDs. Colloidal nanocrystals of these materials, and of $A^{III}B^V$ materials, have been actively studied since the 1990s [34, 35]. The first semiconductor nanocrystals grown in solid dielectric matrices and showing the quantum size effect were demonstrated earlier, in the pioneering work of A. I. Ekimov and A. A. Onushchenko in 1981 [36].

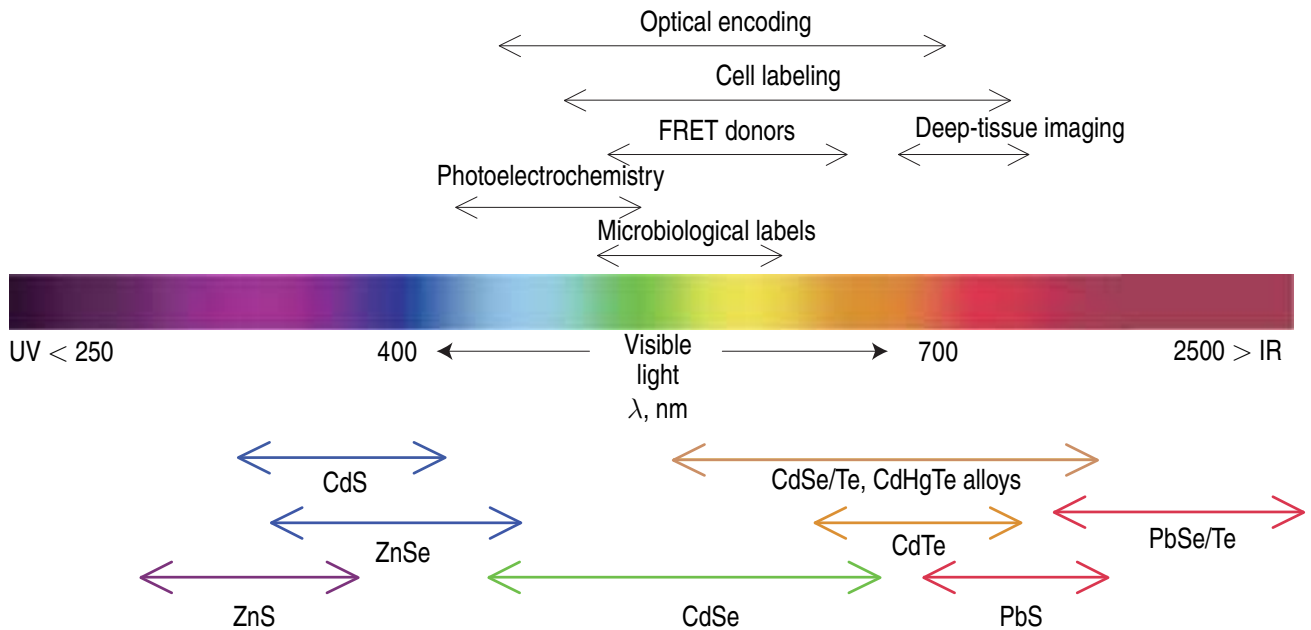


Figure 1.2. Schematic diagram of the photoluminescence tuning ranges of quantum dots of various semiconductor materials and their applications [26]

The diagram in Fig. 1.2 explains, in particular, why QD synthesis based on cadmium selenide, CdSe, has been developed so extensively. By varying the particle size of this material, one can move the PL maximum across almost the entire visible range. Moreover, unlike organic fluorophores, QDs of all sizes can be efficiently excited by a single high-energy source [37].

The main drawback of the materials shown in Fig. 1.2 is the toxicity of their elements, especially of metals such as cadmium, mercury, and lead. Therefore, one of the current directions in colloidal QD synthesis is the search for alternative materials without toxic elements. One promising approach uses chalcopyrite and related systems (CuInS_2 [38], CuInSe_2 [39], $\text{Cu}_2\text{ZnSnS}_4$ [40], AgInS_2 [41], etc.), which have high absorption coefficients. Another approach is doping of wide-gap host nanoparticles with transition metals (for example, ZnS:Mn [42], ZnSe:Mn [43]).

Another direction in improving the synthesis of semiconductor nanostructures for practical use is increasing the PL efficiency: the quantum yield of as-synthesized structures rarely reaches 10%. The reason is surface states. Their influence is much stronger in objects as small as QDs, where the number of surface atoms is comparable to the total number of atoms in the particle. Surface states either quench PL through nonradiative recombination or produce PL bands red-shifted relative to the fundamental absorption edge. Another explanation of the low PL efficiency is slow carrier relaxation through trap states, which reduces the recombination rate. To passivate surface states, the core is coated with a layer of another semiconductor material. The resulting structures are called *core/shell* structures. The most popular are type-I core/shell particles, in which the energy levels of the core and the shell are aligned so that photogenerated carriers of both signs are localized in the same part of the structure, usually the core. Examples of such material pairs are CdSe/ZnS [44] and InP/ZnS [45]. These structures have a high quantum yield of several tens of percent (depending on the technology) and are the most widely used in practice. The optimal shell thickness for the maximum quantum yield is usually 1–2 monolayers. With a thicker shell, the lattice mismatch between the two materials begins to matter. It causes mechanical stress, which relaxes by forming structural defects that reduce the quantum yield. This problem is solved with buffer layers of solid solutions, in which the lattice constant changes gradually from layer to layer [46]. Note, however, that the need for such shells makes nanocrystal synthesis considerably more complex.

In 2015, the group of M. Kovalenko published a paper on ternary perovskite cesium lead halides [13]. These materials are a promising replacement for cadmium chalcogenides: they show a very high PL quantum yield (up to 95%) without additional inorganic shells of wide-gap semiconductors.

1.1.2 Quantum size effects

Let us consider the key physical processes in quantum dots, to better understand the possible technological limitations. Under electronic excitation, for example by external electromagnetic radiation, weakly bound electron–hole pairs form in a semiconductor: Wannier–Mott excitons. The delocalization region of electron–hole pairs is determined by the exciton Bohr radius r_B :

$$r_B = \frac{\hbar^2 \varepsilon}{m e^2}, \quad (1.1)$$

where $\hbar$ is the reduced Planck constant, ε is the permittivity of the material, e is the elementary charge, m is the reduced exciton mass, expressed through the effective masses of the electron and the hole m_e and m_h : $1/m = 1/m_e + 1/m_h$. The exciton is a hydrogen-like quasiparticle, so the natural length scale of the exciton delocalization region is the Bohr radius (the radius of the electron orbit closest to the nucleus in a hydrogen atom):

$$a_0 = \frac{\hbar^2}{m_0 e^2} = 0.0529 \text{ nm}, \quad (1.2)$$

where m_0 is the electron mass. Substituting (1.2) into (1.1) shows that this scale makes it easy to compare r_B for different materials:

$$r_B = \frac{\varepsilon m_0}{m} a_0.$$

For most semiconductors, the reduced exciton mass m is a few tenths or hundredths of m_0 , while ε is always greater than one. Therefore $r_B \gg a_0$ in semiconductors. If the radius r of a semiconductor particle is comparable to r_B , the electronic properties of the particle differ considerably from those of the bulk material. *Confinement* is observed: the exciton is spatially confined, and the positions of the energy levels depend on the particle size (the quantum size effect). In bulk semiconductors, the high carrier concentration screens the Coulomb interaction between electrons and holes, which weakens the excitonic features in absorption spectra. In nanoparticles, there are fewer carriers, and excitonic features are clearly visible in the absorption spectrum even at room temperature.

If the linear size of the crystal is smaller than the exciton Bohr radius ($r \leq r_B$), the confinement is called *strong*. The energy spectrum of the nanoparticle becomes discrete, and the density of states of the 0D object is a set of δ -functions. Decreasing the particle size (with $r \leq r_B$) shifts the excitonic absorption band (and the whole spectrum) to higher frequencies. This shift to shorter wavelengths is called the *blue shift*. The same effect is seen in the position of the PL maximum of QD ensembles (Fig. 1.3). Note that in practice one usually studies the spectra not of single particles but of ensembles, for example

on a substrate or in solution. The absorption and PL spectra are then broadened by the particle size distribution and by the nonzero temperature. The more spectral features the absorption spectrum shows, and the better they are resolved, the narrower the size distribution of the sample.

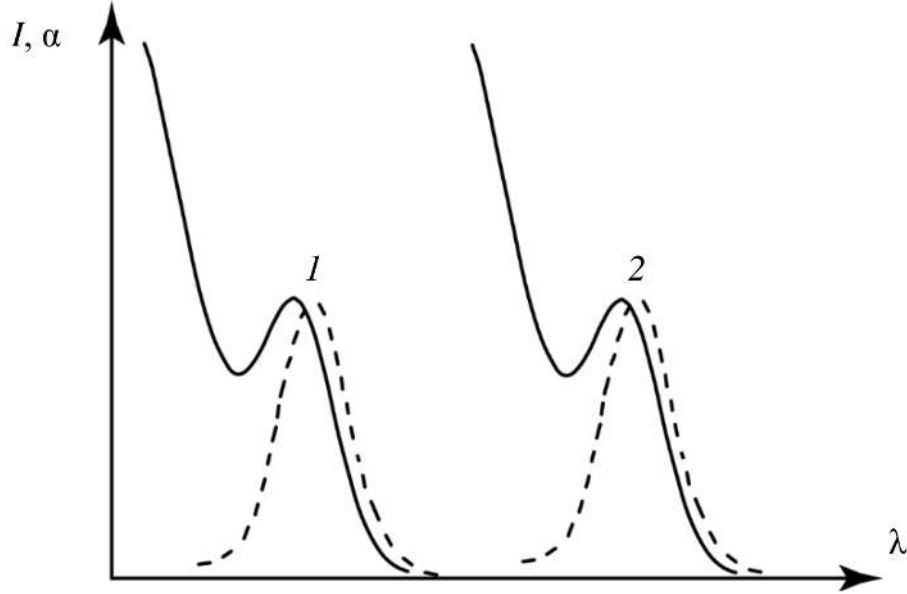


Figure 1.3. Change of absorption (solid lines) and luminescence (dashed lines) spectra with the size of a QD ensemble: $r_1 < r_2 < r_B$ [26]

Because the carrier energy levels are discrete (or partly discrete), band-theory concepts such as forbidden and allowed bands are not quite appropriate for QDs. To stress the difference between the energy spectra of semiconductor nanoparticles and bulk crystals, one speaks of the *effective band gap*, or simply the *energy gap* E_g^* . Since nanoparticles consist of a relatively small number of atoms ($\sim 10^3$), by analogy with molecular systems one often considers the highest occupied (HOMO) and the lowest unoccupied (LUMO) molecular orbitals separated by the energy gap [47].

For many semiconductor materials, the dependence of E_g^* on crystal size is described by the expression found by L. Brus from perturbation theory for small crystals in the effective mass approximation [48]:

$$E_g^* = E_g + \frac{h^2}{8mr^2} - \frac{1.8e^2}{4\pi\epsilon\epsilon_0 r}, \quad (1.3)$$

where h is the Planck constant, r is the particle radius. The last term describes the Coulomb interaction of the electron–hole pair.

Equation (1.3) makes it possible to estimate the mean QD size from the blue shift in the absorption spectrum. Note that expression (1.3) comes from solid-state physics and uses tabulated effective carrier masses of bulk crystals. Strictly speaking, (1.3) is therefore not fully correct for nanocrystals. Nevertheless, experimental results for many materials agree well with it.

1.1.3 Quantum dot synthesis technology

Traditionally, colloidal QD synthesis produces sols of monodisperse particles by condensation of a solid phase from a supersaturated solution. Let us consider the *hot-injection method* used in this work and the general physical and technological features common to most synthesized materials.

In outline, the hot-injection method is as follows. A dispersion medium containing one of the precursors, a source of ions A of the semiconductor compound AB , is continuously stirred and heated to the nucleation temperature T_N (200–300 °C). A solution containing the source of B is injected into it (for metal nanoparticles, the role of component B is played by a reducing agent). For example, for cadmium selenide (CdSe) nanoparticles, a selenium precursor solution is injected into a cadmium precursor solution, or vice versa. The precursors form chemically active atomic or molecular fragments, monomers, which react as $nA + nB = n(AB)$. Because the reaction is fast, the solution quickly becomes locally supersaturated throughout its volume, and *nucleation* follows: nuclei of the solid phase form. At the next stage, the nuclei grow by capturing monomers. Nanocrystal growth from a supersaturated solution is fast, and the solution concentration drops to saturation almost instantly. At each moment there is a critical size: larger particles grow, and smaller ones dissolve. This critical size is not constant; it increases as the supersaturation decreases. The process is diffusion-controlled, so the main process parameter is the growth temperature T_G .

The size of the particles is controlled by the time at which aliquots of the colloidal solution are withdrawn from the reaction medium after nucleation (or, for cesium lead halide nanocrystals, by the temperature T_G). Thus, a single process run can yield a series of samples with different emission wavelengths, by withdrawing particles of different diameters and hence different energy spectra. Monodispersity is ensured by a single nucleation event: particles that nucleate at the same time and grow under the same conditions have the same size

when the process ends. Surfactant molecules in the medium adsorb on the nanoparticle surface and prevent aggregation, while still letting monomers of the synthesized compound condense on the surface.

Immediately after a colloidal solution with semiconductor or metal nanoparticles of the required size is obtained, it is purified from by-products and unreacted components. This stops particle growth, which could otherwise continue slowly at room temperature. Purification is done by adding a strongly polar substance, an *antisolvent*, most often anhydrous methanol. The particles are covered with surfactant molecules whose hydrophobic parts face the solution, so in a polar solvent they flocculate (precipitate), while light molecules and ions of unreacted substances stay in solution. Isolation of the particles is usually accelerated by centrifugation.

If necessary, the same approach is used for size-selective separation. With a smaller amount of antisolvent, the largest particles flocculate first, because their larger surface area gives a stronger van der Waals interaction. For this, antisolvent is slowly added to a purified QD solution until opalescence (light scattering) appears, which marks the onset of flocculation. The precipitate of larger particles is separated by centrifugation from the solution with smaller particles and redispersed in pure solvent. This gives two colloidal solutions, each with a narrower size distribution than the original one. Repeating the procedure gives colloidal solutions with ever narrower size distributions.

Let us outline the synthesis of the core/shell structures mentioned above, using CdSe/ZnS as an example. The starting material is usually a pre-purified solution of core nanoparticles, i.e., CdSe QDs, which serve as seeds for the hybrid structure. The mean radius of the CdSe cores is determined from transmission electron microscopy images or calculated from absorption spectra [49]. Then, from the desired shell thickness and the core size, the volume ratio of ZnS to CdSe is determined, taking tabulated bulk values of the lattice parameters. Shell growth generally repeats the core synthesis procedure described above, except for the nucleation stage. Since the growth seeds are prepared in advance, lower temperatures are usually used for shell growth. Secondary nucleation of ZnS must be avoided, so the zinc and sulfur precursors are usually added in alternating portions.

Note that this section only briefly describes the features of colloidal QD synthesis needed for the rest of the thesis. For a better understanding of the technology, we refer the reader to our monograph [26] and to books by colleagues [50–53].

1.2 Colloidal plasmonic nanoparticles

1.2.1 Optical properties of the electron gas in metals

To introduce the concept of *plasmonic nanoparticles*, let us consider the optical properties of the electron gas in metals. At optical frequencies, the electronic properties of metals can be explained by the plasma model. In this model, the charge carriers are treated as a free electron gas with concentration n_e moving relative to the positively charged crystal lattice. The free electron model ignores the inhomogeneity of the lattice potential and electron–electron interactions. In band theory, the effect of the lattice on electron motion reduces to replacing the free electron mass m_0 with the electron effective mass m_e (the quasi-free electron model). To describe the interaction of charge carriers, the concept of the *plasmon* is introduced: a quasiparticle corresponding to collective oscillations of the free electron gas relative to the positive ion background, excited by an alternating electromagnetic field. These oscillations have a certain resonance frequency ω_p , called the plasma frequency. For a bulk material it is given by:

$$\omega_p = \sqrt{\frac{n_e e^2}{m_e \varepsilon_0}}.$$

If the frequency of external radiation is above ω_p , a bulk metal is transparent to it. At frequencies below ω_p , the electrons oscillate in antiphase with the field of the external radiation, and the radiation is reflected. In nanoparticles, electron motion is confined in all three directions. As a result, the interaction of a metal nanoparticle with external radiation localizes the plasmons.

The localized surface plasmon resonance (LSPR) of metal nanoparticles works as follows. An alternating electromagnetic field with a wavelength much larger than the particle makes the conduction electrons in the particle oscillate, and at the resonance frequency a plasmon is excited (Fig. 1.4). The LSPR properties are determined by the material, shape, and size of the particle, and

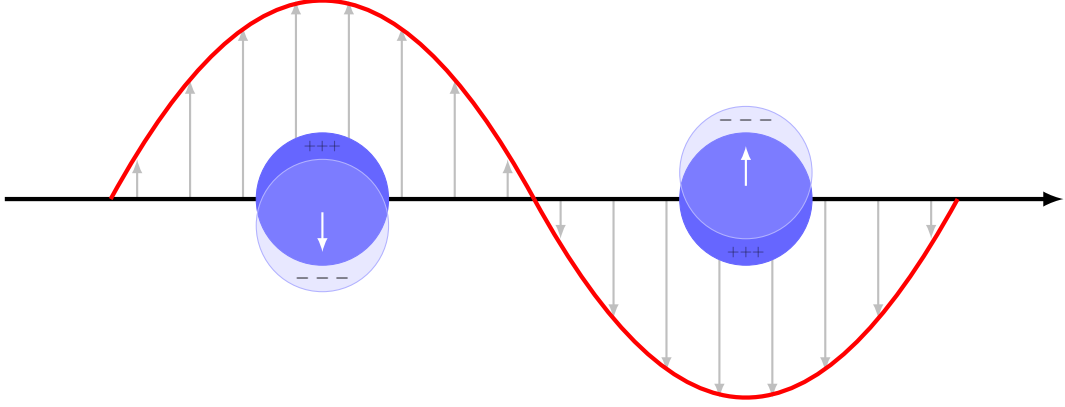


Figure 1.4. Schematic of localized plasmon oscillations in a metal nanoparticle

by the permittivity of the surrounding medium [54].

For $\lambda \gg r$, the electromagnetic field can be considered uniform near the particle. Under an external field, the conduction electrons shift toward the particle surface opposite to the field direction (Fig. 1.4). Because the electrons are spatially confined, negative charge accumulates on one side of the particle and positive charge on the other. This creates an electric dipole whose field opposes the external field and pushes the electrons back to equilibrium. The result is the oscillation of the electron gas.

During the oscillations, the electrons are scattered at the particle boundaries and by the lattice ions. To make up for the lost energy, the plasmons absorb energy from the external field. LSPR is therefore accompanied by resonant absorption, which can be detected by measuring the absorbance (extinction) spectrum. The extinction cross section C_{ext} of a spherical metal particle is [2]:

$$C_{\text{ext}} = \frac{24\pi^2 r^3 \varepsilon_m^{3/2}}{\lambda} \frac{\varepsilon_i}{(\varepsilon_r + 2\varepsilon_m)^2 + \varepsilon_i^2}, \quad (1.4)$$

where ε_m is the relative permittivity of the medium, ε_r is the real part of the metal permittivity, ε_i is the imaginary part of the metal permittivity.

It follows from (1.4) that the extinction cross section increases resonantly under the condition:

$$\varepsilon_r = -2\varepsilon_m. \quad (1.5)$$

Expression (1.4) also shows that, for plasmon oscillations to be sustained in the particle, ε_i must be close to zero. In this model, the frequency dependence of the real part of the permittivity is:

$$\varepsilon_r = 1 - \frac{\omega_p^2}{\omega^2}. \quad (1.6)$$

Equating (1.5) and (1.6) gives the frequency of the localized surface plasmon resonance in metal nanoparticles ω_{pl} :

$$\omega_{pl} = \frac{\omega_p}{\sqrt{1 + 2\varepsilon_m}} \quad (1.7)$$

Expression (1.7) shows that plasmon localization in nanoparticles substantially lowers the plasmon resonance frequency compared with the bulk material. This is why colloidal solutions of noble metals have colors very different from those of the bulk metals.

Fig. 1.5 shows the spectra of the permittivity components of silver in the 200–1000 nm range, calculated from the data of [55]. The real part of the permittivity of silver is negative over the whole range, so the resonance condition (1.5) can be met, while the imaginary part is relatively small in this range.

The resonance condition (1.5) implies that the position of the LSPR extinction maximum also depends on the permittivity of the surrounding medium. The medium is polarized by the field, and charge accumulates at the metal/dielectric interface. This charge partly compensates the charge on the particle surface and weakens the field it creates. The higher the polarizability, and hence the permittivity, of the medium, the stronger the charge compensation and the weaker the field of the particle. Therefore, as the permittivity of the medium around a metal particle increases, the LSPR extinction maximum shifts to longer wavelengths.

In colloidal synthesis, metal nanoparticles are dispersed in a solvent, usually water. A dielectric shell can be deliberately grown on the particle surface, as in this work, for example of silicon dioxide, SiO_2 . Fig. 1.6 shows the relative permittivity spectra of water and silicon dioxide, plotted from the data of [56]. At optical frequencies, the real part of the permittivity of silicon dioxide exceeds that of water. A SiO_2 shell around metal nanoparticles should therefore

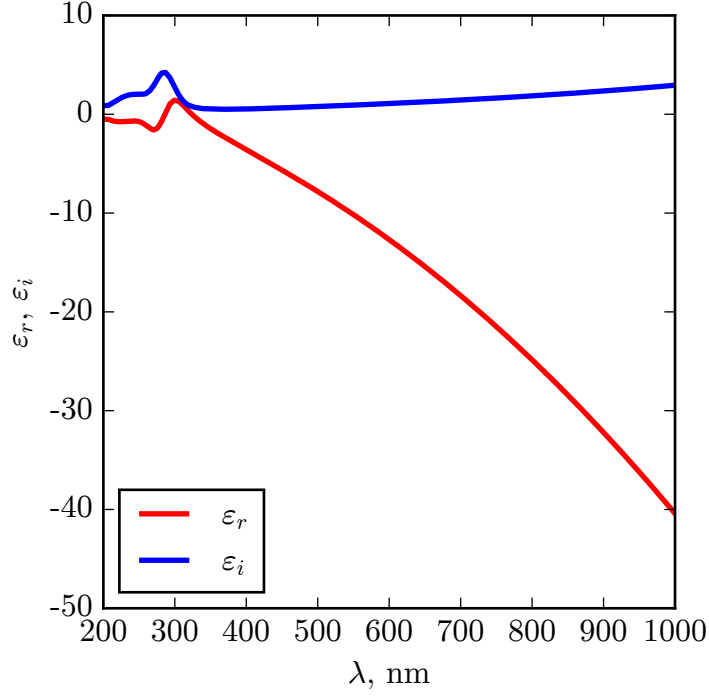


Figure 1.5. Spectra of the real ϵ_r and imaginary ϵ_i components of the permittivity of silver in the 200–1000 nm range

red-shift the extinction maximum.

Fig. 1.7 shows the extinction cross section spectra of silver particles with a radius of 20 nm, calculated from (1.4). For particles in a silicon dioxide matrix, the plasmon resonance wavelength is indeed red-shifted compared with particles dispersed in water. With a thin shell, the plasmon resonance should lie between these two limits. Note also that the extinction cross section is many times (up to 20 times) larger than the geometric cross section of the particle (1300 nm^2). Thus, in a certain frequency range, a silver nanoparticle acts as a kind of lens or antenna for electromagnetic radiation.

This shift of the plasmon resonance maximum with the surrounding medium has been confirmed experimentally. For example, [57] showed that the plasmon resonance shifts to longer wavelengths when the refractive index of the medium n_m changes from 1 (air) to 1.33 (water), and the shift is larger for larger particles. When the surrounding medium has a high permittivity, for example lead chalcogenides, the plasmon resonance frequency should shift strongly. Indeed, [58] reports that coating gold nanoparticles with a lead sulfide (PbS) shell shifts their absorbance maximum from 525 nm to 620 nm.

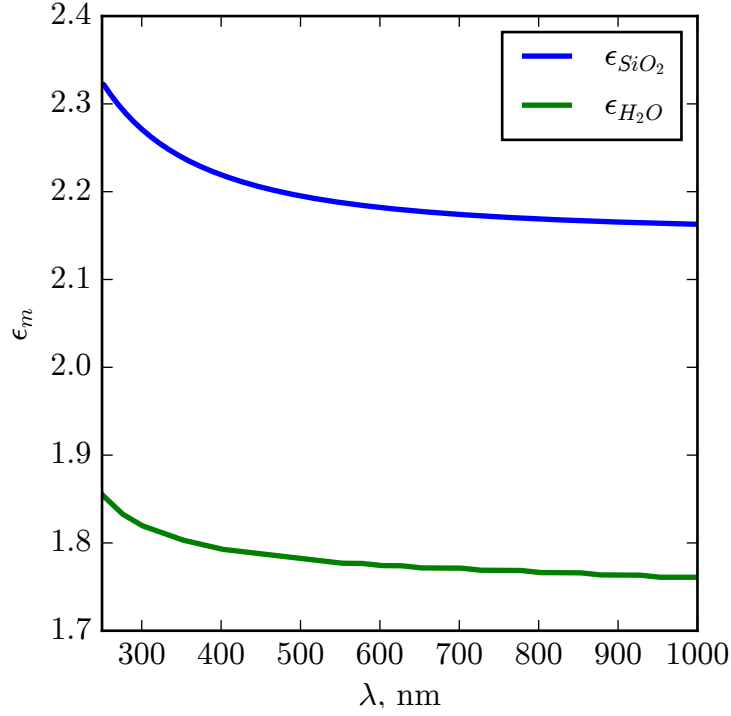


Figure 1.6. Relative permittivity spectra of distilled water and silicon dioxide in the 250–1000 nm range

Plasmon resonance occurs in many metals, including the conductors with the highest conductivity most often used in electronics: silver, gold, copper, and aluminum [59]. Technology for monodisperse silver nanoparticles is of particular interest, because silver has the highest thermal and electrical conductivity of these materials and shows the strongest LSPR. The strong plasmon resonance of silver particles has two causes. First, the strength of LSPR can be described by the quality factor [59]:

$$Q = \frac{\omega}{2\varepsilon_i^2(\omega)} \frac{d\varepsilon_r}{d\omega}$$

Silver has the highest Q of these metals. Second, interband transitions play an important role in damping the plasmon resonance. In silver, the interband transition energies are higher than the LSPR energies and well separated from them, so plasmon damping is minimal. The onset of interband transitions in silver appears in the extinction spectrum as a small rise in absorption starting at about 330 nm (Fig. 1.7) [60]. In gold particles, LSPR usually appears near 500–550 nm, close to the edge of the interband transition region of gold [61]. Copper particles have the weakest LSPR maximum of these metals, which

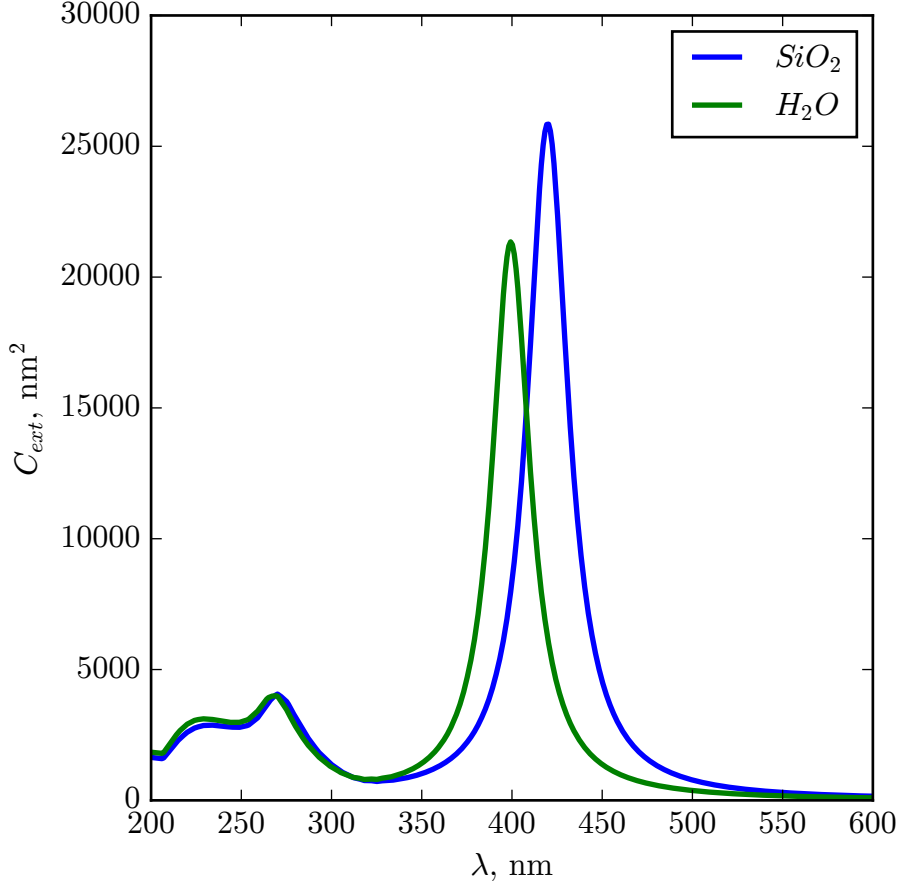


Figure 1.7. Extinction cross section of silver particles with a radius of 20 nm in water and in silicon dioxide, calculated from (1.4)

practically overlaps with the interband transition region [62]. Plasmonic structures of aluminum are hard to synthesize because of its chemistry, which makes spherical aluminum nanoparticles technologically difficult to obtain. However, aluminum disk structures showing LSPR have been fabricated by micro- and nanoelectronics methods [63].

As mentioned above, the position of the LSPR maximum also depends on particle size and shape [64]. Plasmonic properties are found not only in spherical particles but also in particles of more complex shapes, such as triangular prisms and cubes, and in hybrid structures such as metal shells on spherical particles of another material [57]. Depending on the resonator geometry, particles can show additional LSPR frequencies, and the main frequency can shift. For example, in shell structures, changing the geometry tunes the LSPR over a wide visible and infrared range [57]. As particle symmetry decreases, the number of LSPR maxima increases [64, 65], because in less symmetric particles

plasmons can oscillate in more than one direction. Sharp corners also red-shift the LSPR maximum, because they increase charge separation and correspondingly reduce the restoring force of the oscillations [59]. For the same reason, the LSPR maximum of spherical particles red-shifts as their size increases.

In this work, metal and metal–dielectric nanoparticles that show a strong localized surface plasmon resonance are called *plasmonic nanoparticles* (PNPs) for brevity. The name emphasizes the physical effect rather than the particle material (see, for example, the textbook [66]).

1.2.2 Synthesis of monodisperse silver nanoparticles

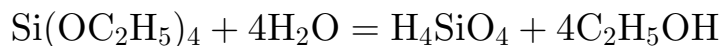
As with quantum dots, the most practically interesting synthesis methods for plasmonic nanoparticles are those that give monodisperse particles. Particles of the same size have the same frequency ω_{pl} , so one can expect the highest quality factor for an ensemble of such particles. For this reason, this work uses, as for quantum dots, colloidal methods that give monodisperse metal particles, i.e., particles whose linear size deviates negligibly from the mean.

The problems of obtaining monodisperse silver particles and ways to solve them are described in [67]. The traditional synthesis of noble metal nanoparticles by sodium citrate reduction gives monodisperse gold nanoparticles [68], but not silver ones. The main reason is secondary nucleation, caused by the slow reduction of Ag^+ ions by sodium citrate, hydrolysis of Ag^+ ions, and gradual oxidation and dissolution of freshly formed silver particles. To solve this problem, a stronger reducing agent is added to the citrate to reach the required supersaturation at the early stages of synthesis. It was demonstrated relatively recently that ascorbic acid [67] or tannic acid [69] can serve as such a strong reducing agent.

It was also shown in [67] that halide ions, as strong adsorbates on the silver surface, can stabilize certain facets and thus direct the growth of metal particles along specified crystallographic directions. Chloride and iodide ions adsorb preferentially on $\{111\}$ facets, which gives quasi-spherical particles. Bromide ions tend to adsorb on $\{100\}$ facets, which elongates the particles in one direction and leads to anisotropic growth.

1.2.3 Shells on metal particles

The most common method for growing a dielectric shell on metal nanoparticles is the modified Stöber method [70], a variant of sol–gel technology [71]. The method gives SiO₂ shells on various metal particles. The shell precursor is usually tetraethyl orthosilicate (TEOS), Si(OC₂H₅)₄, which hydrolyzes in the presence of water. If the hydrolysis of TEOS goes to completion, it can be written as:



As TEOS hydrolysis proceeds, the equilibrium shifts strongly to the right. To shift it to the left and slow down the hydrolysis, the synthesis is usually carried out in the presence of ethanol [71]. Hydrolysis is followed by condensation:



To obtain dense, low-porosity aggregates, the reaction is carried out in an alkaline medium, for example with aqueous ammonia. Typical volume ratios of TEOS, ethanol, and alkali are given, for example, in [72], where gold nanoparticles were coated with silica by this method.

1.3 Plasmonic enhancement of photoluminescence

It is well known that molecules adsorbed on a rough metal surface show surface-enhanced Raman scattering (SERS) [73–75]. Besides SERS, such structures can under certain conditions enhance not only the Raman signal but also the PL signal. The first results on luminescence enhancement of quantum dots on a rough metal surface were apparently reported in 2002 [76].

Luminescence is enhanced near nanostructured metals because of the local enhancement of the electromagnetic field near metal nanoparticles caused by LSPR. Unlike ordinary SERS, the plasmonic enhancement of luminescence depends non-monotonically on the distance between the QD and the PNP. At short distances, when the QD and the PNP are in close contact, QD emission is quenched by resonant energy transfer from the photoexcited QD to the PNP. In addition, charge carriers from electron–hole pairs generated in the semiconductor nanoparticle can tunnel from the QD to the PNP and be captured there, which leads to nonradiative recombination at the QD–PNP interface. Efficient

PL quenching is also observed for organic dye molecules [77].

However, luminescence quenching by a metal particle is a short-range effect. It weakens with distance much faster [78] than the PNP-enhanced electromagnetic field that increases the luminescence of the emitter. Therefore, the QD luminescence intensity should be maximal at some optimal distance of a few to several tens of nanometers. This distance can be fixed by placing a dielectric layer (*spacer*) between the two types of particles; the spacer also acts as a potential barrier for electron tunneling. These features of the effect are illustrated in Fig. 1.8.

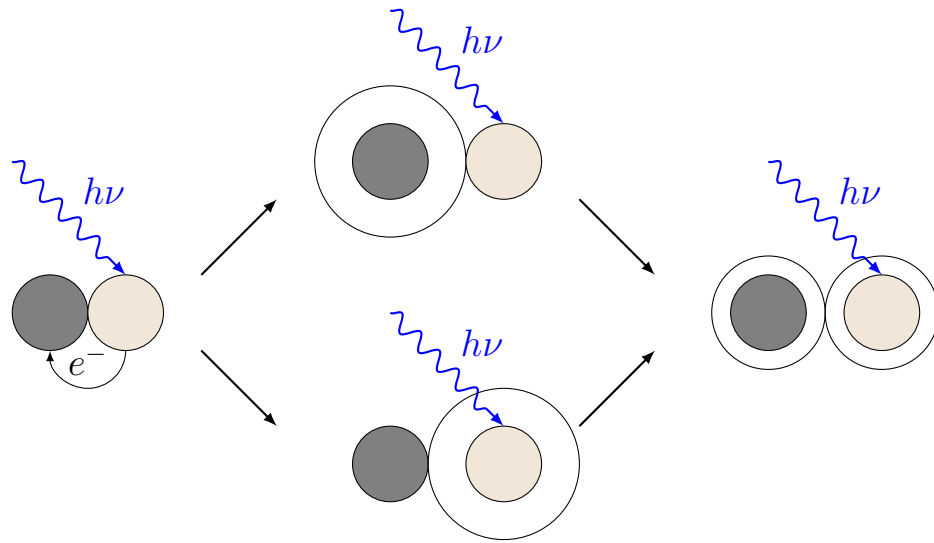


Figure 1.8. Schematic of photoinduced charge separation in a QD near a core-only and a core/shell plasmonic particle [79]

One of the main reasons for the increased PL intensity is the change of the spontaneous emission rate due to the Purcell effect [80], in which the plasmonic nanoparticle acts as the resonator. The lifetime of excited states in spontaneous emission from quantum dots is known to be about 1–10 ns. It was shown in [81] that in a hybrid quantum dot–plasmonic nanoparticle system the lifetime of excited states can be reduced substantially (down to the order of 10 ps). Faster relaxation means that more photons emitted by the QD are observed in the same time interval.

The work [82] studied the luminescence enhancement of CdSe/ZnS QDs on gold nanoparticles as a function of the distance between the semiconductor and metal particles. The separating layer between the QDs and gold particles was made of successively deposited polyelectrolyte layers, which do not absorb in the visible range and do not reduce QD emission. The PL maximum of the

quantum dots was at ~ 645 nm. Gold particles 12–15 nm in diameter with a plasmon resonance at 550 nm were used. The strongest PL enhancement occurs when the excitation frequency ω_{ex} matches the plasmon resonance frequency ω_{pl} of the gold nanoparticles. The maximum enhancement factor of 5 was reached with a separating layer ~ 11 nm thick.

The work [83] showed plasmonic PL enhancement of CdSe/ZnS and CdSe/CdS/ZnS core/shell QDs placed near silver nanoprisms. The emission intensity increased by 25% for CdSe/ZnS QDs (PL maximum at 550 nm) and by 40–50% for CdSe/CdS/ZnS.

QD luminescence can also be enhanced when the PL maximum lies in the infrared. For example, [84] showed enhancement for InAs quantum dots capped with GaAs interacting with gold particles with an LSPR maximum at 529 nm. The enhancement decreased as the GaAs layer became thicker.

Plasmonic particles can also enhance photoluminescence in bulk semiconductors: [85] showed that a thin layer of gold nanoparticles on a GaAs substrate increases its luminescence intensity threefold.

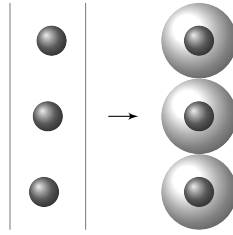


Figure 1.9. Schematic of the change from particles distributed in a layer to a layer of core/shell particles

Note that most of the works on plasmonic enhancement reviewed here used metal particle syntheses that provide neither monodisperse nanoparticles nor a uniform spacer thickness. Therefore, to study the effect of spacer thickness, this work proposes to use layers of core/shell particles instead of Ag particles embedded in a SiO_2 layer (Fig. 1.9), aiming at the smallest possible spread of both the Ag particle diameter and the thickness of their dielectric shell. This approach makes it possible to state with more confidence that the distance from the metal particle surface to the quantum dot is the same for all particles.

1.4 Conclusions to Chapter 1

Based on the literature review of the physical and chemical features of colloidal quantum dots, plasmonic nanoparticles, and plasmonic enhancement, the following specific tasks were set to achieve the objectives stated in the Introduction:

1. To implement colloidal synthesis procedures for metal and semiconductor nanocrystals, and to consider ways of improving their synthesis, characterization, and deposition on solid substrates.
2. To consider, from the physicochemical point of view, the cesium lead halide systems as the most relevant materials for colloidal quantum dots in optoelectronics.
3. To synthesize monodisperse plasmonic silver nanoparticles and to grow dielectric shells of various thicknesses in the 10–50 nm range on them. The shell material is silicon dioxide, SiO_2 , deposited on the metal nanoparticles by the modified Stöber method.
4. To fabricate ordered structures of plasmonic and semiconductor particles and study them by photoluminescence spectroscopy, to determine the optimal dielectric spacer thickness for plasmonic enhancement.

2 Development of New Methods for the Synthesis and Characterization of Nanoparticles and Layers

The second chapter describes the synthesis and characterization methods used in this work, including those developed as part of the thesis. It covers the synthesis of CdSe ($A^{II}B^{VI}$) and Ag nanoparticles, which show quantum confinement and LSPR, and the deposition and characterization of nanoparticle layers.

2.1 Nanoparticle synthesis by hot injection

Fig. 2.1 shows the setup for the colloidal synthesis of metal and semiconductor nanoparticles by the hot-injection method outlined in Section 1.1.3.

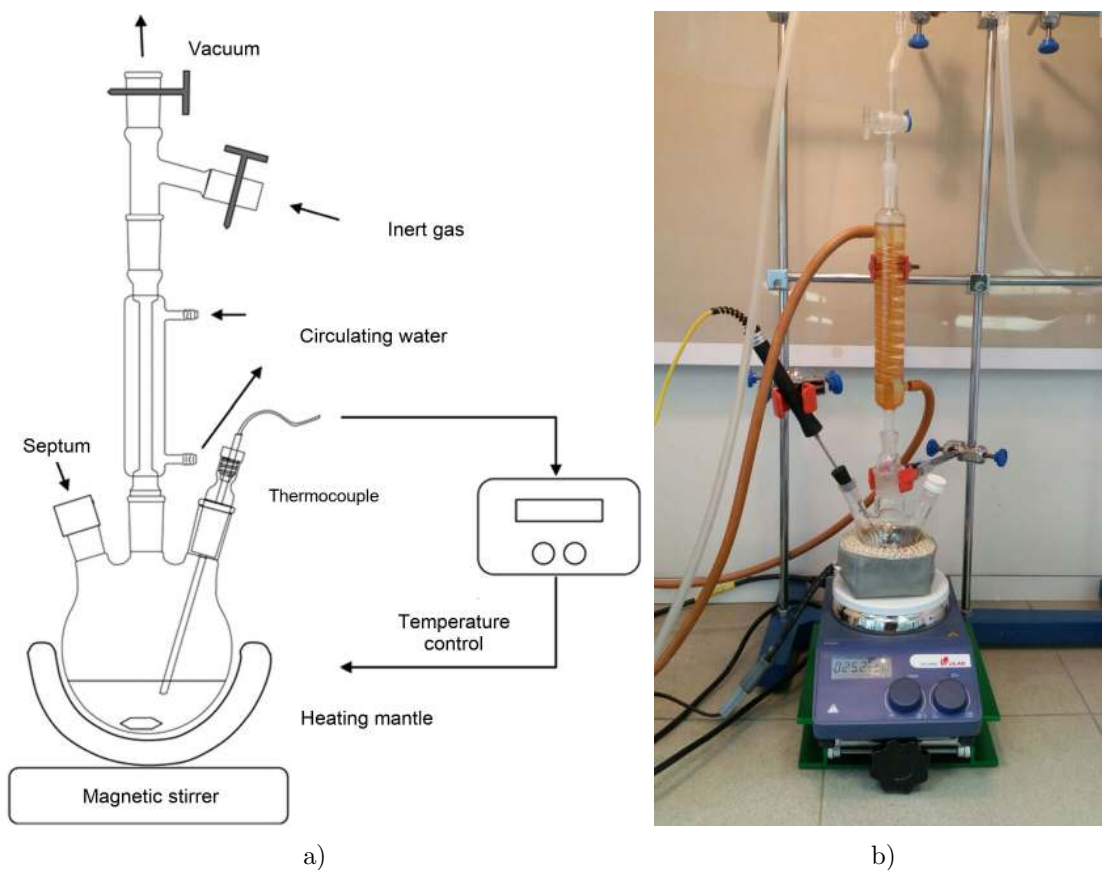


Figure 2.1. Setup for the synthesis of metal and semiconductor nanoparticles: a) schematic, b) photograph

For colloidal synthesis, the precursors needed for the chosen synthesis step and a magnetic stir bar coated with polytetrafluoroethylene (PTFE) are placed in a three-neck round-bottom flask of heat-resistant glass. The flask sits in a

heating mantle that heats liquids uniformly from room temperature to 450°C. The heating mantle with the flask stands on a magnetic stirrer, which rotates the stir bar and ensures a uniform distribution of material and temperature in the reaction medium. A chromel–alumel thermocouple is fixed in one of the flask joints. A microcontroller converts the thermocouple voltage into a temperature value shown on the controller display. With this controller, the operator sets the temperature to be maintained (T_N and T_G).

A reflux condenser is mounted in the middle neck of the flask. Running water circulates in its jacket and condenses the vapors of the heated solution components.

The free joint of the three-neck flask is used to introduce reagents at later stages. At other times it is closed with a glass stopper or a septum, a self-sealing rubber membrane. At the appropriate stage, the material is injected either manually with laboratory syringes or automatically with the removable syringe pump modules of the flow reactor described below.

An inert atmosphere is maintained during synthesis by connecting an inert gas cylinder through a two-stage pressure regulator. Since most colloidal nanoparticle syntheses run at temperatures not above 300°C, nitrogen can be used.

2.2 Flow synthesis of nanoparticles

Hot-injection colloidal synthesis, described in Section 2.1, gives encouraging results for a wide range of colloidal quantum dot and plasmonic nanoparticle materials. Its main drawback is its limited applicability in real production. The key physical and technological parameters of synthesis are the growth temperature and duration. A larger reaction volume leads to nonuniform temperature and mass distribution during mixing, which widens the final nanoparticle size distribution. Scaling up the technology therefore requires more identical setups and more operators.

The solution to this problem is to move from classical colloidal synthesis in a flask to flow synthesis. Instead of a static volume of reaction medium, a liquid flow passes through zones held at the set temperatures T_N and T_G : the nucleation and growth zones. These processes are thus separated, and their durations are set by the flow velocity and the path length in each zone. The background of this part of the work and the relevant literature were described

in our monograph [26]. Here we briefly consider the main principles of nanoparticle synthesis in a liquid flow.

Microfluidic systems are a logical choice for controlling such a flow. However, a continuous flow in a thin channel, especially of viscous liquids, develops a parabolic velocity profile, which widens the particle size distribution. In addition, a continuous flow gradually clogs the channel.

The solution to this velocity distribution problem is to segment the flow, either with an inert gas or with a carrier liquid that is immiscible with the reaction medium [86] and wets the channel walls. The latter approach also avoids reactor fouling, since the reagents moving within the inert carrier liquid do not touch the channel walls.

We propose to adapt this approach by using a PTFE capillary for nanoparticle synthesis. This allows a system architecture that is an alternative to microfluidics. Let us consider whether a reactor of one type can synthesize both metal and semiconductor nanoparticles. Fluorinated polyethers that wet the walls of the PTFE capillary can be used as the carrier liquid in such a reactor.

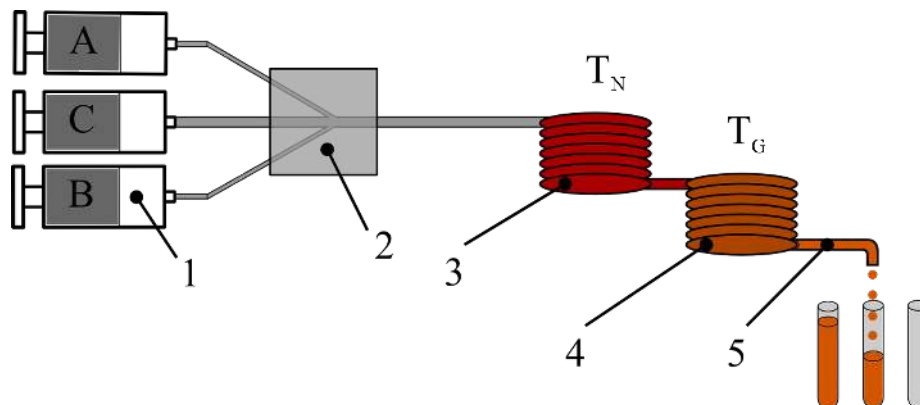


Figure 2.2. Schematic of the flow reactor: 1: syringe pumps with precursors and the carrier liquid; 2: flow mixer; 3: nucleation zone; 4: growth zone; 5: sample collection [87]

The flow reactor (Fig. 2.2) consists of three syringe pumps (for precursors *A* and *B* and the carrier liquid *C*), a mixer, two zones at different temperatures, and a capillary. In this work, the capillary was a thin round PTFE tube with an inner diameter of 0.9 mm and a wall thickness of 0.25 mm. The flows are driven by syringe pumps built from stepper motors and parts made by extrusion 3D printing. A Ψ -shaped junction served as a passive mixer (Fig. 2.3). A capillary 2 was inserted into the tube coming from the carrier liquid pump 1. The capillary had side inlets in the form of hollow stainless steel needles 3 for

independent feeding of precursors A and B . The side inlets are perpendicular to each other. The resulting droplet flow moves along the capillary through the zones at T_N and T_G , which are oil baths with thermocouple temperature control. The total capillary length is 4 m. The parts of the setup, i.e., the three syringe pumps and the temperature-controlled zones, are controlled and synchronized by purpose-built printed circuit boards and software written in NI LabVIEW.

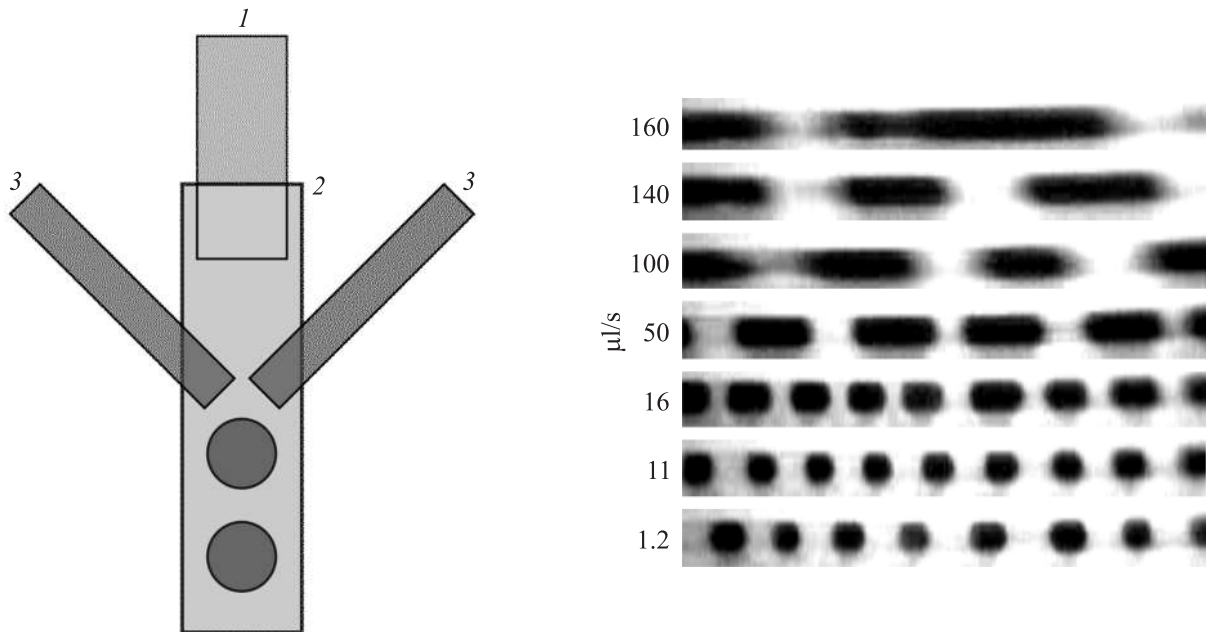


Figure 2.3. Schematic of the mixer of precursor and carrier liquid flows, and selection of flow rates (capillary under a microscope) [86]

Fig. 2.3 shows the gradual transition from continuous flow at high flow rates to segmented flow as the flow rates of both the precursors and the carrier liquid decrease (two water-soluble dyes were used as a model system). On the one hand, lower flow rates give more uniform droplets. On the other hand, at flow rates of $\sim 1 \mu\text{L/s}$, mixing slows down. By processing such images at different ratios of the flow rates of micropumps A, B, and C, flow rates of $10 \mu\text{L/s}$ were chosen as working rates for the water–octadecene liquid pair, as they give droplets of equal volume and uniform mixing. The optimal flow rate is specific to each pair of immiscible liquids.

Note that when the segmented flow leaves the capillary, the carrier liquid and the solution formed by mixing the precursors separate naturally, because the liquids differ in nature and density. The resulting solution is practically

identical in composition to the colloidal solution obtained by standard hot-injection synthesis. The carrier liquid can therefore be reused many times.

To test the setup, semiconductor cadmium selenide (CdSe) nanoparticles (Fig. 2.4) and metal silver (Ag) nanoparticles (Fig. 2.5) were synthesized.

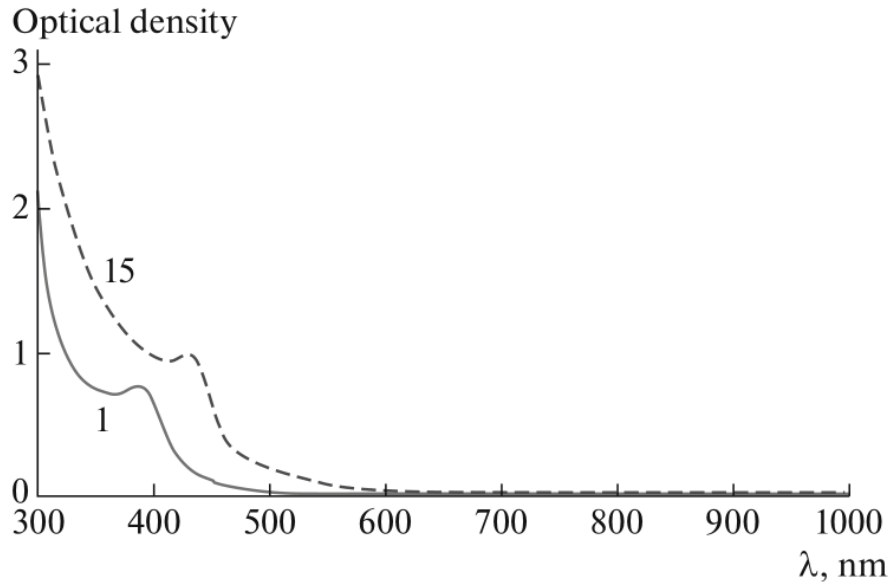


Figure 2.4. Absorbance spectra of CdSe QDs obtained by flow synthesis with different growth times (1 and 15 min) [86]

For the synthesis of cadmium selenide (CdSe) nanoparticles, a mixed solution of cadmium chloride CdCl_2 and L-cysteine (solution *A*) served as the cadmium source and stabilizer. An aqueous solution of sodium hydroselenide NaHSe , prepared immediately before the synthesis (solution *B*), served as the selenium source. Toluene was the carrier liquid. To increase the growth time, the flow was stopped for the growth period, and the solution stayed within the corresponding zone. As the particles grew, the first excitonic absorption peak shifted (Fig. 2.4). The well-defined shape of the excitonic peak indicates a narrow size distribution. There is also no light scattering in the long-wavelength region, i.e., the particles did not aggregate during synthesis. From the positions of the excitonic peaks, following [49], the mean particle diameters were calculated as 1.4 and 1.8 nm for 1 and 15 min of growth, respectively.

Note that flow synthesis is also suitable for core/shell nanostructures. Colloidal solutions of “core” nanoparticles obtained in the flow reactor and then purified can serve as one of the precursors at the shell growth stage, with sources of, for example, zinc and sulfur as the second and third precursors for

a wide-gap shell.

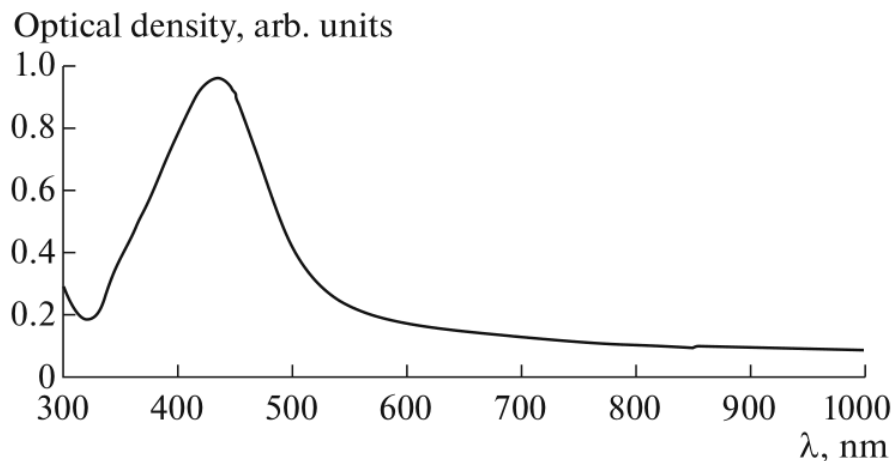


Figure 2.5. Absorbance spectrum of Ag nanoparticles obtained by flow colloidal synthesis [86]

For the synthesis of silver nanoparticles, a $1 \cdot 10^{-3}$ mol/L silver nitrate solution served as the silver ion source (solution *A*). A mixed solution of ascorbic acid and sodium citrate served as the reducing agent and stabilizer (solution *B*). Dodecane was the carrier liquid. The reaction time was 1 min. The absorbance spectrum of the nanoparticles (Fig. 2.5) shows an LSPR with a maximum at $\lambda_{\max} = 430$ nm. For silver particles, $\lambda_{\max}$ depends linearly on the diameter. From the expression given in [88], the plasmon resonance maximum corresponds to silver particles with a mean diameter of 40 nm.

These results were published in [86, 87], and the engineering solution is protected by a Russian utility model patent [21].

2.3 Isolation of nanoparticles from colloidal solutions

The standard procedure for isolating nanoparticles from a solution is centrifugation: fractions of different density are separated by the centrifugal forces arising from acceleration in a centrifuge. During centrifugation, particles sediment if their density is higher than that of the solution. The sedimentation rate depends on the centrifugal acceleration, which is proportional to the square of the angular velocity of the rotor and to the distance between the rotation axis and the particle. Centrifugal acceleration is expressed in units of the gravitational acceleration g . Table 2.1 lists the specifications of the PE-6926 centrifuge used in this work. Fig. 2.6 shows the centrifuge.

The required centrifugation time is given by:

Table 2.1. PE-6926 centrifuge specifications

Parameter	Value
Maximum rotation speed	12000 rpm
Maximum centrifugal acceleration	13400 g
Rotor type	fixed-angle
Standard rotor capacity	12 × 10 mL



Figure 2.6. Centrifuge used to isolate nanoparticles

$$t = \frac{27.4 \cdot \ln(R_{\max}/R_{\min}) \cdot \mu}{N^2 \cdot r^2 \cdot (\rho_{\text{Ag}} - \rho)} \quad (2.1)$$

where t is the separation time in minutes, μ is the dynamic viscosity of the liquid in poise, N is the rotation speed in revolutions per minute, r is the mean particle radius in centimeters, ρ_{Ag} is the density of the particle material in g/cm^3 , ρ is the density of the liquid in g/cm^3 , $R_{\max}$ is the distance from the bottom of the suspension to the rotation axis in centimeters, $R_{\min}$ is the distance from the top of the suspension to the rotation axis in centimeters. For the PE-6926 centrifuge, $R_{\max} = 7$ cm and $R_{\min} = 4.5$ cm.

For example, for the cesium lead halide perovskite nanocrystals discussed in Chapter 3, a two-step precipitation procedure was used. In the first step, the largest particles were separated by centrifugation for 5 min at 10000 rpm. In the second step, the solution was centrifuged for 10 min at 4000 rpm with isopropyl

alcohol added in a 1:1 ratio, to precipitate the particles and remove reaction by-products and unreacted components. After purification, the nanocrystals are usually redispersed in nonpolar volatile solvents, most often hexane or toluene, to avoid changes in their properties in the presence of oxygen.

2.4 Films and ordered structures on substrates

A technological advantage of colloidal synthesis is that it gives monodisperse nanoparticles. Its main drawback compared with solid-state methods of fabricating quantum-size structures is that the particles are initially not bound to any substrate. The active layers of electronic devices are usually solid. Therefore, an important part of nanocrystal technology for electronic applications is depositing the particles onto a solid substrate. As the main deposition methods used in this work, let us consider spin coating and, in more detail, layer-by-layer deposition.

Table 2.2. MiniSpin centrifuge specifications

Parameter	Value
Capacity, mL	$12 \times 1.5/2.0$
Maximum rotation speed, rpm	13400
Maximum acceleration, g	12100
Timer, min	up to 30



Figure 2.7. Centrifuge used to deposit particles on solid substrates

Layers of plasmonic particles and colloidal quantum dots on quartz substrates were made with a MiniSpin centrifuge fitted with a custom substrate

holder. Table 2.2 lists the centrifuge specifications, and Fig. 2.7 shows the device.

Dip coating is one of the well-known methods for making layers of micro- and nanoparticles. The first patent on this method, dated 1939 [89], covers silica films made from the corresponding colloidal solution. The main idea of dip coating [90], used to synthesize solids with a regular structure, is the successive growth of layers of a given chemical composition on a solid substrate by dipping it into a colloidal solution that contains the particles or the film-forming precursors. During deposition, the film forms through a combination of processes such as gravitational draining, solvent evaporation, and condensation. The final film microstructure depends on the substrate material and morphology, the structure of the initial sol, the rates of condensation and aggregation in the sol, the rates of condensation and solvent evaporation, the deposition time, and the capillary and shear forces.

The substrate is usually held vertically (or at an angle to the vertical) and withdrawn from the solution at a set speed, carrying particles from the boundary layer. As the solvent drains and evaporates, an interface forms between the dried film and the solution. When the evaporation rate equals the withdrawal rate, the process reaches a steady state.

The proposed setup consists of an electromechanical part, which positions the substrate holder in space; a temperature control part, which keeps the solutions at user-set temperatures; and a control system for the mechanics and temperature. These parts are controlled and linked by an Arduino Mega microcontroller. It executes the layer deposition routine set by the operator in purpose-built software written in NI LabVIEW. The automated system includes the layer-by-layer thin-film deposition setup placed in an inert glove box, an inert gas supply, and the corresponding temperature, humidity, and oxygen level control systems. The block diagram of the setup is shown in Fig. 2.8.

Fig. 2.9 shows the layer-by-layer deposition setup. The control unit with the electronics is mounted on an acrylic base, together with the horizontal positioning unit, which consists of a stepper motor, guide rails on supports, and a belt drive. The rotation of the motor shaft is converted into the linear motion of a carriage attached to the belt. The carriage carries the vertical positioning unit for the sample. It uses the same components as the horizontal unit, but

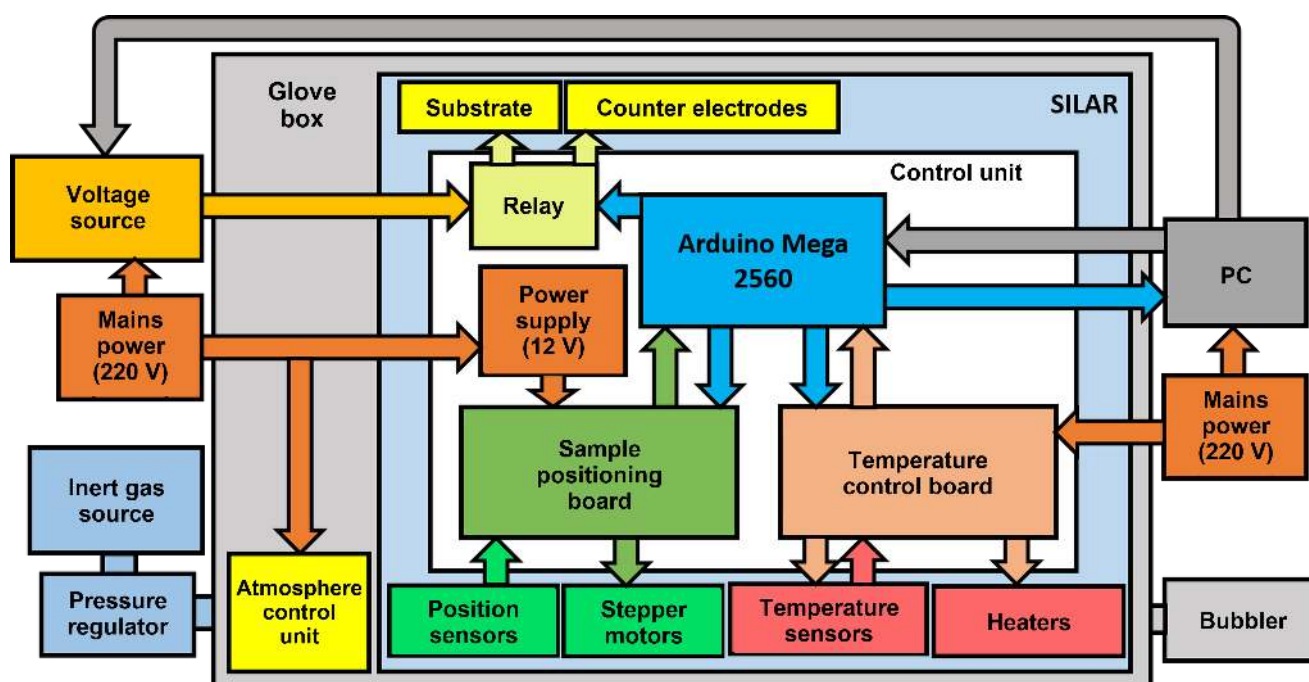


Figure 2.8. Block diagram of the system for layer-by-layer thin-film deposition in an inert atmosphere

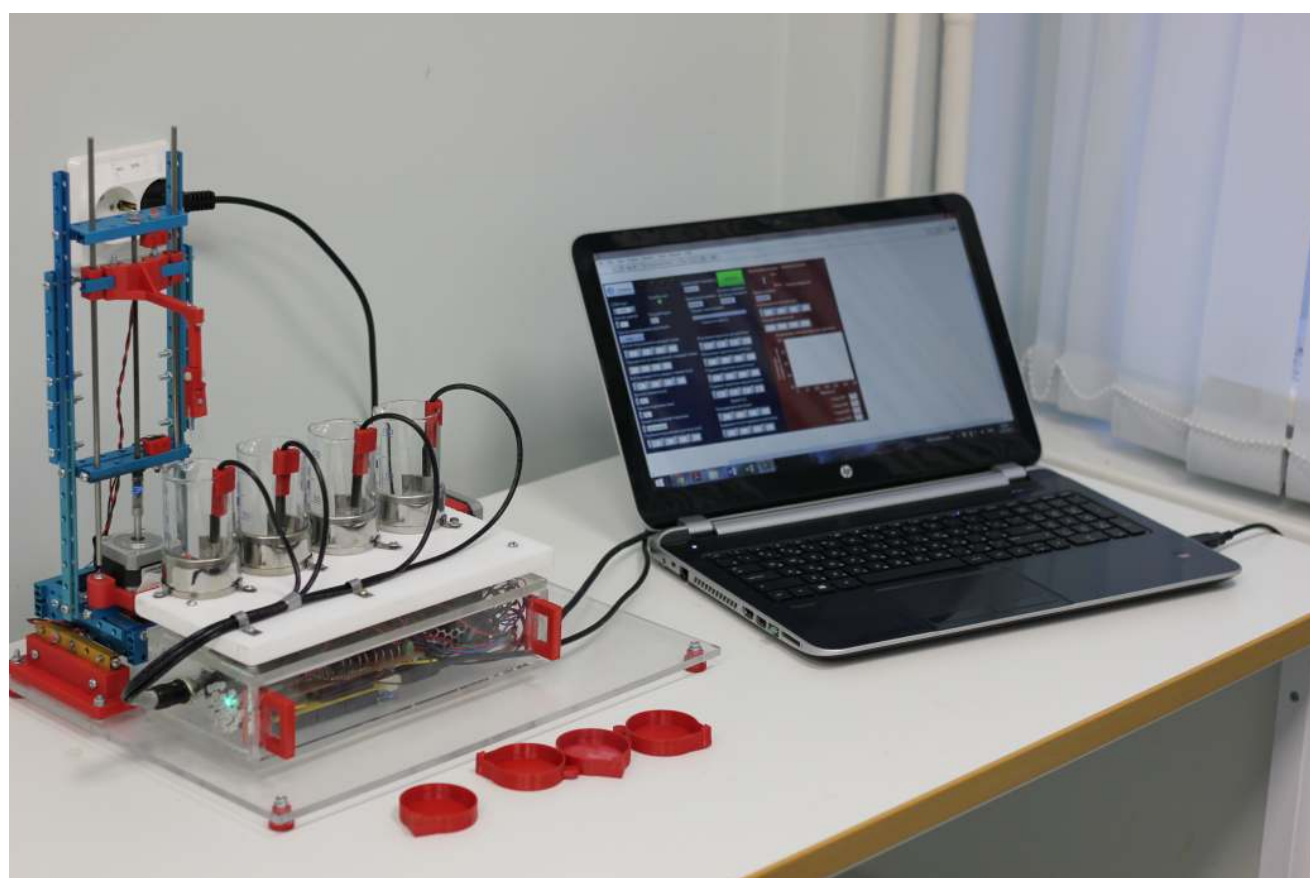


Figure 2.9. Setup for layer-by-layer deposition of nanoparticles from colloidal solutions

with a lead screw drive for more precise positioning. Microswitches at the ends of the vertical and horizontal rails detect the sample position. The rod of the

vertical positioning unit carries a sample holder with an electrical contact, so that an external field can be applied. Above the control unit, on a chemically inert PTFE plate, there are four ring heaters that also hold the solution vessels. The vessels contain removable temperature sensors and counter electrodes.

The control unit contains a microcontroller (MCU), boards for controlling the vessel temperatures and the sample position, a power supply for the stepper motors, and a double-pole relay for switching the polarity of the electrode potentials. Through the temperature control board, the MCU receives the solution temperatures and controls the heaters. The positioning board sends control signals to the stepper motor drivers, which rotate the motors through programmed angles. Through the same board, the MCU monitors the extreme vertical and horizontal positions of the sample by the microswitch signals. The double-pole relay and an external programmable DC voltage source connected to the setup set the voltage between the electrode in the substrate holder and the counter electrode in the vessel. The MCU and the DC voltage source are connected to a computer via USB.

To deposit layers, a pre-cleaned and chemically activated solid substrate is fixed in the holder, and the temperature sensors and electrodes are placed in the vessels, which are filled with solutions. In the software, the operator sets the number of cycles; the number of vessels used (up to four); the potential of the working electrode relative to the counter electrode; the solution temperatures; the times the sample spends inside and outside each vessel; and the number, speed, and depth of dips into each vessel. The sample positioning accuracy relative to the reagents is 20 μm , the substrate speed in solution is 0.5 to 3.0 mm/s, the maximum substrate size is $30 \times 40 \times 2$ mm, the minimum solution volume is 4.5 mL, and the solution temperature range is 20 to 90 $^{\circ}\text{C}$. The setup is shown in Fig. 2.9.

Note that the setup can also deposit films from true and colloidal solutions by other methods: successive ionic layer adsorption and reaction [91], electrochemical atomic layer deposition [92], dip coating [93], and chemical bath deposition [94], as well as form solid porous structures by electrochemical etching [95]. Examples of thin films deposited from solutions of complex compounds were described in [96], and the design and applications of the setup itself for thin films in [97]. The engineering solution is protected by a Russian utility

model patent [22].

2.5 Measurement of transmission and luminescence spectra

Transmission/absorbance spectra were measured on two spectrophotometers, PE-5400UV and SF-256BIK. They are built on single-beam and double-beam schemes, respectively, and cover the 190–1000 nm and 1000–2700 nm ranges.

The general scheme of transmission measurements is shown in Fig. 2.10. Radiation selected by the monochromator M from the spectrum of the light source L (a halogen or deuterium lamp) falls on the sample S . The light transmitted through the sample reaches the photodetector D , whose signal is amplified and processed by the electronics E . The transmittance is determined relative to a reference sample S' : the bare substrate on which the studied layer is deposited, or a cuvette with the pure solvent that serves as the dispersion medium for the nanoparticles. In the single-beam scheme, S' is measured before S ; in the double-beam scheme, S and S' are measured simultaneously.

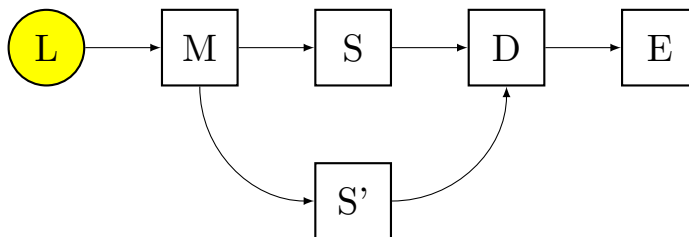


Figure 2.10. Scheme of transmission spectrum measurement

The instruments compare the light flux Φ_0 transmitted through S' with the light flux Φ transmitted through the studied sample S . The photodetector converts Φ_0 and Φ into electrical signals I_0 and I . The dark signal I_d of the unilluminated detector is also measured. From these signals, the transmittance T and the absorbance A are calculated:

$$\tau = \frac{\Phi}{\Phi_0} = \frac{I - I_T}{I_0 - I_T},$$

$$T = \tau \cdot 100\%,$$

$$A = \log_{10} \frac{1}{\tau} = \log_{10} \frac{100\%}{T}.$$

The scheme of luminescence measurements is shown in Fig. 2.11. Comparison with the scheme in Fig. 2.10 shows that luminescence measurement differs in that the monochromator M and the sample S swap places.

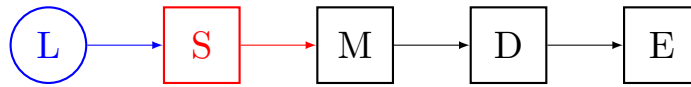


Figure 2.11. Scheme of luminescence spectrum measurement

Photoluminescence spectra are measured with a low-power 405 nm laser (maximum power < 10 mW) from Shawo Hwa Industrial Co, an MDR-206 monochromator, and a silicon photodiode with a preamplifier (LOMO Photonica). For most measurements, a low-power laser is used to avoid changing the physicochemical properties of the materials during measurement. Table 2.3 lists the monochromator specifications, and Fig. 2.12 shows the instrument.

Table 2.3. MDR-206 monochromator specifications

Specification	Value
Spectral range, nm	190–5000
Focal length of the mirror objectives, mm	180
Relative aperture	1:4
Overall dimensions, mm, max	$310 \times 240 \times 170$



Figure 2.12. Spectrofluorometer based on the LOMO Photonica MDR-206 monochromator. A silicon photodiode with an amplifier is mounted at the exit slit

The monochromator uses interchangeable diffraction gratings. Table 2.4 lists the parameters of the grating used in this work.

Photoluminescence spectra of colloidal quantum dot solutions and of solid samples are measured with the holder and the special accessory shown in

Table 2.4. Parameters of the diffraction grating

Parameter	Value
Grooves per millimeter	1200
Working range, nm	200–1250
Blaze wavelength, nm	500
Reciprocal linear dispersion, nm/mm	4.3

Fig. 2.13 and Fig. 2.14, respectively. The accessory for solid samples allows the optical scheme to be adjusted with three degrees of freedom, by linear movement of 1) the sample, 2) the laser, and 3) the mirror. The accessory implements the following excitation scheme: the laser beam hits the mirror, the reflected beam falls on the sample and excites its photoluminescence, and the emitted light enters the entrance slit of the monochromator and is then analyzed as shown in Fig. 2.11.

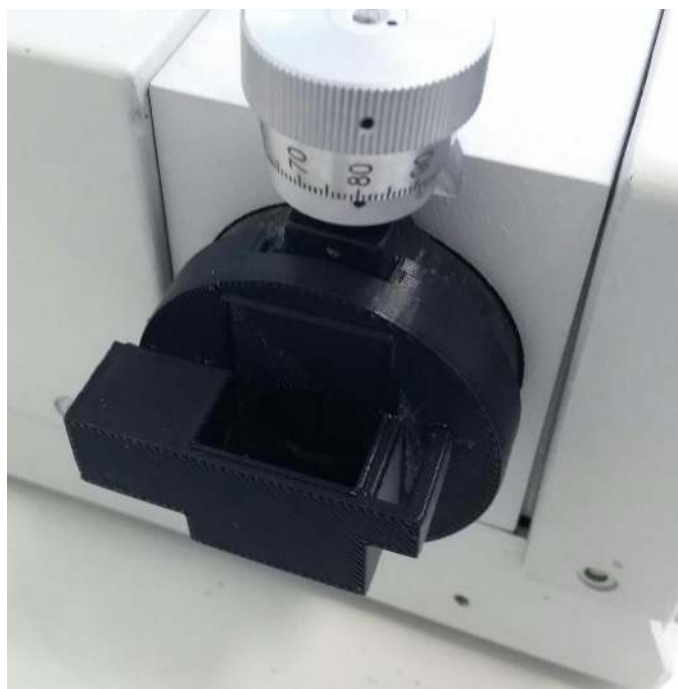


Figure 2.13. Holder for photoluminescence measurements of solutions

To study how the optical properties of the synthesized materials and structures depend on ambient temperature, and to keep samples at a set temperature, the accessory described below was built.

The temperature control accessory sits in the cuvette compartment of the spectrophotometer in the beam path. It consists of two Peltier modules with a hole, pressed tightly against the sample from both sides to heat or cool it. Heat sinks are mounted on the outer sides of the Peltier modules. A DS18S20

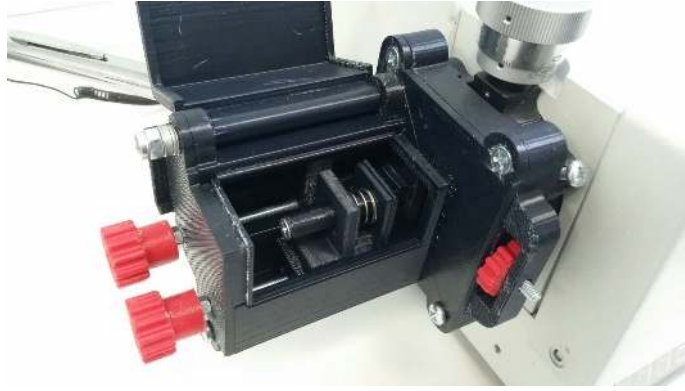


Figure 2.14. Accessory for photoluminescence measurements of solid samples

temperature sensor sits in a purpose-made metal holder. The structural parts of this accessory, like those of the PL accessories described above, were made by extrusion 3D printing [98]. The schematic of the accessory, its appearance, and its position in the spectrophotometer are shown in Fig. 2.15, Fig. 2.16, and Fig. 2.17, respectively.

The accessory works as follows. A programmable power supply applies a voltage of 0.3–5 V to the Peltier elements, and the sample temperature is measured by a DS18B20 sensor. The accessory is controlled by purpose-built software in NI LabVIEW, which uses an Arduino Uno microcontroller to provide feedback between the Peltier power supply and the temperature sensor. The accessory allows measurements at 25–90°C with an accuracy of 1°C.

2.6 Measurement of light-scattering indicatrices

Preparing colloidal solutions and depositing nanoparticle films from them requires continuous monitoring of nucleation/growth and deposition. For this, we proposed and built an experimental setup for studying the synthesis of nanostructured materials by the method of light-scattering indicatrices, i.e., the angular distributions of the intensity of electromagnetic radiation scattered by particles.

Scattering of optical radiation by small objects is usually described by Rayleigh theory [99] for the smallest particles and molecules, or by the general Mie theory [100] for larger objects. Rayleigh theory is a limiting case of Mie theory and gives the following angular dependence of the scattered intensity:

$$\frac{I_s}{I_0} = \frac{8\pi^4 d^2}{r^2 \lambda^4} (1 + \cos^2 \theta), \quad (2.2)$$

where I_s and I_0 are the intensities of the scattered and incident light, respectively, with wavelength λ , θ is the scattering angle, d is the size of the scattering particles. Note the quadratic dependence of the scattered intensity on particle size and the symmetric shape of the dependence in polar coordinates. However, if d is comparable to λ , expression (2.2) no longer holds, and a more rigorous treatment by Mie theory is needed.

The setup is shown schematically in Fig. 2.18. Sixteen radially arranged photodiodes serve as photodetectors. They record the intensity of light scattered by the sample at different angles to the laser beam. With the detector arrangement shown in Fig. 2.18, scattering indicatrices can be built for both reflected and transmitted light, so the setup works with both transparent and reflecting samples (Fig. 2.19).

Scattering indicatrices are measured as follows. Radiation from the laser 1 is directed at the sample 2. The radiation scattered by the sample is received by the radially arranged photodiodes 3. The photodiode signals are amplified by a circuit on the printed circuit board 4 and passed to the microcontroller board 5, which acts as an analog-to-digital converter and sends digital data to the computer 6.

The experimental setup is shown in Fig. 2.20. The axis of the arc coincides with the laser beam. Rotating the arc with the photodiodes gives three-dimensional scattering indicatrices, which is of interest for studying solid anisotropic materials.

The radiation source is a KLM-H650-40-5 continuous-wave laser module with a wavelength of 650 nm and an output power of 40 mW. A laser module was chosen because such a setup needs a directional beam pattern and a stable beam intensity. On the choice of wavelength: measuring ever smaller objects requires ever shorter wavelengths, but longer-wavelength radiation, with its lower photon energy, has less physical effect on the sample. In particular, the method can then be applied to a wider range of materials without exciting the sample's own fluorescence. For the same purpose of minimizing the effect on the sample, a low-power laser was used, and the measurement accuracy was achieved through the sensitivity of the detectors.

The detectors of scattered radiation are fast silicon SFH 203 P PIN photodiodes (OSRAM, Germany), connected through an operational amplifier and

a resistor in a current-to-voltage converter circuit. The output voltage of the operational amplifier is thus proportional to the incident light intensity. An Arduino Mega 2560 microcontroller platform serves as the analog-to-digital converter; it has 16 analog inputs with 10-bit resolution and a 0–5 V input range. The setup is housed in a black ABS plastic case made by 3D printing [98]. The case includes interchangeable holders for different measurement configurations and a cover that shields the sample from ambient light during measurement.

Experimental data are processed automatically by purpose-built software written in NI LabVIEW with the LabVIEW Interface for Arduino. For this, code that provides a virtual interface between the board and the software is loaded into the MCU once. The program reads the voltage of each photodiode and plots it in polar coordinates: the scattered intensity at each angle is plotted at the known angular position of the corresponding photodiode. The experimental data are interpolated with cubic splines in a Python program using the Matplotlib, NumPy, and SciPy libraries. These results were published in [101], and the engineering solution is protected by a Russian utility model patent [23].

Fig. 2.21*a* shows an example set of indicatrices for the growth of silica nanoparticles in solution by the Stöber method. This set shows the transition from the shape of Rayleigh scattering at the nucleation stage to the shape described by the Mie model in later indicatrices. After 60 min of synthesis, the shape and size of the indicatrices stop changing, which indicates that nanoparticle growth has ended.

Fig. 2.21*b* shows an example of monitoring the chemical bath deposition of lead sulfide layers [102], which form a continuous opaque film on a glass substrate, with light-scattering indicatrices. As the film grows, the indicatrix area changes monotonically, because the growing layer of narrow-gap semiconductor gradually absorbs the radiation.

For a more detailed description of the physicochemical processes involved, we refer the reader to [101].

2.7 Conclusions to Chapter 2

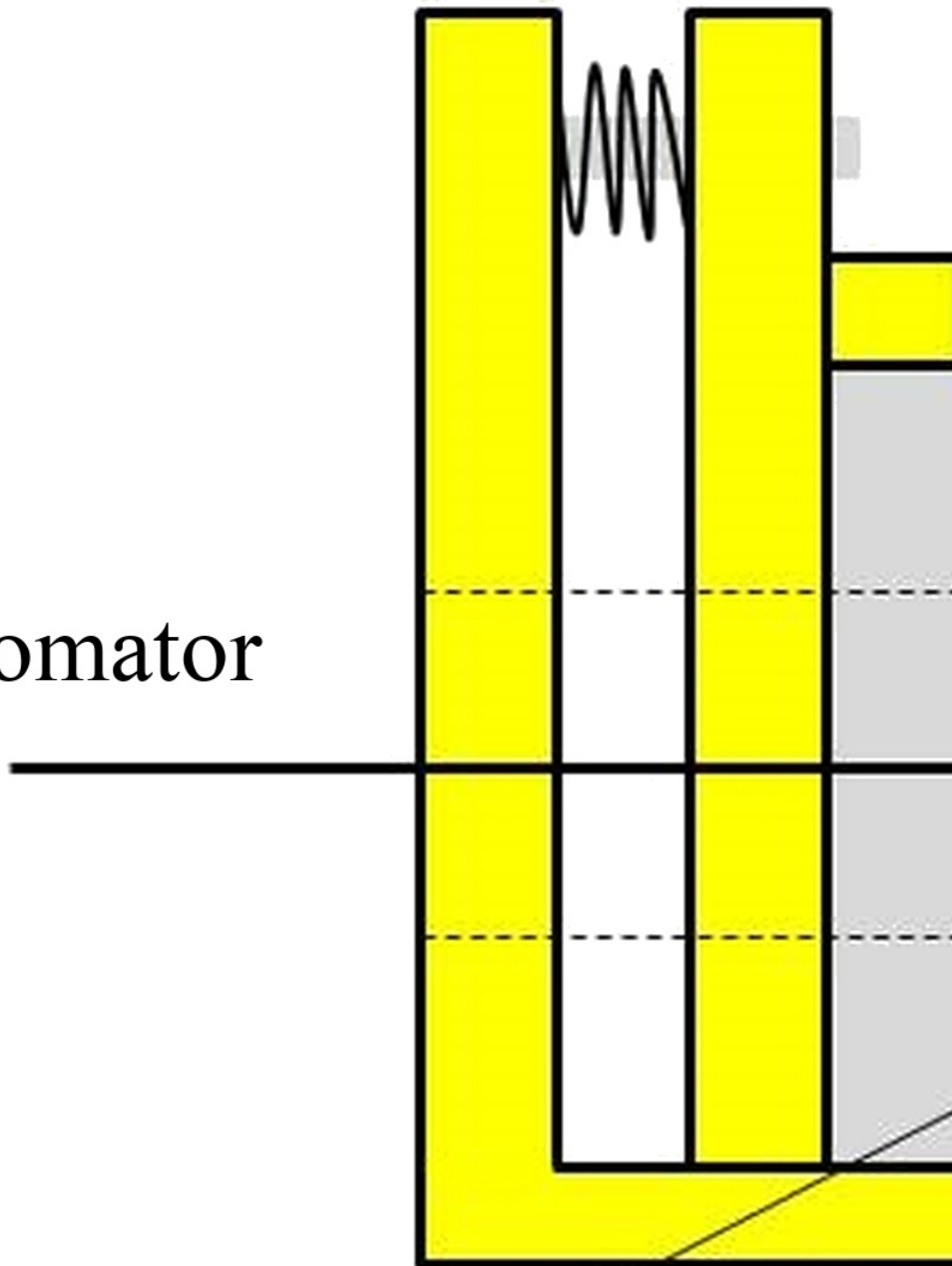
1. The colloidal synthesis setup described above, which provides uniform heating of the solution and synthesis in an inert atmosphere, was chosen for

particle synthesis. For continuous synthesis of semiconductor and metal nanoparticles, a reactor for nanoparticle synthesis in a flow of immiscible liquids was proposed and built.

2. Centrifugation was proposed for purifying particles, isolating them from colloidal solutions, and forming ordered structures of plasmonic particles and quantum dots. As an additional method for depositing ordered layers, an automated setup for layer-by-layer deposition of nanoparticles from colloidal solutions was designed and built.
3. Optical methods were chosen as the main measurement methods: transmission and photoluminescence spectroscopy. A low-power 405 nm laser was chosen as the main excitation source; its wavelength is close to the LSPR of the small (≈ 20 nm) silver particles (403 nm).
4. Measurement of light-scattering indicatrices was proposed and implemented as a method for continuous monitoring of nanoparticle synthesis and deposition.

Peltier elements

Monochromator



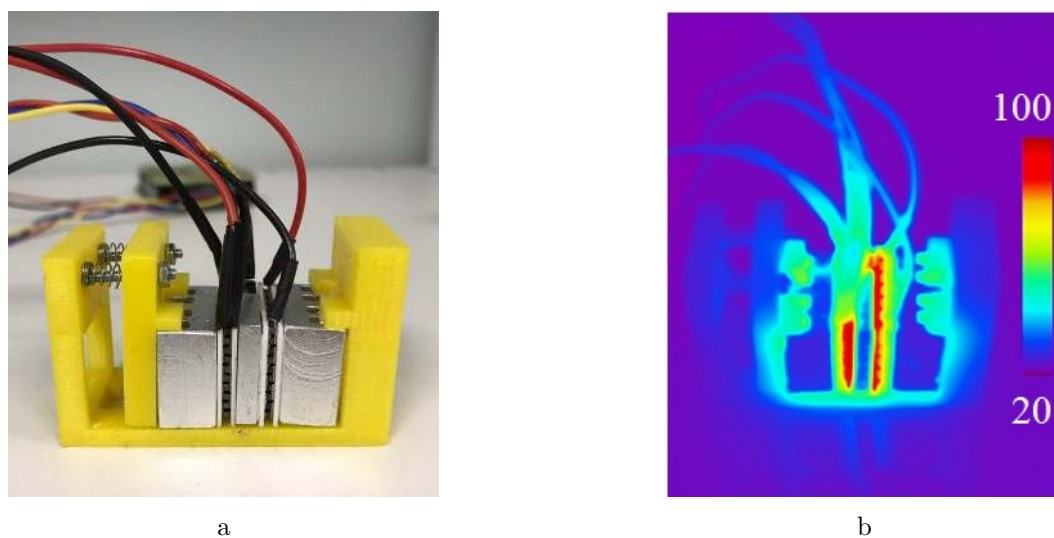


Figure 2.16. Temperature control accessory: a) photograph, b) thermal image



Figure 2.17. Temperature control accessory in the PE-5400UV spectrophotometer

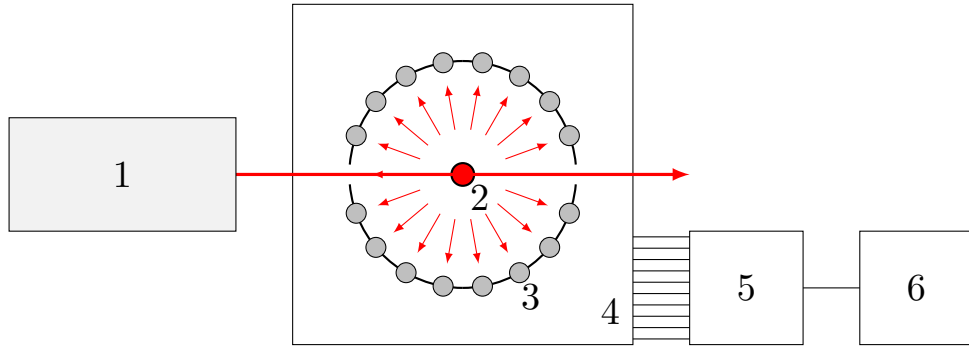


Figure 2.18. Schematic of the setup for measuring light-scattering indicatrices [101]

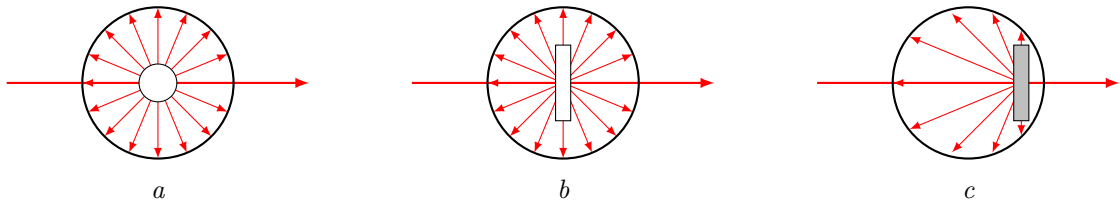


Figure 2.19. Light-scattering indicatrix measurement schemes: a) colloidal solution, a layer on a transparent (b) and a reflecting (c) substrate [101]

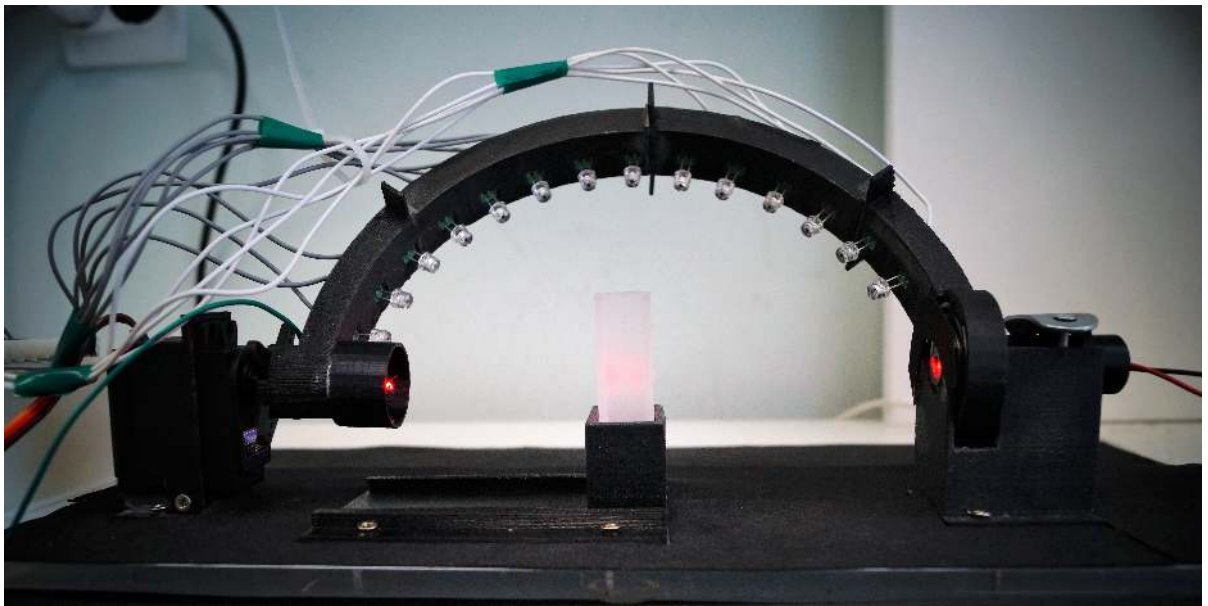
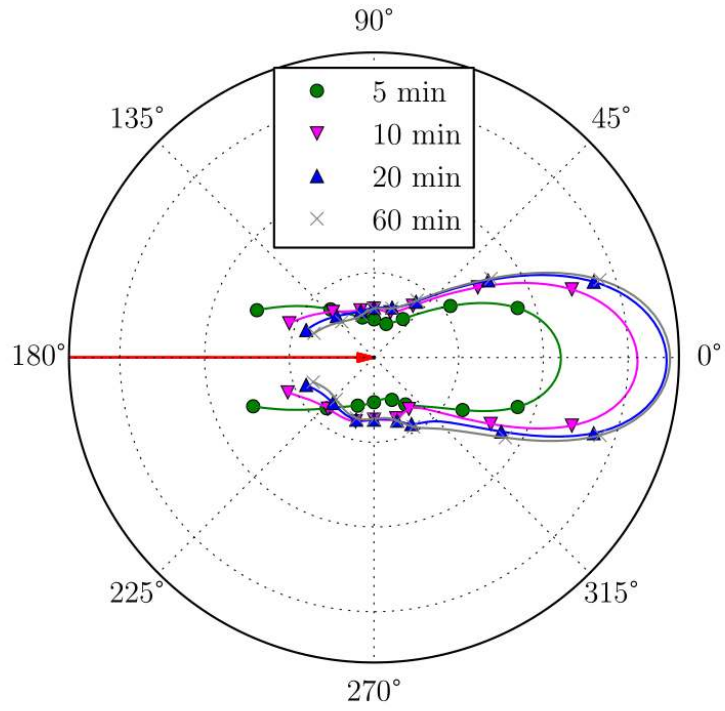
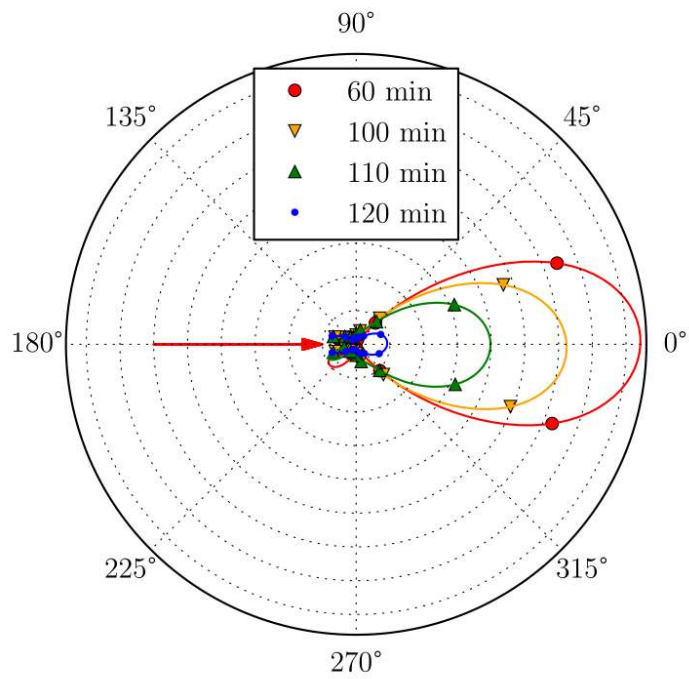


Figure 2.20. Experimental setup for measuring light-scattering indicatrices



a)



b)

Figure 2.21. Sets of light-scattering indicatrices for a) the growth of SiO_2 particles by the Stöber method, b) the growth of a PbS layer by chemical bath deposition [101]

3 Perovskite QDs of Ternary Cesium Lead Halides and Their Solid Solutions

Besides various technological improvements, the efficiency of devices based on colloidal quantum dots can be increased by searching for previously unstudied materials, in particular ternary compounds. Let us consider how such materials are analyzed, using perovskite nanocrystals of the cesium–lead–halogen system as an example. Practical interest in this group of materials comes, on the one hand, from the advances in perovskite photovoltaics [103] and, on the other hand, from the possible use of such colloidal quantum dots in LEDs [104] and lasers [105].

3.1 Analysis of ternary compounds. Triangulation of the Cs–Pb–X system

Fig. 3.1 shows the cubic perovskite crystal lattice with the general formula ABX_3 . In cesium lead halides, the A site is occupied by the cesium ion Cs^+ , B by the lead ion Pb^{2+} , and X by a halide ion (iodide, bromide, chloride). In the organic–inorganic perovskites popular in solar energy, the A site is occupied by an organic group, most often methylammonium, CH_3NH_3^+ . For solar cells, the CsPbX_3 system ($X = \text{Cl}, \text{Br}, \text{I}$) therefore has the advantage of being less prone to structural degradation, since it cannot suffer from deprotonation of the organic group [106] typical of organic–inorganic perovskites. The inorganic nature of cesium lead halides also explains their better thermal stability compared with hybrid perovskites [107].

One of the criteria used to assess the stability of the crystal structure is the Goldschmidt tolerance factor:

$$t = \frac{r_A + r_X}{\sqrt{2}(r_B + r_X)},$$

where r_A , r_B , r_X are the ionic radii of the corresponding ions. The lattice geometry (Fig. 3.1) implies that $t = 1$ in an ideal cubic crystal. Cesium lead halides have calculated t values in the range $t = 0.71$ – 0.9 , where other crystal symmetries are possible: besides the cubic perovskite, the cesium–lead–halogen system also shows tetragonal, orthorhombic, and monoclinic crystal systems [108]. The

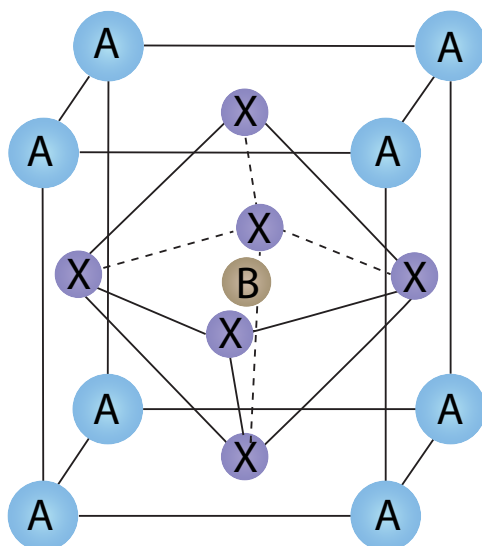


Figure 3.1. Perovskite crystal lattice [108]

cubic lattice, as the most symmetric of these, is the highest-temperature and least stable phase. Møller noted this [109] when he synthesized the first bulk crystals of CsPbCl_3 and CsPbBr_3 from aqueous solutions in the 1950s; he also demonstrated the photoconductivity of these materials.

Note that, from the physicochemical point of view, many ternary systems, unlike binary ones, are relatively poorly studied, because the necessary analysis is difficult. The cesium lead halides are no exception. To study multicomponent inorganic compounds, physicochemical analysis has developed ways to divide complex phase diagrams into simpler elementary diagrams using quasi-binary sections [110]. For ternary systems, this procedure is called singular triangulation. Quasi-binary sections are polythermal sections that have the properties of binary phase diagrams: every composition of the liquid phase corresponds to an equilibrium composition of the solid phase lying on the same section. A complete triangulation of the Cs–Pb–X systems is currently difficult to construct. We therefore present our considerations and discuss the physicochemical features of these ternary systems (Fig. 3.2 and Fig. 3.3).

In the triangulation method, the composition of a ternary system is represented by a point inside an equilateral triangle, the Gibbs triangle. The sum of the perpendiculars dropped from an interior point to the sides of an equilateral triangle equals its height. Therefore, each point inside the triangle corresponds to a certain composition of the system; in particular, the vertices correspond

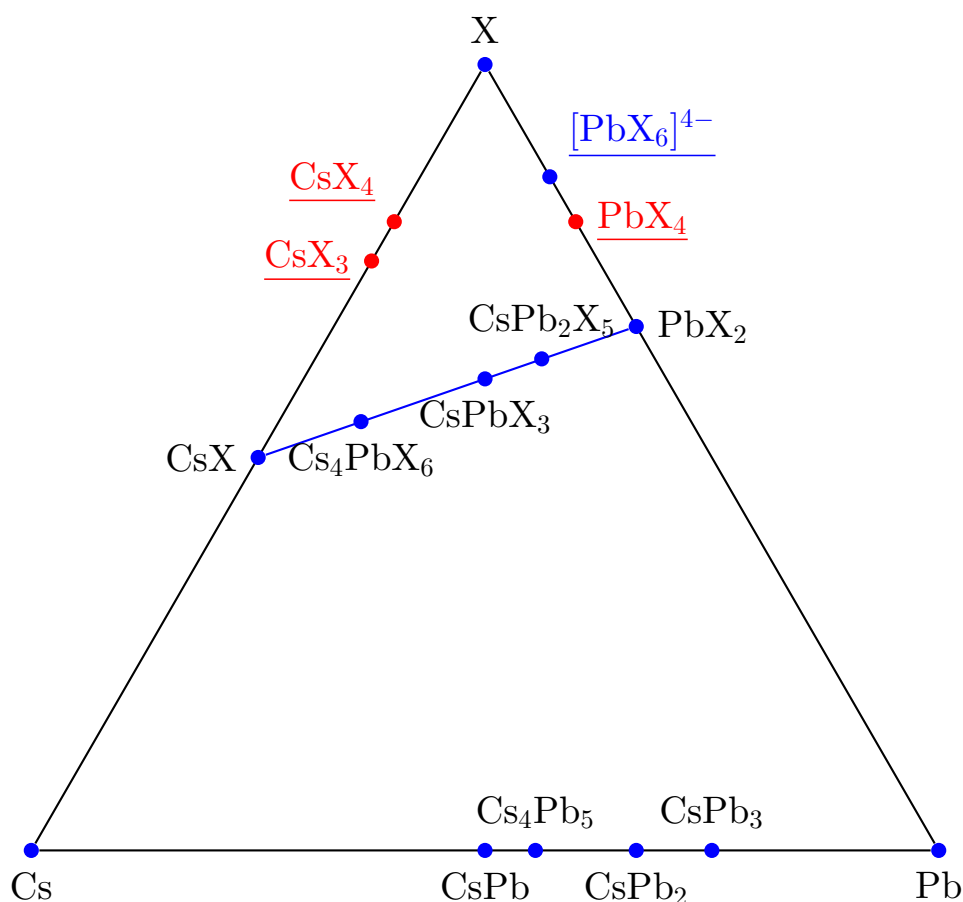
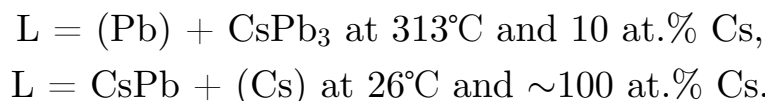


Figure 3.2. Main compounds of the ternary cesium lead halide systems on the Gibbs triangle (the underlined binary compounds do not exist for every halogen X among Cl, Br, I) [108]

to the pure components.

First, let us plot the composition points of the binary compounds on the corresponding sides of the Gibbs triangle. For the binary cesium–lead system, a number of intermetallic compounds are known [111]. The equiatomic compound CsPb melts congruently (653°C). The other compounds on this side form peritectically: Cs₄Pb₅ melts at 592°C, CsPb₂ at 560°C, and CsPb₃ at 345°C. Two eutectics crystallize in the system:



For the binary metal–halogen systems, the following cesium and lead(II) halides are known: CsCl, CsBr, CsI, PbCl₂, PbBr₂, PbI₂. Lead(IV) chloride, PbCl₄, decomposes to PbCl₂ and Cl₂ on the slightest heating, and PbBr₄ and PbI₄ do not exist. The only lead(IV) halide stable at room temperature is the tetrafluoride, PbF₄.

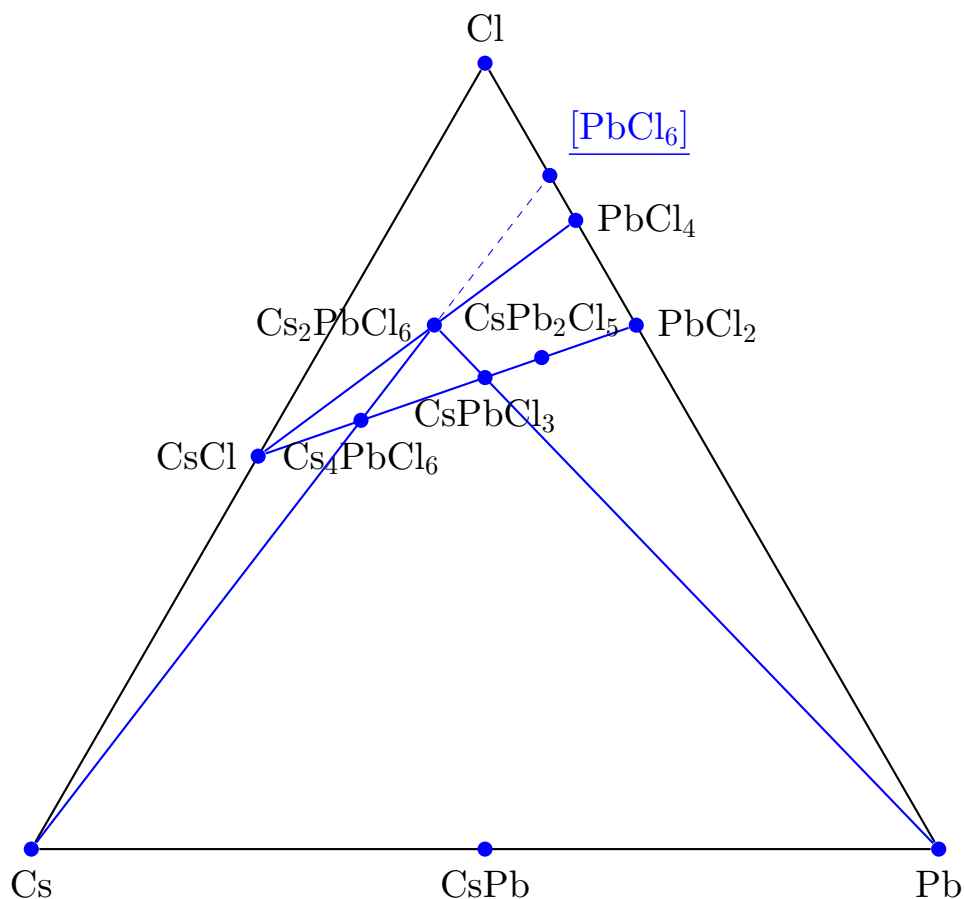


Figure 3.3. Partial triangulation of the system, using CsPbCl_3 as an example [108]

Bromine and iodine form the coordination polyhalide compounds CsBr_3 , CsI_3 , and CsI_4 . The trichloride ion Cl_3^- is unstable, and its compounds with cesium are unknown. The stability and number of possible coordination complexes increase with the atomic number of the halogen [112].

Thus, for none of the halogens taken separately does the triangulation contain the sections $\text{CsX}_3\text{--PbX}_4$ or $\text{CsX}_4\text{--PbX}_4$. The main section, present in all the systems, is CsX--PbX_2 . This section contains the composition point of the CsPbX_3 perovskites, as well as the points of the thermodynamically less stable compounds Cs_4PbX_6 and CsPb_2X_5 [113]. These composition points correspond to different molar ratios CsX:PbX_2 : 1:1, 4:1, and 1:2, respectively (Fig. 3.2).

Note that the polyhalides matter for crystal growth in solution and for Grätzel cells based on ternary compounds [114] and their solid-state analogues. Polyhalides are in fact coordination compounds. So is the structural unit $[\text{PbX}_6]^{4-}$, which is geometrically an octahedron of a central lead ion coordinated

by six halogen atoms (Fig. 3.1). This element, unusual for classical triangulation, may play a decisive role in the growth of perovskite crystals from solution. In addition, the $[\text{PbX}_6]^{4-}$ octahedron is responsible for efficient light absorption and luminescence in this group of materials: the band-edge states that define the energy gap of the material are formed by lead and halogen orbitals [13].

Comparing the Gibbs triangles in Fig. 3.2 and Fig. 3.3, one can see that a hypothetical section starting at the Cs point and passing through the composition points of Cs_4PbCl_6 and Cs_2PbCl_6 reaches the composition point of $[\text{PbCl}_6]$. This suggests that a low-temperature route to perovskite-based solid solutions is possible, which was confirmed in 2015 by the group of M. Kovalenko [13].

The $[\text{PbX}_6]$ composition point indicates that in liquid-phase methods the structural elements are preassembled as $[\text{PbX}_6]^{4-}$ anionic octahedra, which may contain halide ions of one or several types. In the presence of charge-balancing Cs^+ cations, the structure assembles into the perovskite lattice. Note that, because of this ionic bonding, the CsPbX_3 materials have a drawback: they are sensitive to moisture.

The precursors for CsPbX_3 nanocrystal synthesis are a cesium oleate solution (usually obtained by decomposing cesium carbonate Cs_2CO_3 in the presence of oleic acid in octadecene) and, for lead and halogen, a solution of the lead halide PbX_2 in octadecene. The injection is carried out at 140–200 °C. A 1:1 molar mixture of oleic acid and oleylamine is used to solubilize PbX_2 and to stabilize the QD surface. The nanocrystals form within the first few seconds after injection (faster for CsPbX_3 with a higher molecular mass). Because the growth stage is so fast, the QD size, unlike that of metal chalcogenides, is determined more by the synthesis temperature than by the growth time. Energy-dispersive X-ray spectroscopy (EDX) and Rutherford backscattering spectrometry show that the elements in CsPbX_3 nanocrystals are in a 1:1:3 ratio, including mixed-halide compositions.

To obtain solid solutions during synthesis, a mixed solution of PbX_2 and PbY_2 sources is added to the cesium oleate as the precursor, where X, Y are Cl, Br or Br, I. (For Cl, I, nanocrystals do not form at every ion ratio, apparently because of the large difference between the ionic radii of chloride and iodide.)

This group of materials allows the PL spectrum to be tuned over 410–700 nm, with a band FWHM of 12–42 nm, a high PL quantum yield (up to

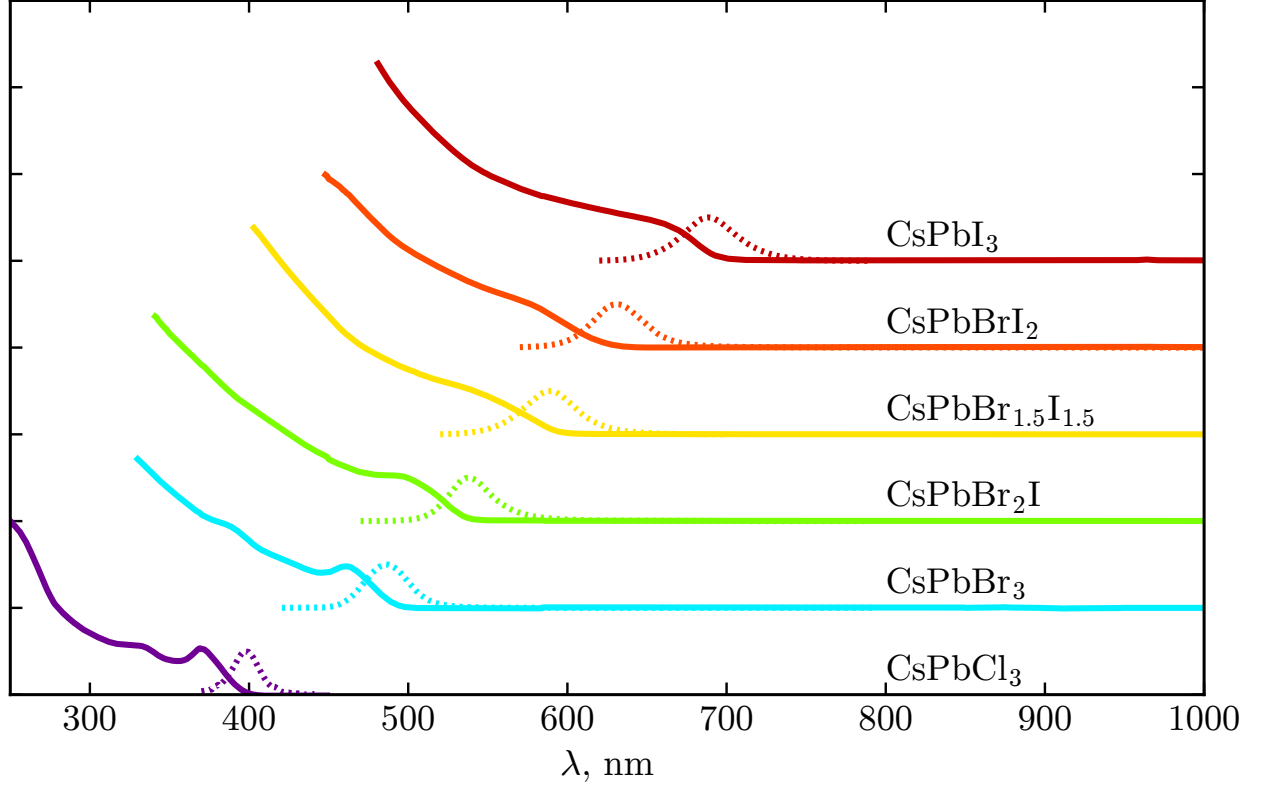


Figure 3.4. Tuning of the normalized absorption (solid lines) and photoluminescence (dashed lines) spectra of cesium lead halide nanocrystals with composition (measured at 300 K) [108]

95%), and PL lifetimes of 1–29 ns. Fig. 3.4 shows the absorption and PL spectra, and Fig. 3.5 the PL spectra alone, of CsPbX_3 colloidal quantum dots of different composition synthesized at 170°C . (Here and below, the colors in the families of spectra correspond to the emission of the respective nanocrystals under UV illumination.) The PL FWHM decreases gradually from CsPbI_3 to CsPbCl_3 . The samples luminesce both as colloidal solutions and as solid films. The absorption spectra of colloidal CsPbX_3 nanocrystal solutions (Fig. 3.4) show features corresponding to excitonic states, most pronounced in CsPbCl_3 nanocrystals.

Note that researchers usually use 1-octadecene as the reaction medium. In our work, however, we noticed that nonpolar organic media with a lower boiling point (for example, dodecane) give a narrower PL band of the final particles, possibly because of the lower viscosity of these solvents. Unlike $\text{A}^{\text{II}}\text{B}^{\text{VI}}$ and $\text{A}^{\text{IV}}\text{B}^{\text{VI}}$ quantum dots, CsPbX_3 nanocrystals grow much faster (seconds versus minutes to hours), so the diffusion conditions of the components are decisive.

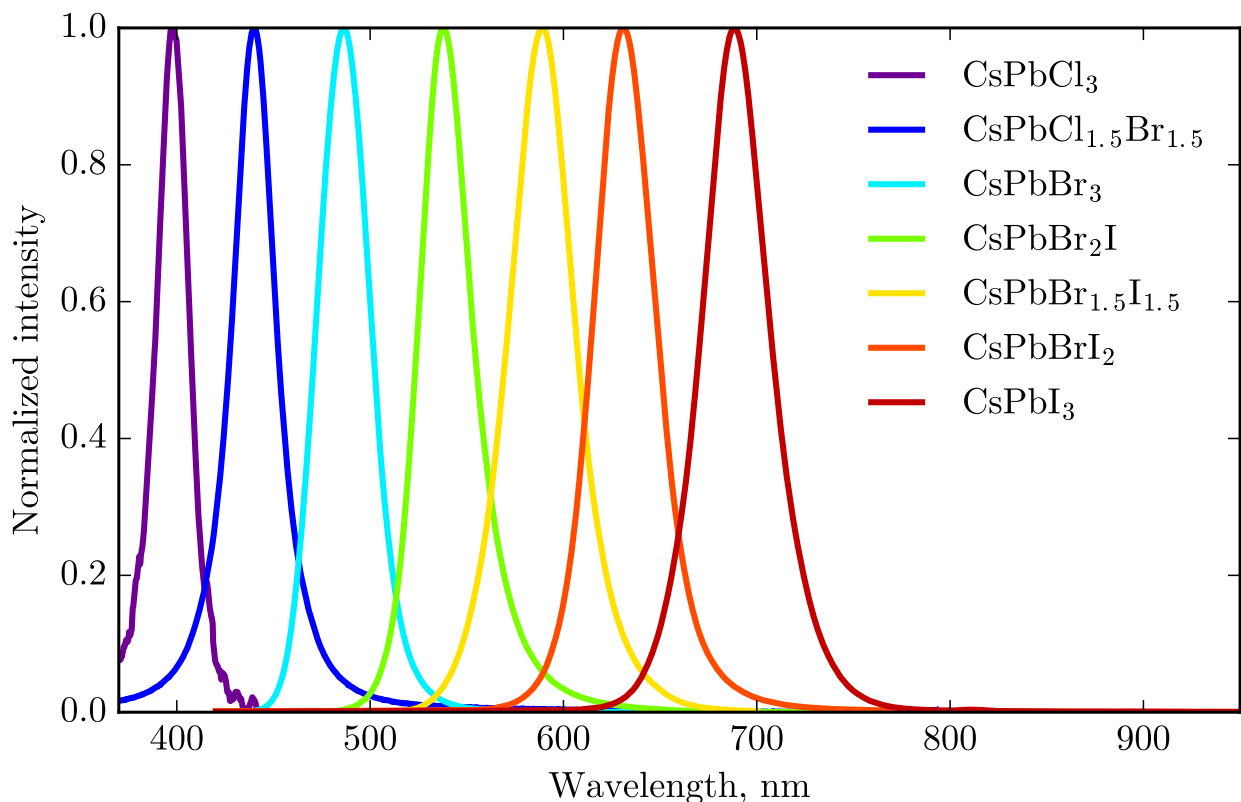


Figure 3.5. Tuning of the normalized photoluminescence spectra of cesium lead halide nanocrystals with composition (measured at 300 K) [108]

Another important difference of this system from the QD groups mentioned above is the dynamic binding of the organic ligands, oleic acid and oleylamine, to the nanocrystal surface. These bonds apparently arise from Coulomb interaction between halide anions and the δ^+ charge of the amine group of oleylamine, and between cesium cations and the δ^- charge of the carboxyl group of oleic acid. This interaction has a low binding energy. This limits the purification procedure: adding a strongly polar antisolvent to the solution can strip the ligands and dissolve the nanoparticles. For this reason, of the possible antisolvents, anhydrous isopropyl alcohol was used for CsPbX_3 nanocrystals.

To analyze the crystal structure of the nanoparticles, powder X-ray diffraction (XRD) analysis of the nanocrystals was performed on a DRON-3 X-ray diffractometer using Cu-K_α radiation. Powder X-ray diffraction showed that, of the possible crystal modifications, the cubic perovskite structure is realized (Fig. 3.6). The diffraction peaks are broadened by the small nanocrystal size

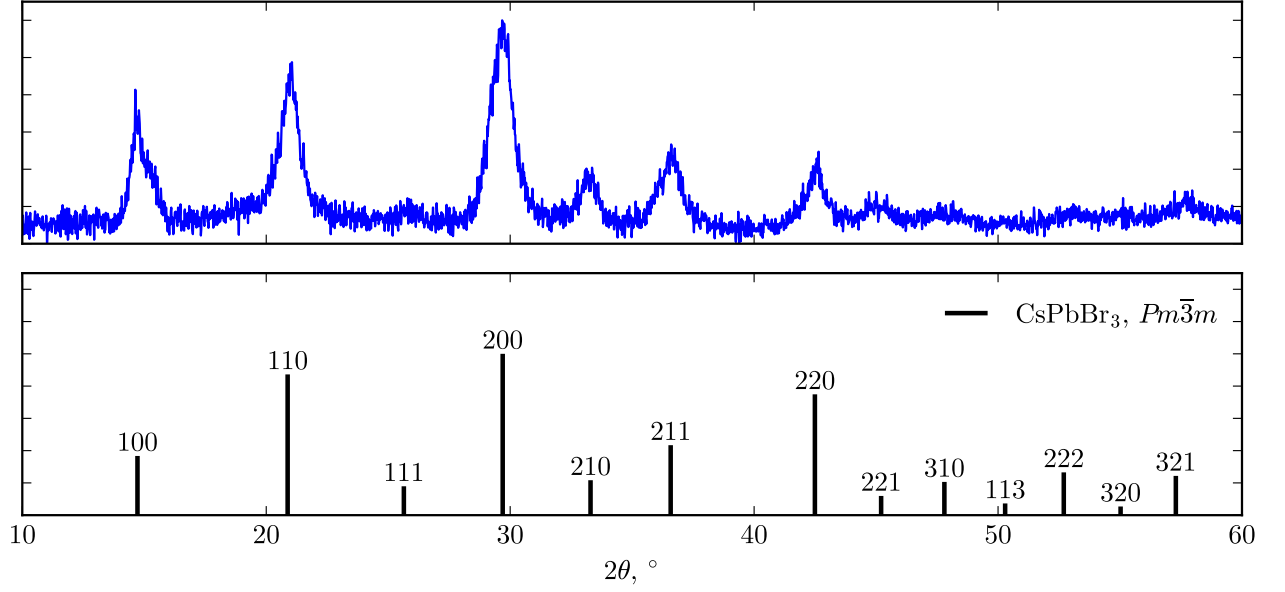


Figure 3.6. Powder X-ray diffraction pattern of cesium lead halide nanocrystals, using CsPbBr_3 as an example [108]

and by microstrain [115]. Taking both factors into account, to a first approximation the integral breadth of the resulting peak β can be taken as the sum:

$$\beta = \beta_S + \beta_D,$$

where β_S and β_D are the reflection breadths due to the size effect and to microstrain, respectively:

$$\beta_S = \frac{\lambda}{D \cos \theta},$$

$$\beta_D = \frac{4\varepsilon}{\cot \theta},$$

where λ is the wavelength of the X-ray source (Cu- K_α), D is the mean nanocrystal size, ε is the mean microstrain of the interplanar spacing ($\Delta d/d$).

The two effects can be separated by their different dependences of the integral peak breadth on the diffraction angle. Assuming that both the size broadening function and the microstrain broadening function are Lorentzian (for Gaussian functions, squared values are used), let us consider $\beta \cos \theta$ as a function of $\sin \theta$:

$$\beta \cos \theta = \frac{\lambda}{D} + 4\varepsilon \sin \theta.$$

The slope of this linear dependence gives 4ε , and the intercept with the ordinate gives λ/D . The approximate values found are $D \approx 10$ nm and $\varepsilon = 0.5\text{--}2\%$. The available data, unfortunately, do not allow the anisotropy of the microstrain to be determined reliably. However, the mean size from XRD agrees with electron microscopy data, which show cubic nanocrystals with an edge length of 13 nm.

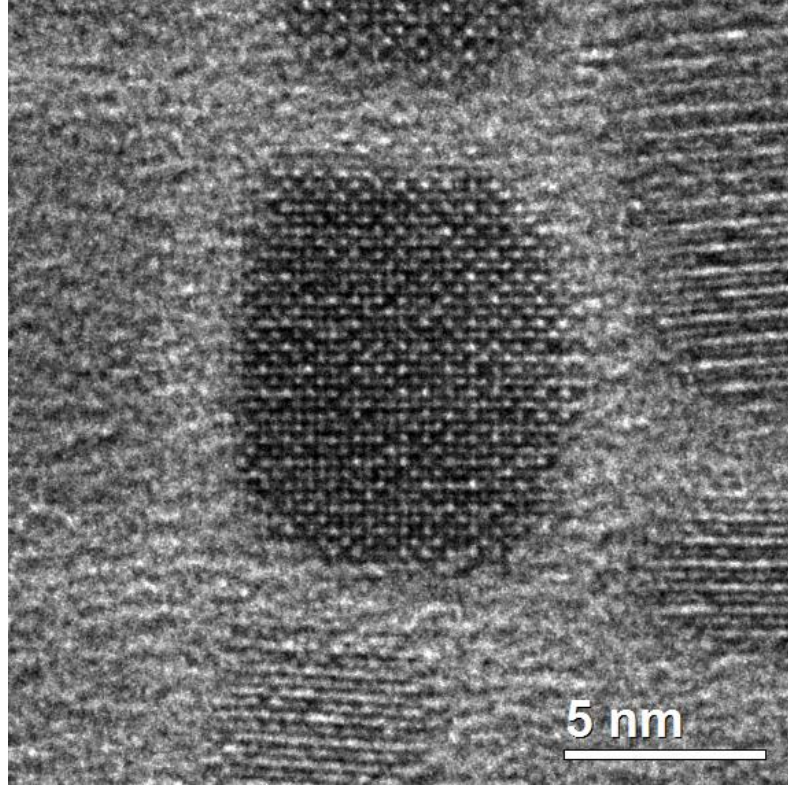


Figure 3.7. High-resolution transmission electron microscopy image of a CsPbBr₃ nanocrystal

Nanocrystals synthesized at a lower temperature, and therefore smaller on average, were studied by high-resolution transmission electron microscopy (TEM) images (Fig. 3.7). The image shows periodic variations of electron density corresponding to atoms of different elements. The period of the structure matches the tabulated lattice period of CsPbBr₃.

Note that for CsPbI₃ nanocrystals the perovskite lattice is metastable: in bulk crystals, the transition to the cubic lattice occurs above 315 °C. The orthorhombic phase observed for bulk CsPbI₃ at room temperature does not luminesce. The authors of [13] showed that the small size of such crystals

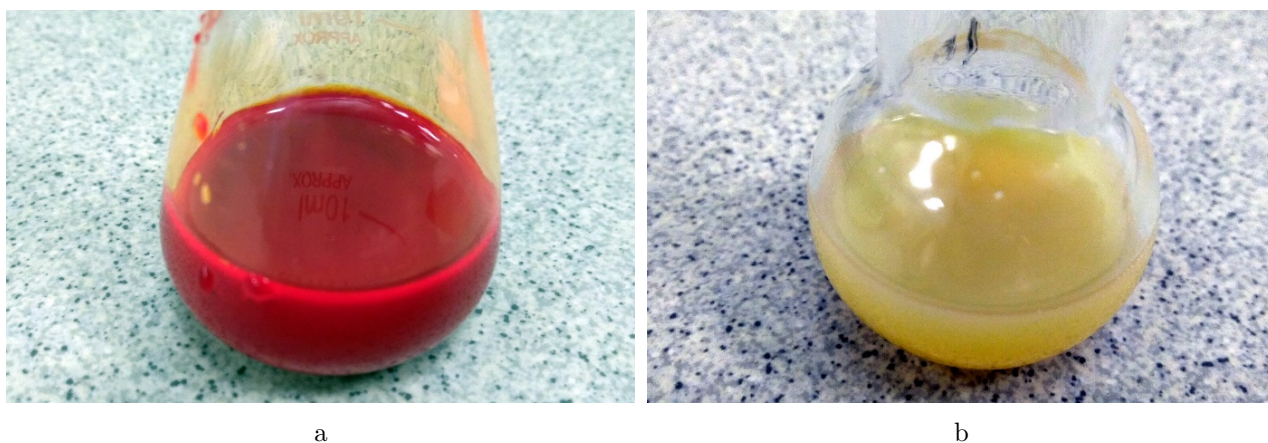


Figure 3.8. Colloidal solutions of CsPbI_3 nanocrystals with a predominant a) perovskite phase, b) non-perovskite orthorhombic phase

(5–15 nm) plays an important role in stabilizing the cubic phase of CsPbI_3 . Our experiments confirm this. Nevertheless, a common observation is that CsPbI_3 nanocrystals still turn into the non-luminescent state over time (within a month) (Fig. 3.8). To solve this problem, the literature proposes various ways to stabilize the perovskite phase of CsPbI_3 : incorporating other ions into the structure, such as rubidium cations substituting cesium [116] or chloride anions [117], or using bidentate ligands that change the surface energy of the nanocrystals [118].

Since halide perovskite nanocrystals of different composition are synthesized by the same technology, the PL band in this system is tuned mainly by changing the composition rather than the QD size. The quantum size effect is nevertheless present in this system. The Bohr diameters of Wannier–Mott excitons and the exciton binding energies are: CsPbCl_3 5 nm, 75 meV; CsPbBr_3 7 nm, 40 meV; CsPbI_3 12 nm, 20 meV.

CsPbX_3 QDs show bright photoluminescence (Fig. 3.9) even without inorganic shells of wide-gap materials. This shows that, unlike in core-only cadmium chalcogenide QDs, surface defects do not create recombination centers within the band gap. In the shape of their PL spectra, the ternary CsPbX_3 compounds and their solid solutions compare favorably with ternary chalcopyrites, which have broad PL bands [119]. The ionic character of the chemical bond in CsPbX_3 and the large differences in ionic radii and charge between Cs^+ and Pb^{2+} ensure a stoichiometric composition and an ordered arrangement of the elements. Multicomponent metal chalcogenides, by contrast, show a strongly



Figure 3.9. Photoluminescence of colloidal solutions of CsPbX_3 nanocrystals of different composition under a UV lamp

nonuniform distribution of cations and anions. These local deviations from stoichiometry lead to a high density of donor–acceptor states associated with point defects, and to broadband luminescence from the combination of photon emission mechanisms through various defect electronic states.

3.2 Fast anion exchange

One of the features of halide perovskites is fast anion exchange [120]: the formation of solid solutions through the chemical reaction of a cesium lead halide with a source of another halogen. For these perovskites, the process occurs under ambient conditions.

To study the dynamics of anion exchange, a concentrated colloidal solution of CsPbBr_3 nanoparticles, purified from reaction products beforehand, was mixed with a comparable amount of PbCl_2 precursor (Fig. 3.10) or PbI_2 precursor (Fig. 3.11). After mixing, the PL spectrum of the solution was measured continuously. In both cases, a 405 nm laser was used to excite PL.

In Fig. 3.10, during the first moments the spectrum shifts and the PL band broadens as anion exchange proceeds. Then, in the solid solution, the line narrows and the PL signal intensity increases. Over time, the intensity falls again, because the absorption edge of the material approaches the excitation wavelength and the excitation becomes less efficient. The total time from mixing until the PL spectrum stopped changing substantially was 180 min.

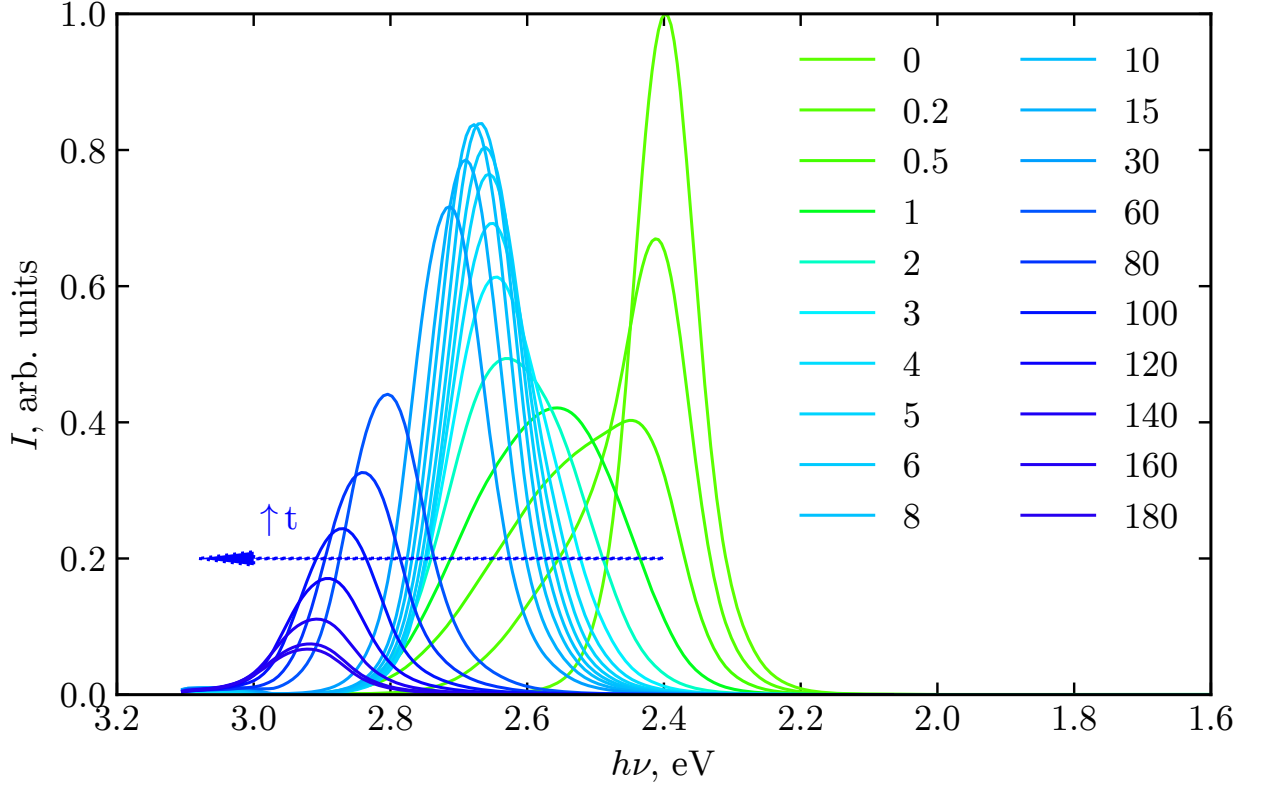


Figure 3.10. Evolution of the photoluminescence spectrum of $\text{CsPb}(\text{Br}_{1-x}\text{Cl}_x)_3$ nanocrystals during anion exchange. The legend gives the time in minutes after the solutions were mixed

Fig. 3.11 shows the results of a similar experiment. The spectra change in a similar way, but without the intensity drop at the end, because in this case the energy gap gradually “narrows”. The total time from mixing until the PL spectrum stopped changing substantially was 60 min, three times shorter than in Fig. 3.10. This difference may be explained as follows: the excess energy transferred to the nanocrystals of the forming $\text{CsPb}(\text{Br}_{1-x}\text{I}_x)_3$ solid solution may additionally heat them and accelerate the diffusion of iodide ions in the original structure. Thus, laser irradiation can accelerate anion exchange.

Fig. 3.12 shows the dynamics of the PL spectrum, i.e., the energy of the PL maximum and the line FWHM, obtained by processing the data of Fig. 3.11. Both dependences split into two segments with a break point at the same time, 120 s. The shape of these dependences suggests that the key change results from processes at the surface of the perovskite nanocrystals. The smooth shift of the PL spectrum and the absence of distinct features during the shift indicate strong sharing (hybridization) of the corresponding electronic states.

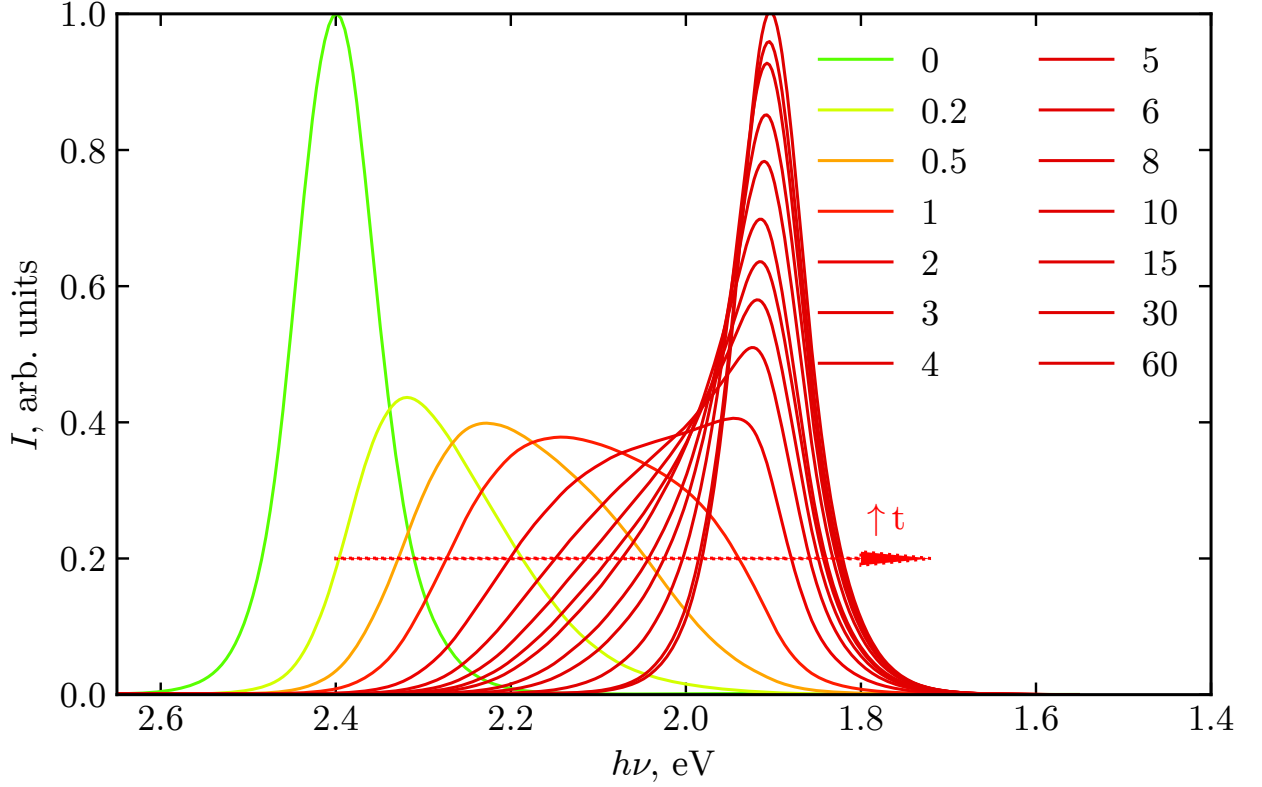


Figure 3.11. Evolution of the photoluminescence spectrum of $\text{CsPb}(\text{Br}_{1-x}\text{I}_x)_3$ nanocrystals during anion exchange. The legend gives the time in minutes after the solutions were mixed

Anion exchange is technologically important. Nanocrystals of a certain chemical composition can be synthesized at elevated temperature, and then the photoluminescence maximum can be moved to a desired position by adding a source of another halogen, without further heating. On the other hand, anion exchange limits the possibility of making heterojunctions of solid solutions based on these ternary compounds, since the compositions of materials in contact should equalize.

Fig. 3.13 shows spectral tuning with a step of ~ 5 nm for 60 samples obtained by anion exchange from the same starting CsPbBr_3 nanocrystals. Similar results can also be obtained with the flow reactor described in Section 2.2, using mixed halides with different molar ratios as precursor B .

Note that the type of chemical compound used as the halogen source is critical for anion exchange. For example, adding I_2 instead of PbI_2 in a similar experiment only etches the nanocrystal surface and gradually quenches the PL, as we showed in [121].

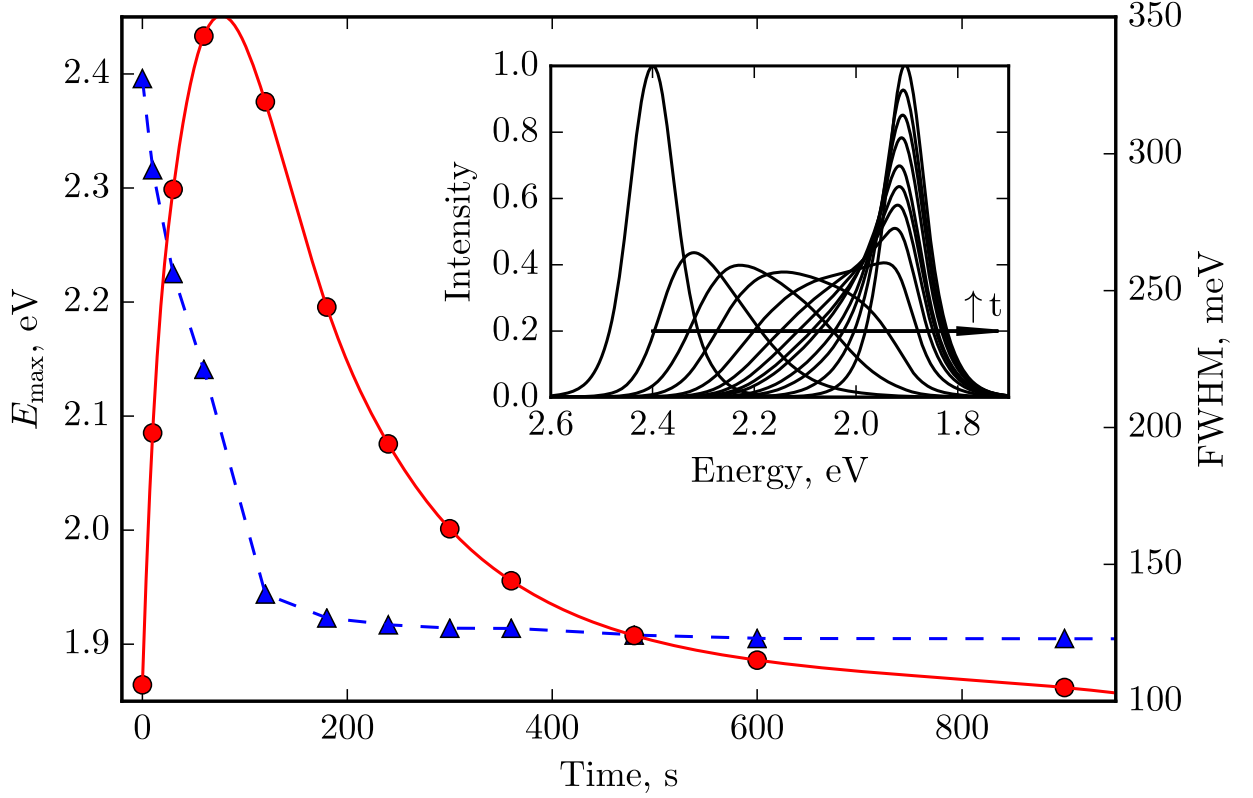


Figure 3.12. Dynamics of the photoluminescence spectra during Br-I anion exchange: triangles: time dependence of the energy of the PL maximum; dots: time dependence of the PL line FWHM; inset: evolution of the PL spectrum during anion exchange [108]

3.3 Manganese doping of CsPbX_3 nanocrystals

A debated question in the technology of these materials is doping, in particular doping with the magnetic impurity manganese [122, 123], and the position of the impurity energy levels relative to the perovskite host. To clarify this question, doped perovskite nanocrystals of different chemical composition were synthesized.

The synthesis was based on the method described above. Component *A* was a solution of cesium oleate in octadecene, and component *B* was a mixed precursor containing lead chloride and manganese chloride in octadecene. To vary the dopant-to-lead ratio, the dopant sources were taken in different volume ratios to the lead halide sources.

The absorption and photoluminescence spectra of $\text{CsPbCl}_3\text{:Mn}$ nanocrystals (nominal Mn:Pb ratio 1:10) are shown in Fig. 3.14. The absorption spectrum has a pronounced feature near 400 nm due to the excitonic absorption

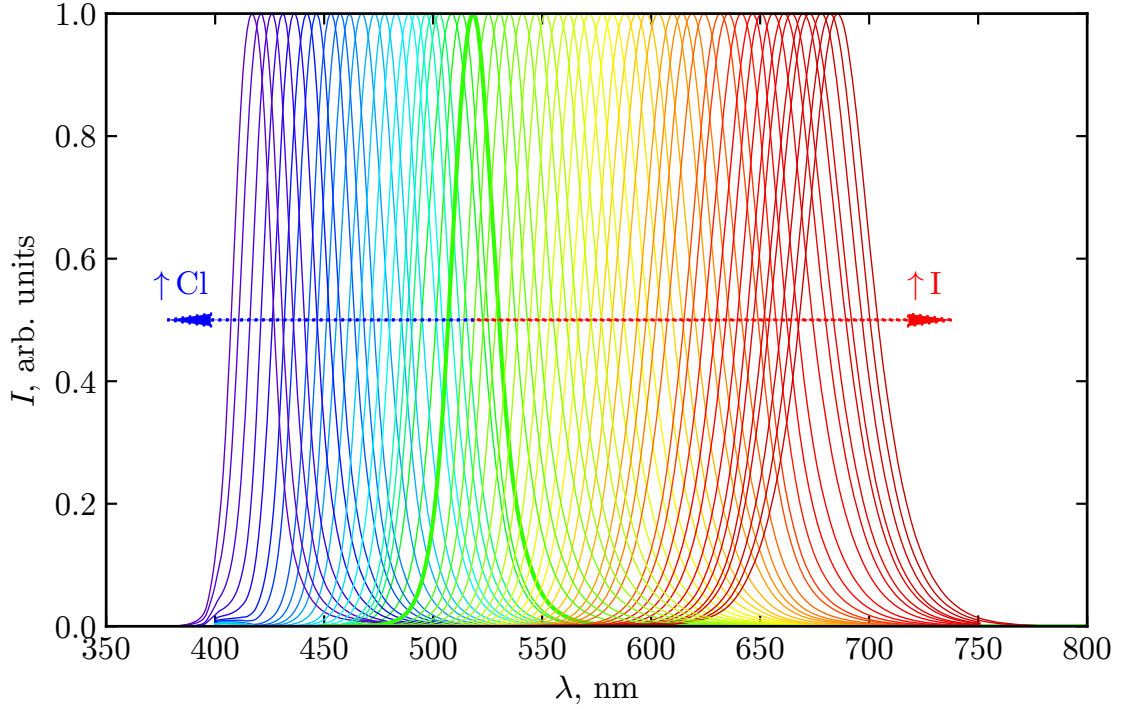


Figure 3.13. Photoluminescence spectra of a series of samples obtained by anion exchange [108]

typical of CsPbCl_3 . The rest of the spectrum shows no significant features and no additional scattering, which indicates a narrow size distribution of the nanocrystals and colloidal stability of the solution. Comparison with other experimental data shows that the mean particle size (the edge length of the cubic nanocrystal) is ~ 10 nm.

The PL spectrum (Fig. 3.14) shows two key features: a narrow peak in the short-wavelength region with a maximum at 410 nm, and a broad band (FWHM = 0.2 eV) in the long-wavelength region with a maximum at 600 nm. The short-wavelength feature corresponds to excitonic recombination shifted by a Stokes shift of 83 meV; it is also present in samples without manganese. The long-wavelength feature appears only when manganese is introduced, and its intensity grows with the dopant concentration. This luminescence corresponds to intracenter electron transitions within the point emission center, the Mn^{2+} ion. Similar features of manganese doping of wide-gap materials are observed in $\text{A}^{\text{II}}\text{B}^{\text{VI}}$ semiconductors, ZnS and ZnSe [124].

As noted above, of particular interest is the position of the impurity energy levels relative to the levels of the perovskite host surrounding the man-

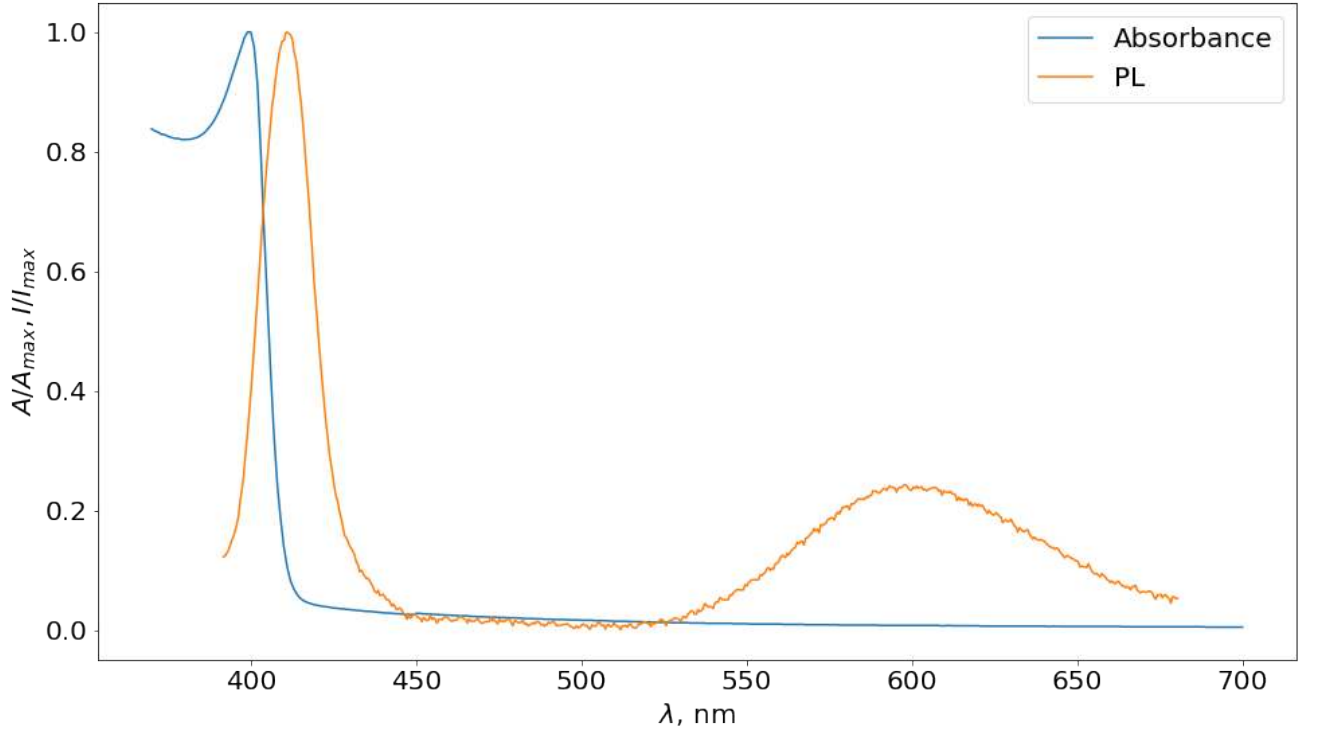


Figure 3.14. Absorption and photoluminescence spectra of a $\text{CsPbCl}_3\text{:Mn}$ sample

ganese ion. To study the level positions, we used an approach based on fast anion exchange. Fig. 3.15 shows how the photoluminescence spectra change as increasing amounts of lead bromide precursor are added. The short-wavelength peak red-shifts, and the relative intensity of the intracenter manganese luminescence decreases without a change in its position on the abscissa.

The corresponding change in the luminescence color of the sample under UV light is shown in Fig. 3.16. As the bromine fraction increases, the absorption spectrum of the sample also shifts to longer wavelengths, so the luminescence is excited more and more efficiently (this is why the spectra in Fig. 3.15 are normalized).

Fig. 3.15 shows that the long-wavelength peak stays at 600 nm, which corresponds to 2.06 eV. Based on these observations and on calculated energy level positions in pure CsPbCl_3 and CsPbBr_3 , we proposed the model shown in Fig. 3.17. The host energy levels were taken from the calculations of [125].

In the model illustrated in Fig. 3.17, the energy positions of the manganese levels and the gap between them are assumed to be constant. The positions of and intervals between the other levels change because of solid solution formation. Note that, according to [125], the level that shifts most when the

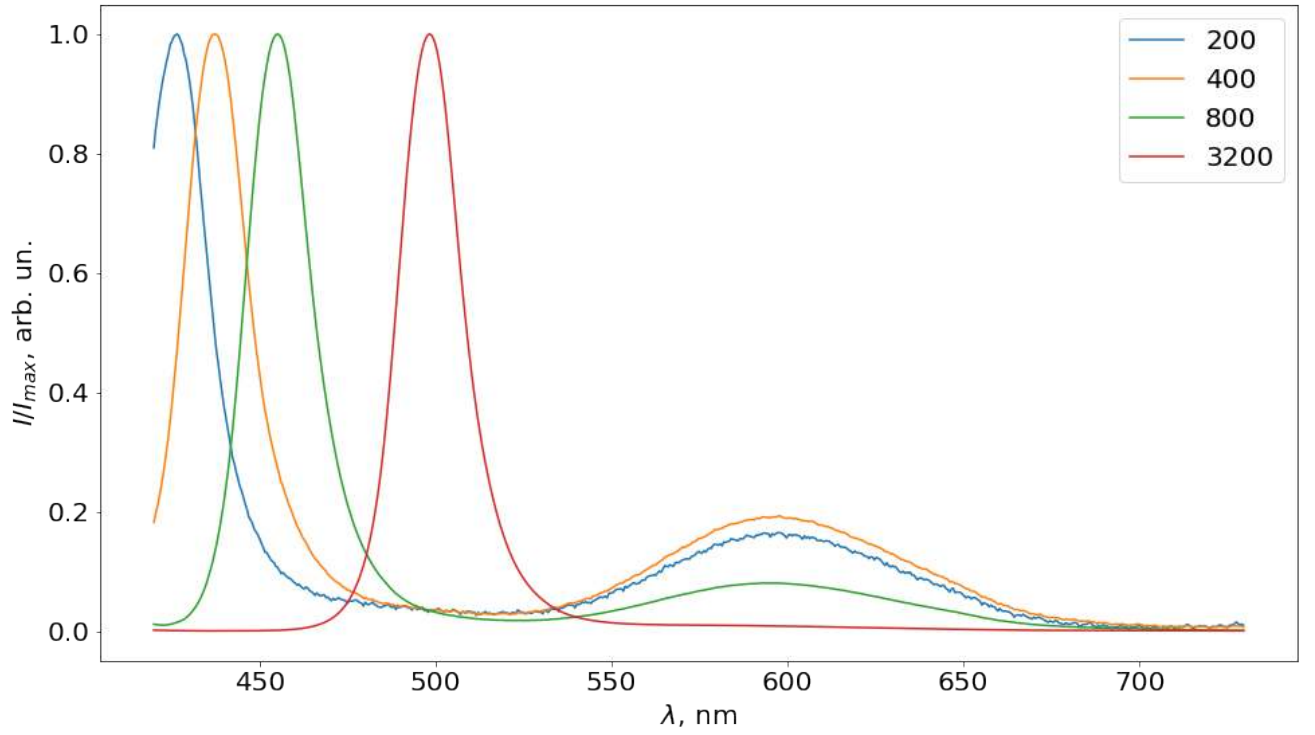


Figure 3.15. Evolution of the photoluminescence spectra of a colloidal solution of $\text{CsPb}(\text{Cl}_{1-x}\text{Br}_x)_3 : \text{Mn}$ nanocrystals as different amounts of lead bromide precursor are added

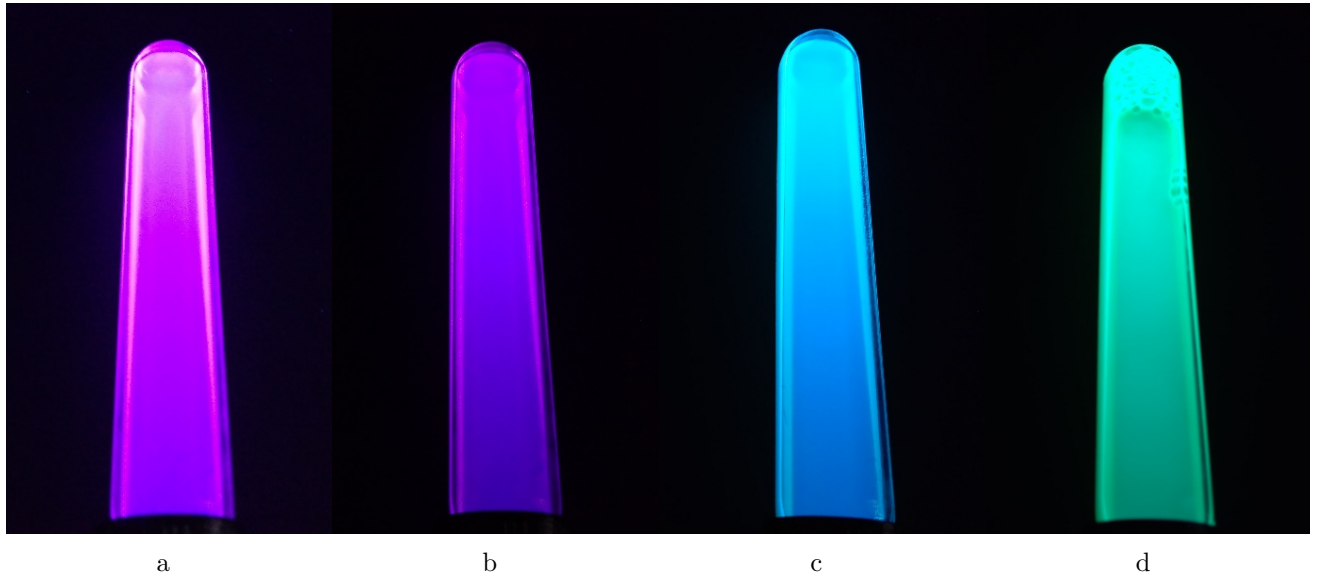


Figure 3.16. Colloidal solutions of $\text{CsPb}(\text{Cl}_{1-x}\text{Br}_x)_3 : \text{Mn}$ nanocrystals with different concentrations under UV light (the manganese concentration increases from a to d)

solid solution forms is the one corresponding to the ground state. The strong decrease in manganese luminescence may be explained by thermal filling of the unexcited ground levels of manganese with equilibrium charge carriers from the host. When the manganese 6A_1 level is occupied, radiative transitions by this

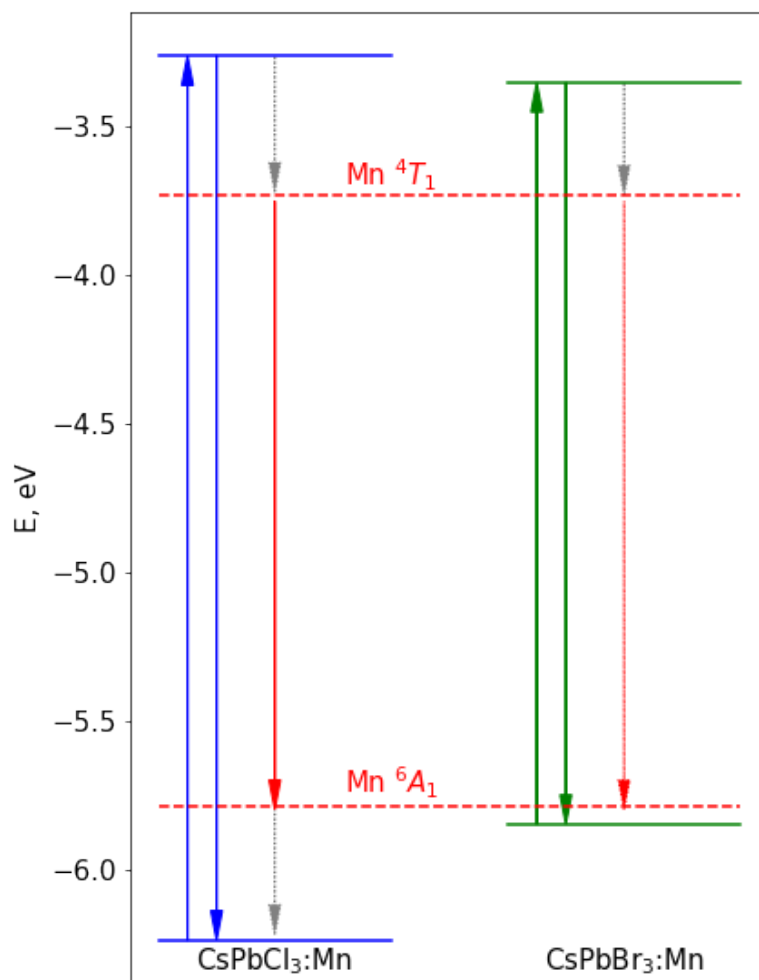


Figure 3.17. Band diagram illustrating the proposed model of the intensity ratio change in $\text{CsPbX}_3 : \text{Mn}$. Blue and green denote the states and transitions associated with the electronic levels of the CsPbCl_3 and CsPbBr_3 hosts, respectively; red denotes the states and intracenter transitions of manganese; gray arrows denote nonradiative transitions

mechanism become less probable, and their intensity drops sharply.

3.4 Ordering of nanocrystal ensembles on solid substrates

The monodispersity of cesium lead halide nanoparticles, their cubic shape, and the weak binding of organic ligands to the nanocrystal surface make oriented attachment possible. The first evidence of this was the observation that, without any additional technological measures, monodispersity and cubic shape lead to peculiar cube-to-cube domains when the nanoparticles are deposited on silicon (Fig. 3.18).

This is further confirmed by TEM data (Fig. 3.19) for a layer of CsPbBr_3 nanocrystals. (The images were taken on a JEM-2100F microscope at an accelerating voltage of 200 kV and a resolution of 0.19 nm.) The samples were

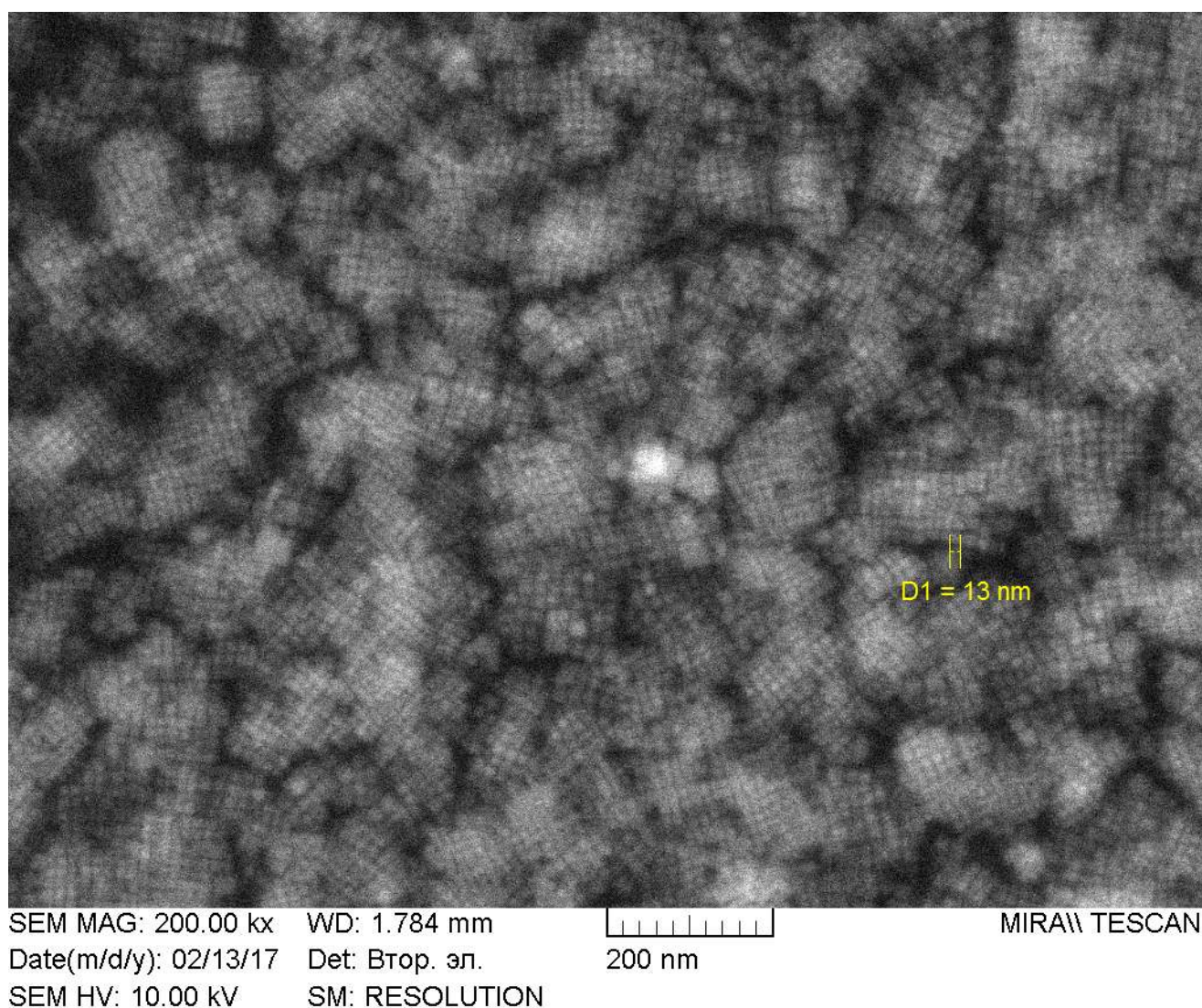


Figure 3.18. Scanning electron microscopy image of nanocrystal ensembles deposited on a silicon substrate

prepared by dipping a copper TEM grid coated with an ultrathin graphene film into the colloidal nanoparticle solution.

Fig. 3.19 shows, first, the near-cubic shape of the nanocrystals and, second, light “gaps” between the nanocrystals whose width correlates with the length of the capping molecules, oleic acid and oleylamine. The contrast arises because, unlike the organic shell, the inorganic core of the nanocrystal consists of much heavier elements. The different contrast of individual nanoparticles in the TEM image may be explained by the different number of lattice planes they contain.

The setup described in Section 2.4 was used for layer-by-layer deposition of a CsPbBr₃ nanocrystal film. A glass substrate was dipped alternately for

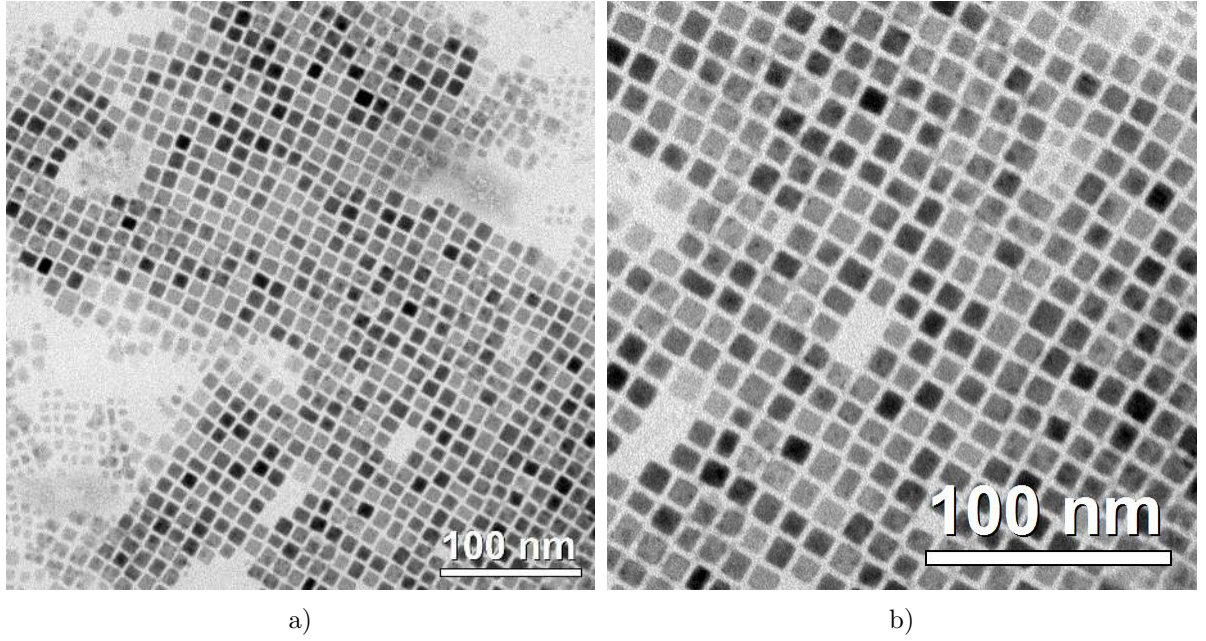


Figure 3.19. Transmission electron microscopy images of CsPbBr₃ nanocrystal ensembles

60 s in a concentrated solution of CsPbBr₃ nanocrystals and for 10 s in methyl acetate, which partly strips the ligands and removes unbound ligands. To passivate the surface states created by ligand removal, a small amount (<1 wt.%) of lead nitrate was added to the methyl acetate.

Fig. 3.20 shows the absorbance spectra of thin CsPbBr₃ films obtained with one and three repetitions of this procedure. Repeated deposition multiplies the absorbance, which confirms nanocrystal deposition. The feature at 530 nm, usually attributed to the first excitonic peak, does not shift. In the long-wavelength part, a thicker film scatters more light and correspondingly increases the absorbance there. For this reason, the setup described in Section 2.6 can be used as one way to monitor deposition. Our results on layer deposition and the possibility of making electronic structures were described in [126].

Repeating this procedure many times gave ordered layers ~100 nm thick on glass substrates, which were studied by XRD (Fig. 3.21). Comparison of the experimental diffraction pattern with the calculated values shows that the layers remain strongly (100)-oriented as they grow thicker: only reflections of the (h00) family are observed, as is typical of textured polycrystalline films.

3.5 Radiation hardness of CsPbX₃ nanocrystals

To compare the stability of samples under hard radiation, undoped CsPbBr₃ nanocrystals were synthesized, purified, and dispersed in 1 mL of hexane. Com-

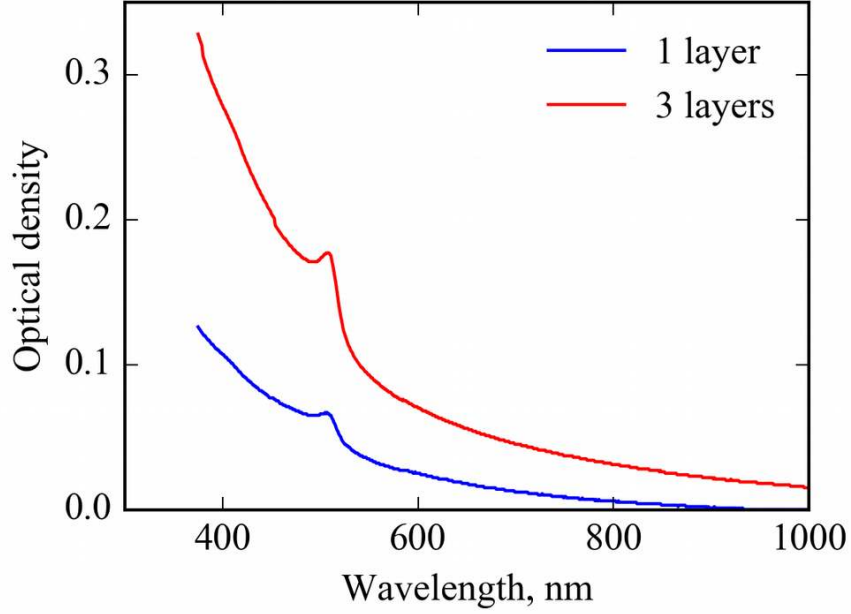


Figure 3.20. Absorbance spectrum after one and three depositions of a CsPbBr₃ nanocrystal layer [126]

mercial colloidal CdSe/ZnS quantum dots with an octadecylamine-stabilized surface served as the reference. Our results on the effect of γ -irradiation on CdSe/ZnS nanocrystals were also published in [127].

Table 3.1. Irradiation times and accumulated γ -radiation doses for CsPbBr₃ and CdSe/ZnS QD samples

Sample	Measurement	t_{Σ} , s	D_{Σ} (H ₂ O), Gy
CsPbBr ₃	0 (initial)	0	0
	1	444600	755820
	2	905100	1538670
	3	1362000	2315400
CdSe/ZnS	0 (initial)	0	0
	1	456900	776730

The concentration of both types of nanoparticles was ~ 1 mg/mL. The gamma irradiator used cesium-137 radionuclide sources. The number and arrangement of the sources and the chamber geometry produced a uniform flux of γ -photons with an energy of 661 keV. Irradiation took place in a chamber inside a shielded enclosure, at an absorbed dose rate (in water) of ~ 1.7 Gy/s.

The samples were placed in the chamber as 1 mL of colloidal solution in glass vials and irradiated until the absorbed dose D (in water) listed in Table 3.1

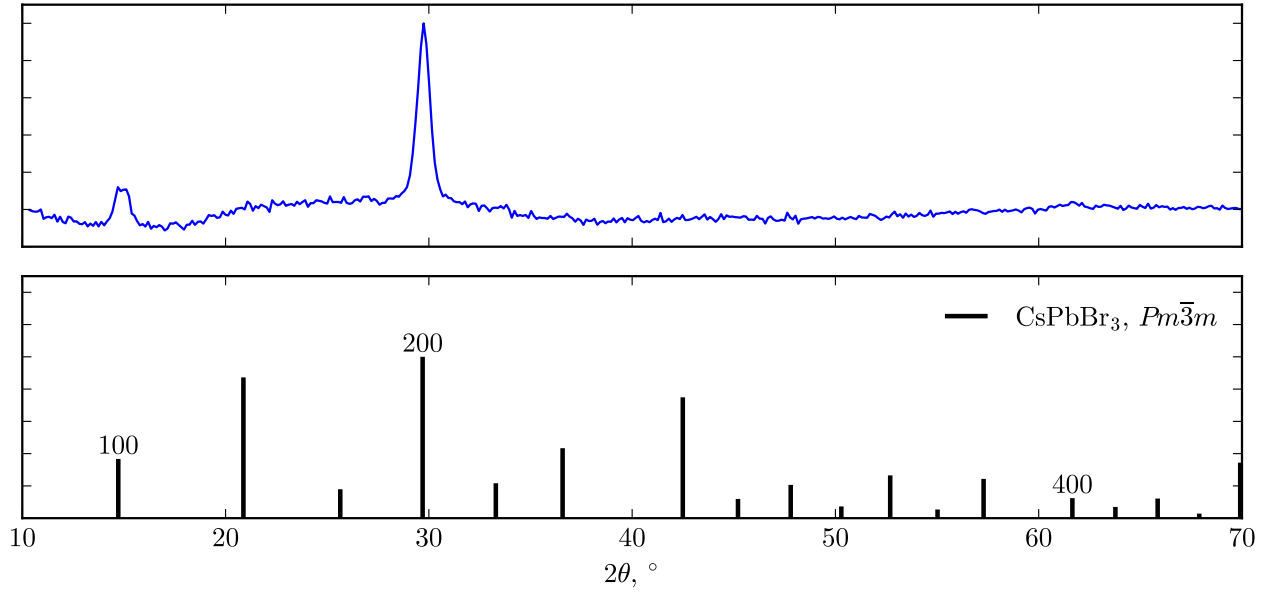


Figure 3.21. Experimental diffraction pattern of an ordered structure deposited on glass, and the calculated stick pattern of CsPbBr_3 $Pm\bar{3}m$ with the reflections observed in the experimental pattern highlighted

was accumulated. About a day passed between exposures, during which the samples were prepared and their spectra measured.

The non-irradiated CsPbBr_3 control sample showed no noticeable changes in its absorption and photoluminescence spectra during the experiment. With increasing absorbed dose, the luminescence intensity decreased gradually, and at 2.3 MGy it dropped substantially, with an additional short-wavelength PL band appearing (Fig. 3.22).

The photoluminescence spectrum suggests that the nanocrystal shape changes substantially during irradiation, which shows up as both short- and long-wavelength “tails” in the spectra. At the same time, the luminescence drop is caused not only by the change in the size distribution but also by nonradiative recombination centers created by the high-energy irradiation of the material. Of particular interest is the comparison of the CsPbBr_3 results with the irradiation of CdSe/ZnS core/shell colloidal quantum dots (Fig. 3.23 and Fig. 3.24).

Comparing our data with the literature for a similar system and a γ -source of similar energy [128], we see similar results: a rapid decrease in absolute intensity and a smearing of the PL band at much lower doses than those we observe for CsPbBr_3 . Even at a three times higher dose, the CsPbBr_3 sample

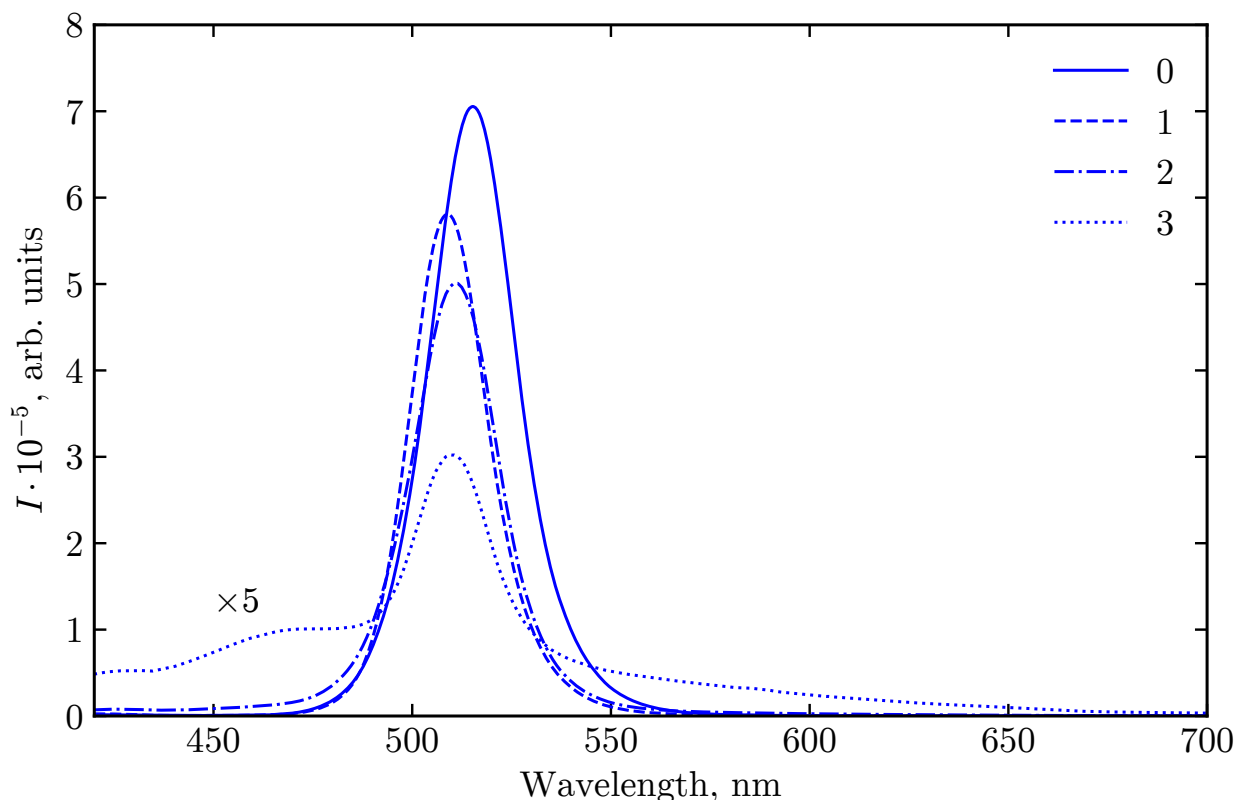


Figure 3.22. Photoluminescence spectra of a colloidal solution of CsPbBr₃ QDs as the γ -radiation dose accumulates (legend: measurement number from Table 3.1; curve 3 is multiplied by 5) [127]

luminesces 20 times brighter than the CdSe/ZnS sample with a similar initial PL.

The physicochemical features of the cesium lead halide system discussed above, and the absence of defect-related bands in the PL spectra, suggest that the system may be resistant to defect formation. The fast anion exchange indicates that the system tends to restore the anionic octahedra, the main structural units responsible for PL. The strong decrease and smearing of the PL signal in CdSe/ZnS QDs may be due to the accumulation of defects at the core/shell interface under high-energy irradiation.

Thus, the experiment showed that the PL of CsPbBr₃ is more stable than that of commercial CdSe/ZnS nanocrystals (Fig. 3.24). These results show that CsPbBr₃ nanocrystals are promising as a luminescent material in high-radiation environments, including, with a suitable protective shell, as biomarkers for anticancer radiotherapy. Note, however, that such applications require inert shells that prevent the nanocrystals from dissolving in aqueous media [129].

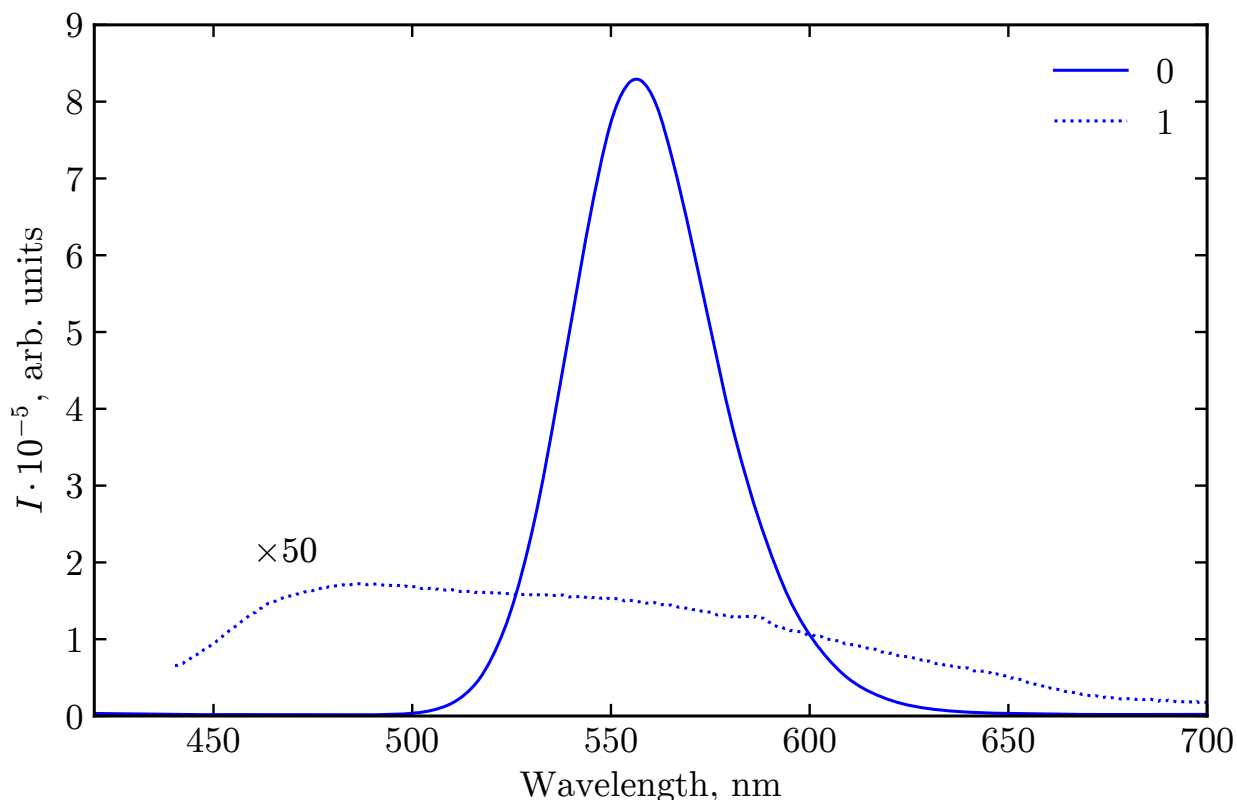


Figure 3.23. Photoluminescence spectra of a colloidal solution of CdSe/ZnS QDs as the γ -radiation dose accumulates (legend: measurement number from Table 3.1; curve 1 is multiplied by 50) [127]

3.6 Conclusions to Chapter 3

As a result of the research described in the third chapter, the following main statements were formulated:

1. The triangulation of the Cs–Pb–X system ($X = \text{Cl}, \text{Br}, \text{I}$) and the growth of CsPbX_3 nanocrystals are governed not only by the binary compounds on the edges of the Gibbs triangle, but also by the composition points that correspond to the $[\text{PbX}_6]$ anionic octahedra.
2. The physicochemical features of the Cs– $[\text{PbX}_6]$ – $[\text{PbY}_6]$ interaction ($X, Y = \text{Cl}, \text{Br}$ or Br, I) predetermine the efficient formation of cesium lead halide solid-solution nanocrystals and the change of their photoluminescence spectra during anion exchange.
3. A model of photoluminescence in cesium lead halide perovskites, in which the photoluminescence is determined by the anionic octahedra as structural units, explains their radiation hardness. This effect gives CsPbBr_3

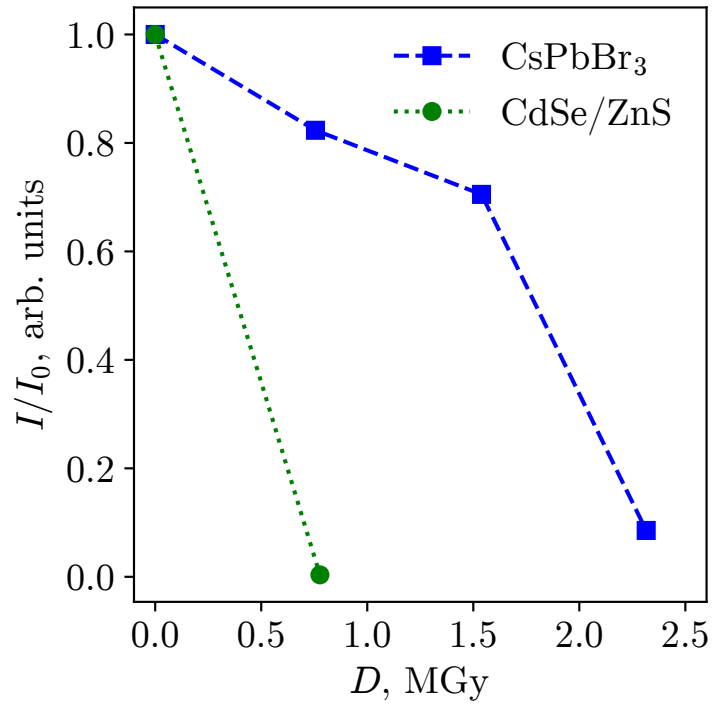


Figure 3.24. PL intensity of colloidal QDs of different composition versus γ -radiation dose nanocrystals an advantage over CdSe/ZnS quantum dots in high-radiation environments.

4 Plasmonic Enhancement

The fourth chapter addresses the increase in the photoluminescence intensity of metal chalcogenide quantum dot layers through the interaction of QDs with plasmonic nanoparticles.

4.1 Synthesis of Ag nanoparticles

Silver nanoparticles were synthesized under an inert gas in the setup described in Section 2.1. A mixed solution of silver nitrate, sodium citrate, and sodium chloride was added to an ascorbic acid solution heated to 95 °C. Heating was maintained for 30 min until the reaction was complete, which was judged qualitatively by the change in solution color. In this synthesis, ascorbic acid is the reducing agent, sodium citrate stabilizes the particle surface, and sodium chloride gives the particles a quasi-spherical shape. Table 4.1 lists the component concentrations in the final solution.

Table 4.1. Component concentrations for the synthesis of silver particles

Component	Final concentration C, mol/L
Ascorbic acid $C_6H_8O_6$	$1 \cdot 10^{-4}$
Silver nitrate $AgNO_3$	$7 \cdot 10^{-4}$
Sodium citrate $Na_3C_6H_5O_7$	$3 \cdot 10^{-4}$
Sodium chloride $NaCl$	$8 \cdot 10^{-4}$

An example spectrum of a colloidal solution of the synthesized silver particles is shown in Fig. 4.1. The absorbance spectrum shows that the particles have a plasmon resonance at 403 nm. The resonance FWHM (54 nm) is relatively small for silver particles. It indicates a narrow size distribution and is close to the smallest values reported in the literature [2, 69]. Fig. 4.2 shows scanning electron microscopy data for silver nanoparticles deposited on single-crystal silicon. The particles have practically identical sizes, with a diameter of ~ 20 nm, which agrees with the size of Ag nanoparticles calculated from optical spectra following [88].

To test whether larger silver nanoparticles can be synthesized, we proposed a two-step method in which small silver particles obtained as described above serve as seeds for the growth of larger particles. In the second step, the colloidal solution of first-step particles is added to distilled water heated

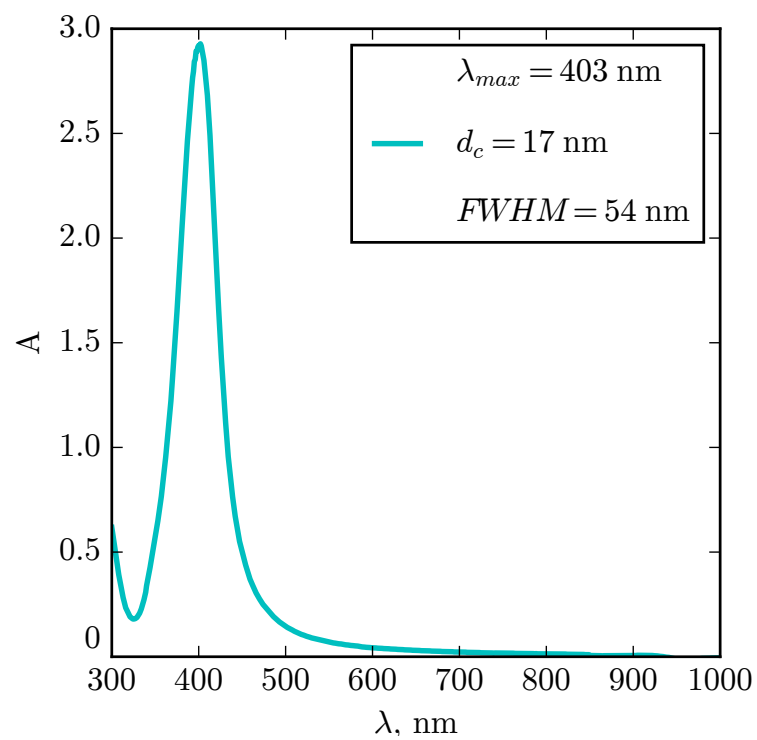


Figure 4.1. Absorbance spectrum of silver particles 17 nm in diameter

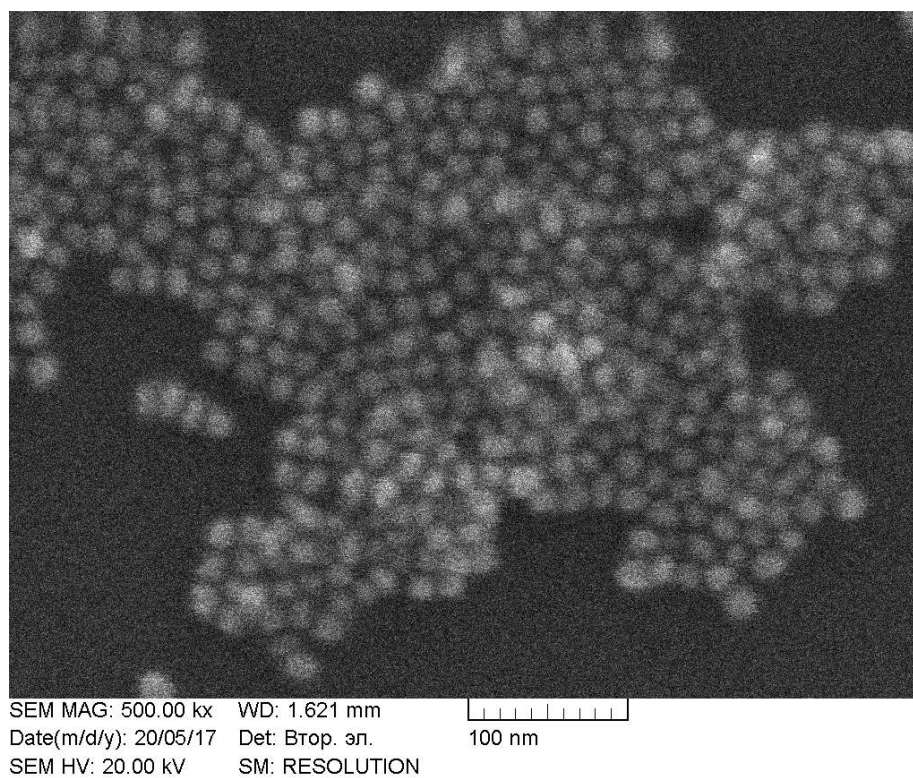


Figure 4.2. Scanning electron microscopy image of silver nanoparticles deposited on a silicon substrate

to 80 °C under stirring. After 5 min, a silver source solution is added; it is the same as in the synthesis of the seed particles, but without sodium chloride. This approach did give larger particles with a mean diameter of up to ~ 60 nm, but with a wider size distribution, as shown below for the Ag/SiO₂ structures.

4.2 Isolation and purification of silver particles by centrifugation

After synthesis, the silver particles are in a solution that contains unreacted starting components and reaction by-products, which must be removed. For this, the colloidal solution was centrifuged. The required centrifugation time was calculated from (2.1). The solutions were centrifuged for 30 min at 9000 rpm. After centrifugation, the supernatant containing the by-products was removed, and the particles were redispersed in distilled water. Fig. 4.3 shows the absorbance spectra of the colloidal silver nanoparticle solution before and after centrifugation. We verified that a single precipitation is enough for purification: further centrifugation does not change the spectrum of the samples.

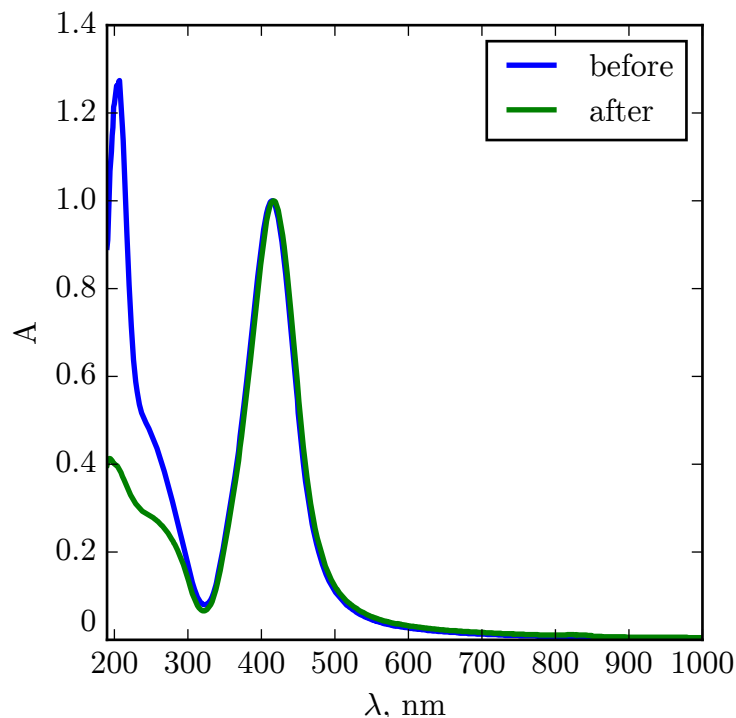


Figure 4.3. Absorbance spectra of silver particles before and after centrifugation

As Fig. 4.3 shows, centrifugation reduces absorption in the ultraviolet, because the reaction by-products are removed from the solution. Fig. 4.4 shows

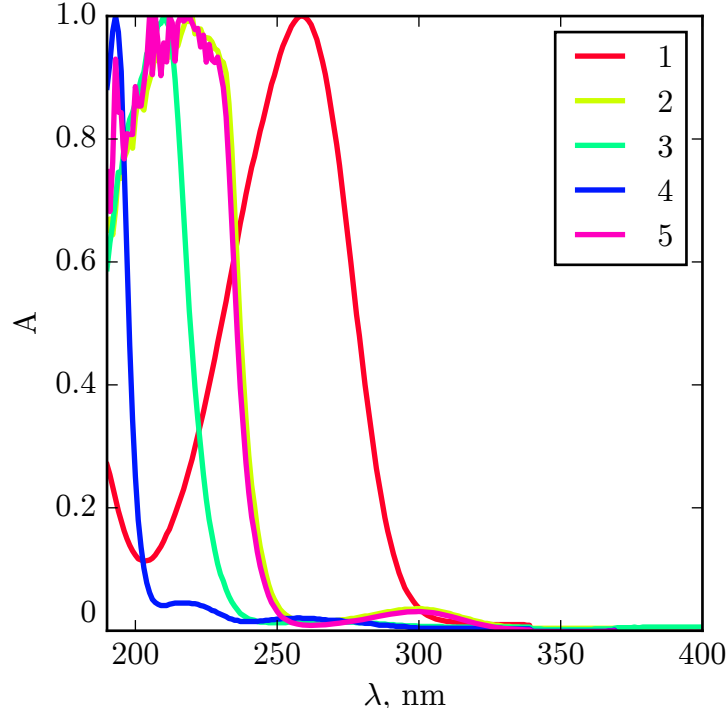


Figure 4.4. Normalized absorbance spectra of the synthesis components: 1: ascorbic acid ($C_6H_8O_6$); 2: silver nitrate ($AgNO_3$); 3: sodium citrate ($Na_3C_6H_5O_7$); 4: sodium chloride ($NaCl$); 5: nitric acid (HNO_3)

the normalized absorbance spectra of the synthesis components. The largest contribution of unreacted components is expected below 300 nm, which is confirmed by the results in Fig. 4.3.

4.3 Precursor volume for a shell of given thickness

As shown in the literature review (Section 1.3), plasmonic enhancement is observed at an optimal spacer thickness of a few to several tens of nanometers. The volume of tetraethyl orthosilicate must therefore be chosen to give the concentration that corresponds to the required silica shell thickness. To find an expression for the TEOS concentration, let us determine the TEOS volume V_{TEOS} needed for a SiO_2 shell of given thickness h_s . Besides h_s and reference data (molar masses M and densities ρ), the known parameters are the volume of the colloidal silver solution V_{sol} , the plasmon resonance wavelength λ_{max} determined from the absorption spectrum of the colloidal silver solution, and the molar concentration of the silver salt C_{Ag} . It is convenient to start by expressing the TEOS volume through the amount of substance:

$$V_{\text{TEOS}} = \frac{m_{\text{TEOS}}}{\rho_{\text{TEOS}}} = \frac{\nu_{\text{TEOS}} \cdot M_{\text{TEOS}}}{\rho_{\text{TEOS}}}. \quad (4.1)$$

In the hydrolysis reaction, not all TEOS turns into SiO_2 . Taking into account the practical yield of the shell formation reaction, η_{SiO_2} , we can write:

$$\nu_{\text{TEOS}} = \nu_{\text{SiO}_2} \cdot \eta_{\text{SiO}_2} = \frac{m_{\text{SiO}_2}}{M_{\text{SiO}_2}} \cdot \eta_{\text{SiO}_2} = \frac{\rho_{\text{SiO}_2} \cdot V_{\text{SiO}_2}}{M_{\text{SiO}_2}} \cdot \eta_{\text{SiO}_2}, \quad (4.2)$$

where V_{SiO_2} is the total volume of silica on the surface of all silver particles. In the approximation of monodisperse particles, this is

$$V_{\text{SiO}_2} = N_c \cdot V_s, \quad (4.3)$$

where N_c is the number of cores, V_s is the volume of a single shell. The number of particles is the ratio of the total mass of reduced silver to the mass of a single particle, taking into account the practical yield of the core formation reaction (silver particles), η_{Ag} :

$$N_c = \frac{m_{\text{Ag}} \cdot \eta_{\text{Ag}}}{m_c} = \frac{V_{\text{sol}} \cdot C_{\text{Ag}} \cdot M_{\text{Ag}} \cdot \eta_{\text{Ag}}}{\rho_{\text{Ag}} \cdot V_c} = \frac{6 \cdot V_{\text{sol}} \cdot C_{\text{Ag}} \cdot M_{\text{Ag}} \cdot \eta_{\text{Ag}}}{\pi \cdot \rho_{\text{Ag}} \cdot d_c^3}. \quad (4.4)$$

The shell volume is the difference between the volumes of the core-shell particle V_{cs} and the core V_c . For spherical particles:

$$V_s = V_{\text{cs}} - V_c = \frac{\pi}{6} (d_{\text{cs}}^3 - d_c^3) = \frac{\pi}{6} ((d_c + 2h_s)^3 - d_c^3). \quad (4.5)$$

The diameter of a silver particle d_c can be determined from the plasmon resonance position λ_{max} . For silver nanoparticles, this dependence is linear [88]:

$$d_c = \frac{\lambda_{\text{max}} - 382.6 \text{ nm}}{1.18}. \quad (4.6)$$

Thus, substituting (4.4) and (4.5) into (4.3), and then (4.3) $\rightarrow$ (4.2) $\rightarrow$ (4.1), we obtain the TEOS volume for spherical silver nanoparticles:

$$\begin{aligned} V_{\text{TEOS}} &= \frac{\eta_{\text{Ag}}}{\eta_{\text{SiO}_2}} \cdot \frac{M_{\text{Ag}} \cdot M_{\text{TEOS}}}{M_{\text{SiO}_2}} \cdot \frac{\rho_{\text{SiO}_2}}{\rho_{\text{Ag}} \cdot \rho_{\text{TEOS}}} \cdot \frac{(d_c + 2h_s)^3 - d_c^3}{d_c^3} \cdot C_{\text{Ag}} \cdot V_{\text{sol}} = \\ &= \frac{\eta_{\text{Ag}}}{\eta_{\text{SiO}_2}} \cdot \frac{M_{\text{Ag}} \cdot M_{\text{TEOS}}}{M_{\text{SiO}_2}} \cdot \frac{\rho_{\text{SiO}_2}}{\rho_{\text{Ag}} \cdot \rho_{\text{TEOS}}} \cdot \left(\left(1 + \frac{2h_s}{d_c} \right)^3 - 1 \right) \cdot C_{\text{Ag}} \cdot V_{\text{sol}}. \end{aligned} \quad (4.7)$$

Note that solution (4.7) is easily generalized to any core-shell particles synthesized by similar reactions:

$$V_p = \frac{\eta_c}{\eta_s} \cdot \frac{M_c M_p}{M_s} \cdot \frac{\rho_s}{\rho_c \rho_p} \cdot \frac{V_{\text{cs}} - V_c}{V_c} \cdot C_c \cdot V_{\text{sol}},$$

where the subscript p denotes the parameters of the shell precursor, c those of the cores or their material, and s those of the shells; η_c and η_s are the percentage yields of the core and shell material formation reactions, respectively. Since η_{SiO_2} and η_{Ag} are of the same order and cancel each other in (4.7), neglecting them in practice is justified.

Next, we can find the ratio of the SiO_2 shell thickness h_s to the particle diameter d_c as a function of the TEOS concentration:

$$\frac{h_s}{d_c} = \frac{1}{2} \left[\sqrt[3]{\frac{M_{\text{SiO}_2} \cdot \rho_{\text{Ag}} \cdot \rho_{\text{TEOS}} \cdot C_{\text{TEOS}}}{(\rho_{\text{TEOS}} - M_{\text{TEOS}} \cdot C_{\text{TEOS}}) \cdot M_{\text{Ag}} \cdot \rho_{\text{SiO}_2} \cdot C_{\text{Ag}}}} + 1 - 1 \right] \quad (4.8)$$

where the TEOS concentration is:

$$C_{\text{TEOS}} = \frac{\nu_{\text{TEOS}}}{V_{\text{sol}} + V_{\text{TEOS}}} = \frac{\rho_{\text{TEOS}} \cdot V_{\text{TEOS}}}{M_{\text{TEOS}} \cdot (V_{\text{sol}} + V_{\text{TEOS}})}$$

The dependence (4.8) is plotted in Fig. 4.5 for the concentration range used in this work. The dependence is nonlinear.

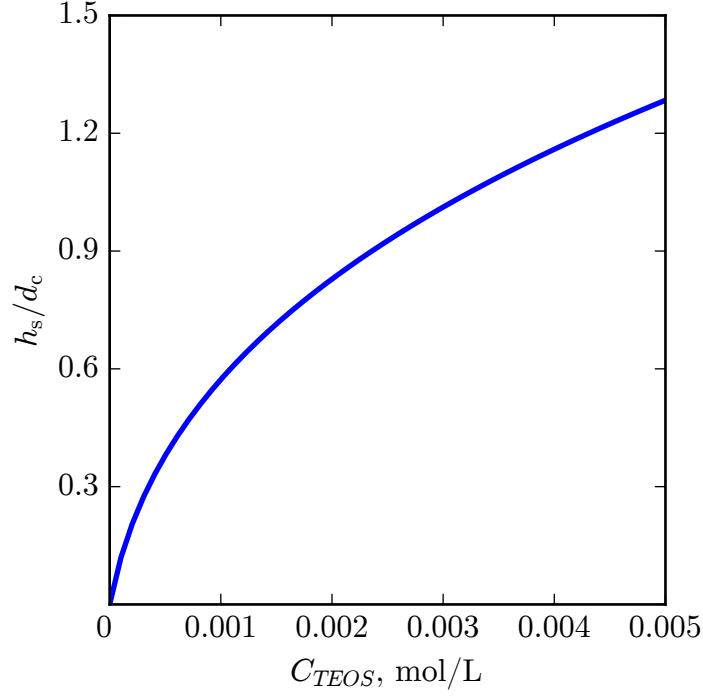


Figure 4.5. Ratio of the shell thickness to the core diameter, h_s/d_c , versus TEOS concentration

4.4 Synthesis of Ag/SiO₂ nanoparticles

Silver nanoparticles with a silica shell were synthesized by the modified Stöber method. Ethanol and various amounts of TEOS, calculated from (4.7), were added to the colloidal silver nanoparticle solution from the previous step to obtain particles with different shell thicknesses. After 5 min of stirring, aqueous ammonia was added to change the pH. The particles then grew for 1 h under continuous stirring, after which the process was stopped by adding ethanol. To isolate the particles, the colloidal solutions were centrifuged at 9000 rpm for 30 min; the supernatant was removed, and the particles were redispersed in ethanol for further work.

For shell growth on larger particles, the required TEOS volume was calculated as follows. The wavelength of the LSPR maximum of the synthesized silver particles was determined from the absorbance spectra. The diameter d_c and the number of particles N_c were found from (4.6) and (4.4). Assuming that the number of particles does not change in the second synthesis step (the particles grow larger, but their number stays the same), the SiO₂ volume was calculated from (4.3). Here the shell volume V_s was found as in (4.5), but with

the diameter of the larger particles d_{c_2} instead of d_c :

$$V_s = \frac{\pi}{6}[(d_{c_2} + 2h_s)^3 - d_{c_2}^3]$$

The amount of TEOS was found as:

$$\nu_{\text{TEOS}} = \frac{\rho_{\text{SiO}_2} \cdot V_{\text{SiO}_2}}{M_{\text{SiO}_2}}$$

The TEOS volume was then found from (4.1).

Absorbance spectra of Ag/SiO₂ particles with different shell thicknesses are shown in Fig. 4.6 for $d_c \approx 20$ nm and in Fig. 4.7 for $d_c \approx 60$ nm. The shifts of the plasmon resonance maxima are shown enlarged in Fig. 4.8 and Fig. 4.9, respectively.

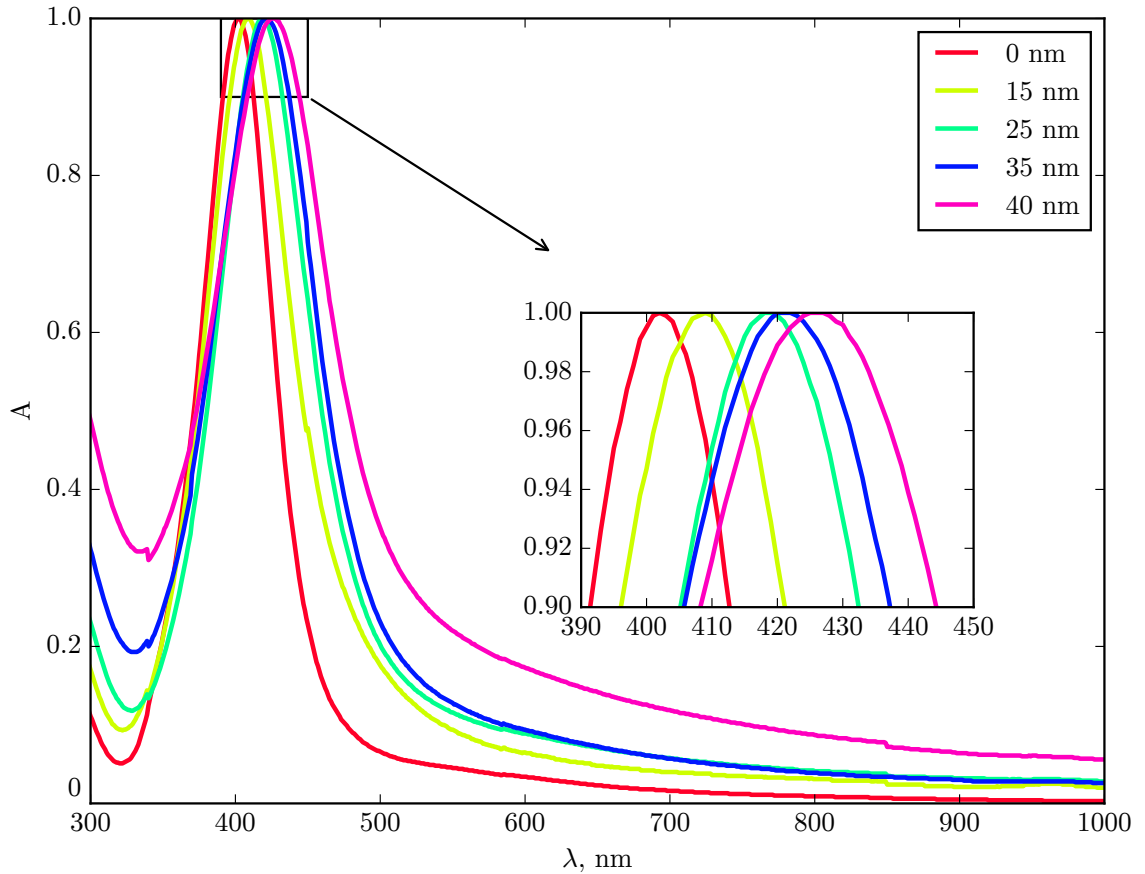


Figure 4.6. Normalized absorbance spectra of colloidal solutions of Ag/SiO₂ particles with $d_c \approx 20$ nm and different shell thicknesses (0 nm corresponds to particles before shell growth) [130]

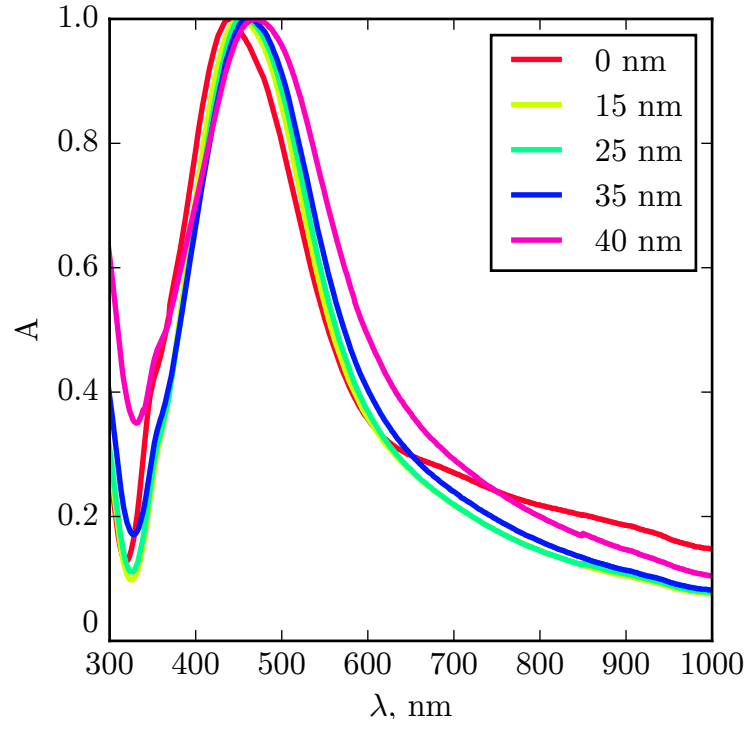


Figure 4.7. Normalized absorbance spectra of colloidal solutions of Ag/SiO₂ particles with $d_c \approx 60$ nm and different shell thicknesses (0 nm corresponds to particles before shell growth)

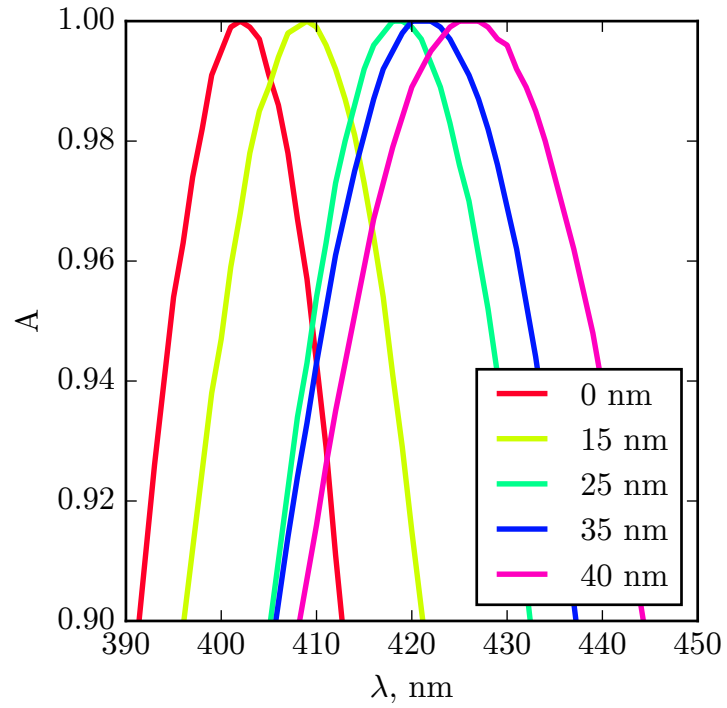


Figure 4.8. Enlarged part of Fig. 4.6 near the absorbance maxima of Ag/SiO₂ particles

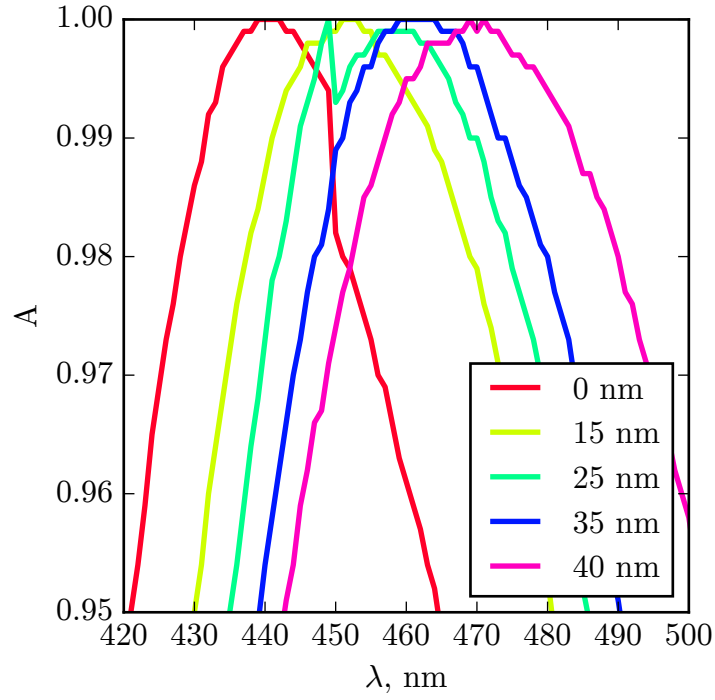


Figure 4.9. Enlarged part of Fig. 4.7 near the absorbance maxima of Ag/SiO₂ particles

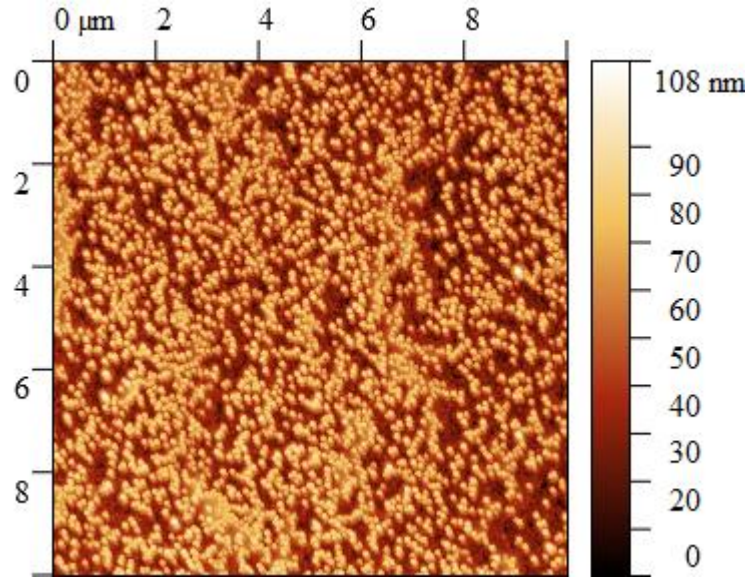


Figure 4.10. Surface topography of an array of Ag/SiO₂ nanoparticles with a calculated core diameter of 20 nm and a calculated shell thickness of 40 nm (atomic force microscopy)

The spectra in Fig. 4.6 show that small particles remain monodisperse after shell growth. As the shell becomes thicker, the absorbance in the long-wavelength part of the spectrum increases because larger particles scatter more light. The spectra in Fig. 4.7 show that the larger silver nanoparticles (calcu-

lated diameter 60 nm) have a wider size distribution and, correspondingly, a broader spectrum.

Fig. 4.8 and Fig. 4.9 show that as the shell becomes thicker, the LSPR shifts to longer wavelengths because the permittivity of the surrounding medium changes. For such thin shells, the effective permittivity of the surrounding medium lies between those of silica and water, since the field near the surface of a metal nanoparticle with a thin shell extends beyond the shell. This also supports the possibility of observing plasmonic enhancement with dielectric spacers a few tens of nanometers thick.

The synthesis results were also checked by atomic force microscopy. Fig. 4.10 shows an array of Ag/SiO₂ nanoparticles deposited on a polished single-crystal silicon substrate. The Ag core diameter was 20 nm and the calculated SiO₂ shell thickness 40 nm, so the Ag/SiO₂ nanoparticle diameter should be about 100 nm. This agrees with the maximum z-values of the image. In addition, the Ag/SiO₂ particles in the image have similar sizes, which also agrees with the optical spectra described above.

4.5 Quantum dot systems studied

Plasmonic enhancement of photoluminescence was studied both on quantum dots without wide-gap shells, CdS and PbS, and on CdSe/ZnS core/shell structures. Initially, the quantum dots were in the form of colloidal solutions of semiconductor nanocrystals whose surface was stabilized by organic molecules: octadecylamine for CdSe/ZnS QDs and oleic acid for CdS and PbS. Toluene was the dispersion medium in all three systems.

Fig. 4.11, Fig. 4.12, and Fig. 4.13 show the absorption and photoluminescence spectra ($\lambda_{\text{ex}} = 405$ nm) of the colloidal quantum dot solutions used in the experiment. The spectral shapes and peak positions stay the same for nanoparticles deposited on substrates.

All types of QDs share some features. First, the photoluminescence spectrum has a narrow symmetric line, especially for core/shell quantum dots. Second, the absorption spectra are broad, and the absorbance increases toward shorter wavelengths. Therefore, one and the same short-wavelength source can excite photoluminescence in any of the samples; in our case, a 405 nm laser. All types of QDs show excitonic states in the absorption spectrum as a local maximum separated from the PL maximum by the Stokes shift.

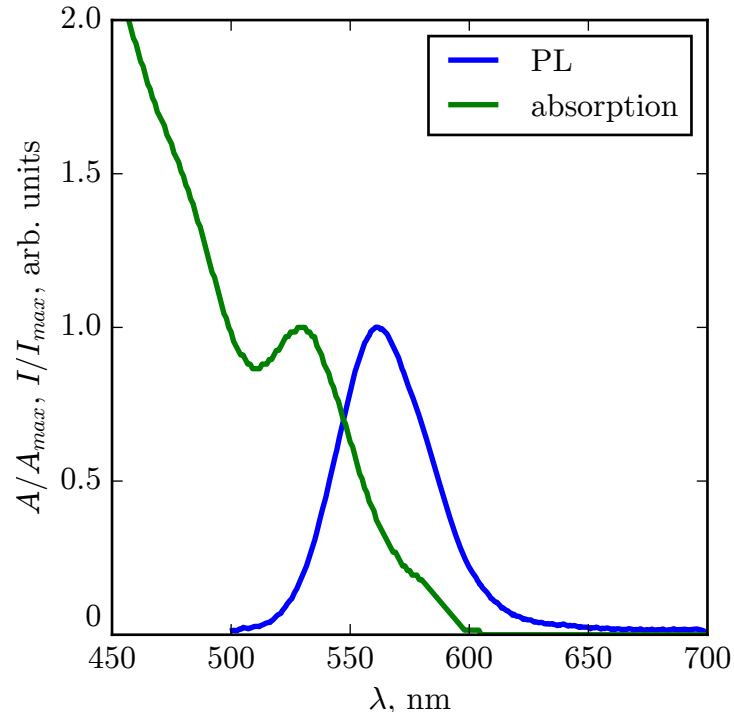


Figure 4.11. Normalized absorption and photoluminescence spectra of CdSe/ZnS QDs ($\lambda_{\text{ex}} = 405 \text{ nm}$)

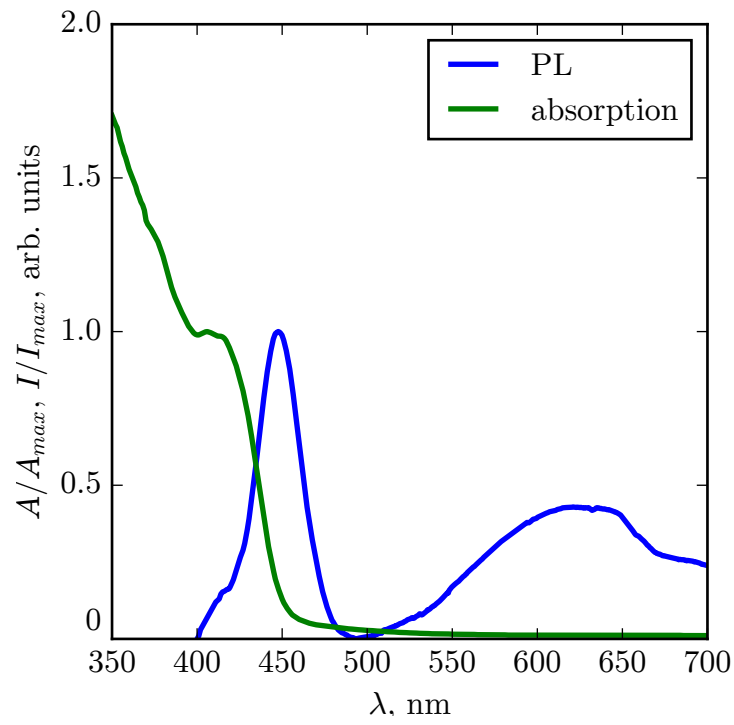


Figure 4.12. Normalized absorption and photoluminescence spectra of CdS QDs ($\lambda_{\text{ex}} = 405 \text{ nm}$)

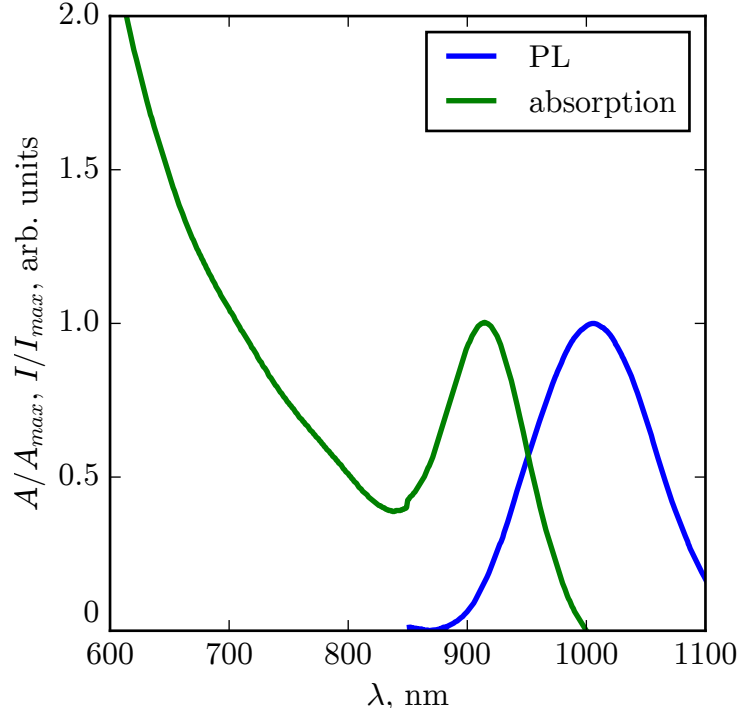


Figure 4.13. Normalized absorption and photoluminescence spectra of PbS QDs ($\lambda_{\text{ex}} = 405 \text{ nm}$)

Note that the plasmon resonance frequencies of the Ag and Ag/SiO₂ plasmonic particles described above lie at higher energies than the first excitonic peak for all these QD types. These QD materials were chosen because of interest in increasing their luminescence intensity: their PL quantum yield is far below that of the perovskite QDs described in Chapter 3. However, similar studies have also been reported for perovskite QDs [131].

Note that QDs with poor surface passivation (Fig. 4.12) show, besides the fundamental transition with a narrow emission band, an additional broad PL band in the long-wavelength region. It is associated with surface states, through which radiative recombination also occurs. PbS QDs, in which oleic acid apparently passivates the electronic surface states better, show no such features.

4.6 Structures for studying plasmonic enhancement of luminescence

To study plasmonic enhancement of photoluminescence, structures of the general form shown in Fig. 4.14 were fabricated. For this, 20 μL of a solution of the synthesized Ag/SiO₂ plasmonic particles was spin-coated onto a quartz

substrate at 1000 rpm for 30 s. The substrate was then placed in an oven at 200°C for 5 min to remove the solvent. The same procedure was then repeated for the QDs, but at a lower temperature (100°C). To obtain a uniform layer of plasmonic particles, the procedure was repeated 5 times. The solution concentrations and the number of layers in the hybrid structure were chosen so that the plasmonic nanoparticles formed a continuous layer covering the substrate, and so that the measured PL signal was dominated by the interaction of individual quantum dots with plasmonic nanoparticles.

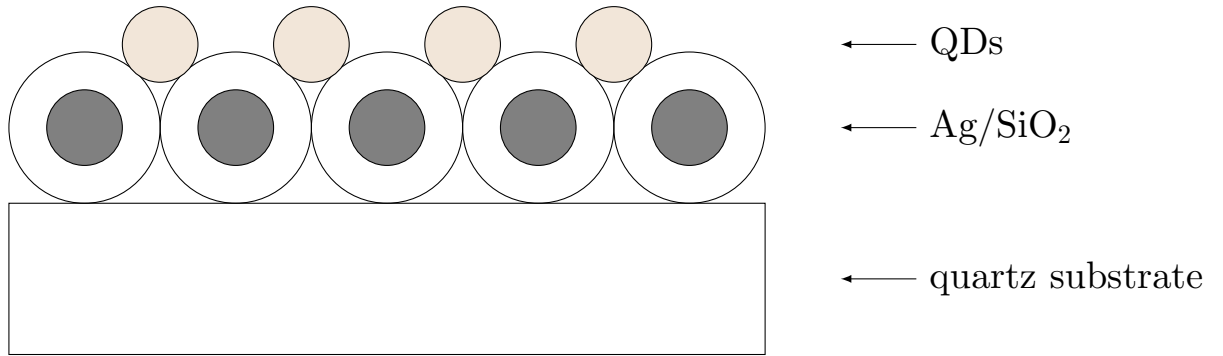


Figure 4.14. Schematic of the structures for studying plasmonic enhancement of luminescence [130]

In PL measurements, the structure was excited from the QD layer side.

4.7 Plasmonic enhancement study

To describe the enhancement and compare plasmonic enhancement for different hybrid system materials, we introduce the enhancement factor K :

$$K = \frac{\max(I(\lambda))}{\max(I_0(\lambda))}$$

where $I(\lambda)$ and $I_0(\lambda)$ are the photoluminescence spectra of the QD layer with and without the plasmonic particle layer, respectively. That is, the factor is the ratio of the luminescence maxima with and without plasmonic particles.

4.7.1 Plasmonic enhancement of photoluminescence of CdSe/ZnS quantum dots

Let us consider plasmonic enhancement in CdSe/ZnS QD structures with a layer of plasmonic particles with a relatively small core ($d_c \approx 20$ nm). Fig. 4.15 shows the photoluminescence spectra of structures with plasmonic particles of

different shell thicknesses, normalized to the PL maximum of the structure without plasmonic particles. These results are described in detail in [130].

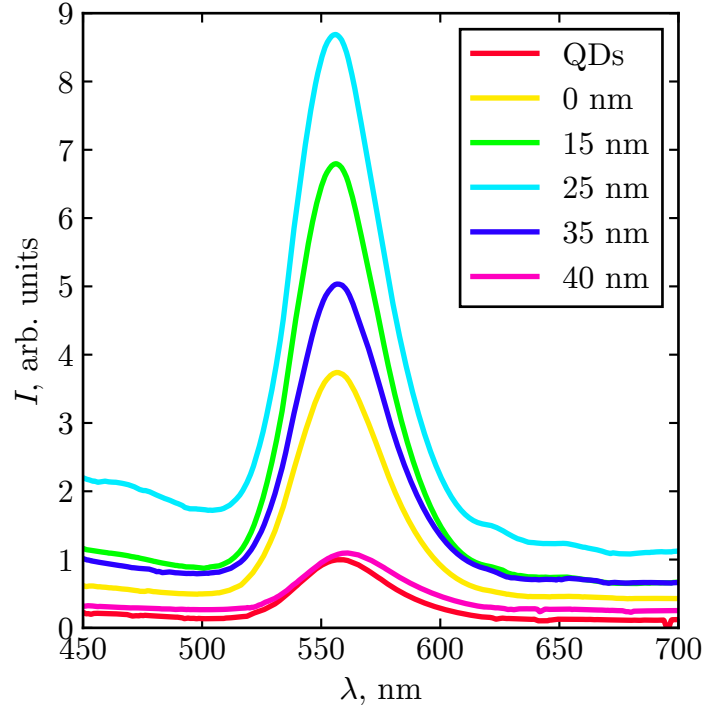


Figure 4.15. Photoluminescence spectra of CdSe/ZnS QDs without and near Ag/SiO₂ plasmonic particles with $d_c \approx 20$ nm and different SiO₂ shell thicknesses (here and below, a shell thickness of 0 nm corresponds to silver nanoparticles without a SiO₂ shell) [130]

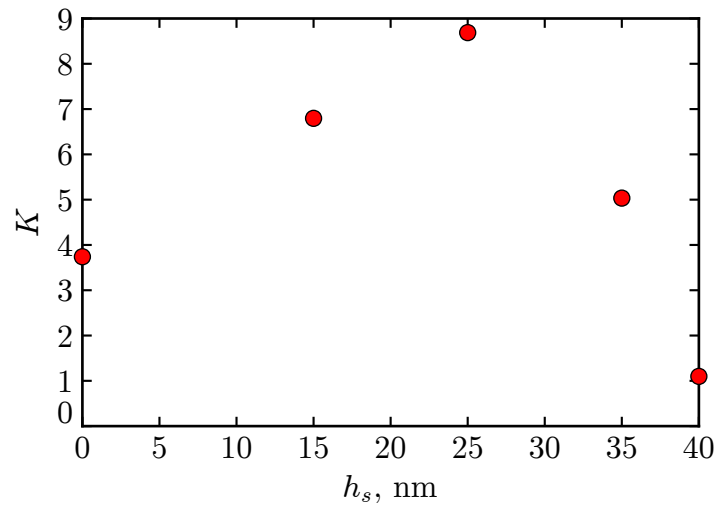


Figure 4.16. Photoluminescence enhancement factor versus SiO₂ shell thickness in hybrid structures of CdSe/ZnS QDs and Ag/SiO₂ plasmonic particles with $d_c \approx 20$ nm [130]

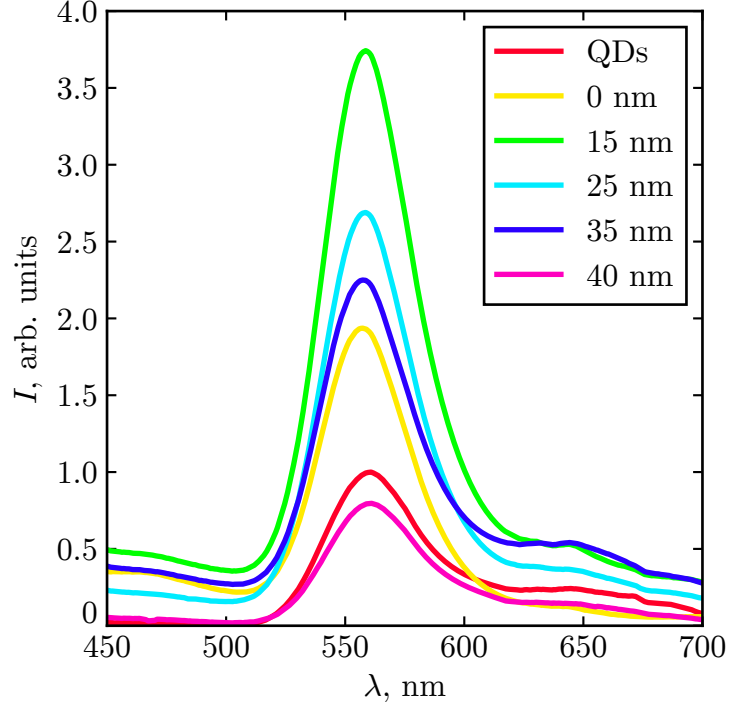


Figure 4.17. Photoluminescence spectra of CdSe/ZnS QDs without and near Ag/SiO₂ plasmonic particles with $d_c \approx 60$ nm and different SiO₂ shell thicknesses

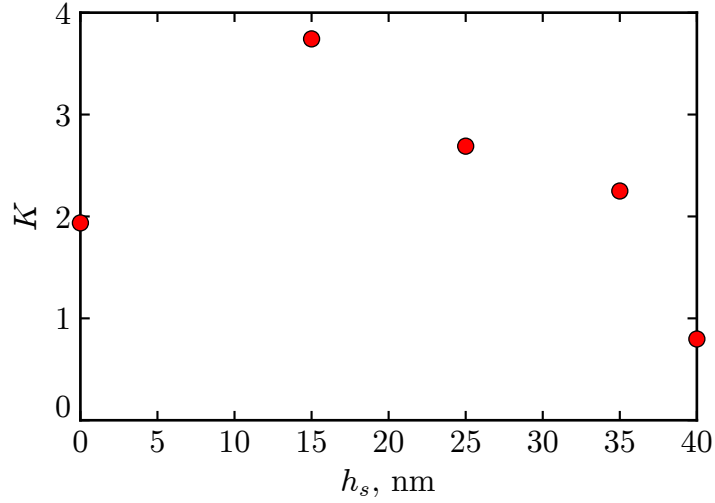


Figure 4.18. Photoluminescence enhancement factor versus SiO₂ shell thickness in hybrid structures of CdSe/ZnS QDs and Ag/SiO₂ plasmonic particles with $d_c \approx 60$ nm

The spectral shape and peak position practically do not change in the presence of plasmonic nanoparticles. Note that metal particles without a dielectric shell do not quench the photoluminescence; instead, they enhance it, though less than particles with a dielectric spacer. Note also that the lumines-

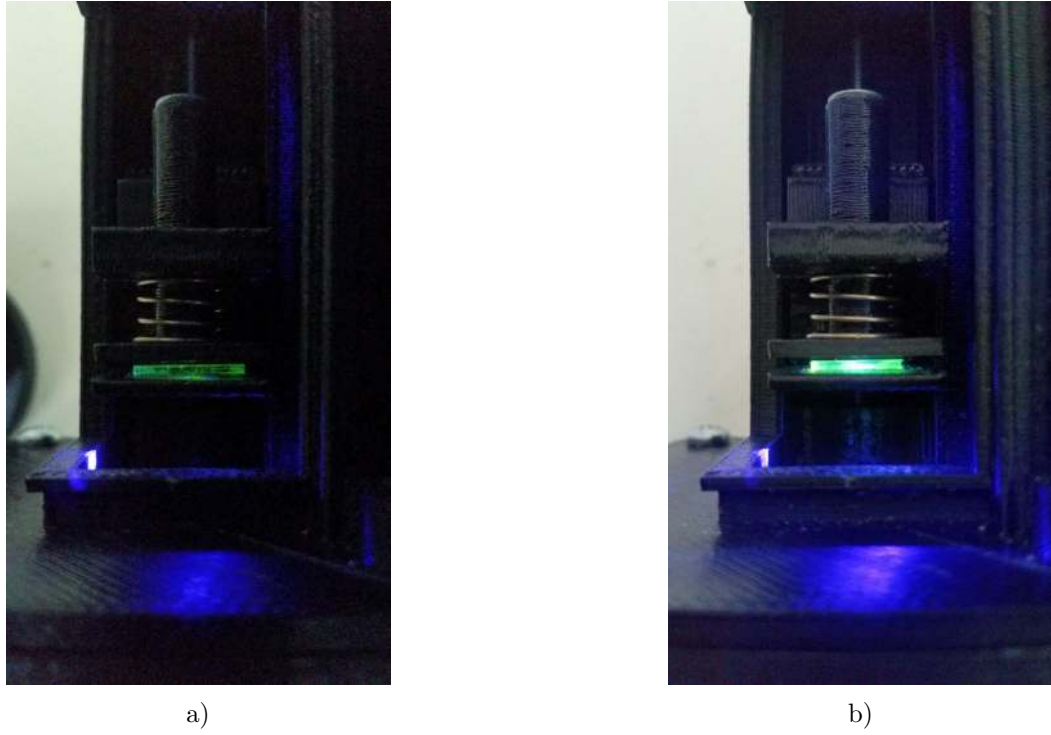


Figure 4.19. Photoluminescence of CdSe/ZnS QDs for samples a) without a plasmonic particle layer, b) with a layer of small plasmonic particles with a 25 nm SiO₂ shell

cence intensity grows together with the background scattering caused by the plasmonic nanoparticles in the structure. This scattering is apparently one of the reasons for the PL increase.

For clarity, the dependences $K(h_s)$ on the shell thickness were also plotted (Fig. 4.16). The largest enhancement for these structures, $K = 8.7$, is observed at a shell thickness of $h_s = 25$ nm. At $h_s = 40$ nm, the photoluminescence intensity falls almost to the initial level. For this QD type, this enhancement factor is the highest among the studied structures. The difference in photoluminescence for the structures with the highest enhancement factor was visible by eye (Fig. 4.19).

The photoluminescence maximum at an optimal shell thickness h_{opt} and the overall shape of the enhancement factor dependence on the shell thickness, $K(h_s)$, agree with the literature review (Section 1.3). The rapid decrease of the enhancement factor for shells thicker than h_{opt} reflects the near-field nature of plasmonic luminescence enhancement.

For larger particles (diameter ~ 60 nm), the largest enhancement, $K = 3.7$, is observed at a shell thickness of $h_s = 15$ nm (Fig. 4.17 and Fig. 4.18). At $h_s = 40$ nm, the PL signal falls below the initial level. The smaller enhancement factor for larger particles may be explained by the fact that excitation was

always with a 405 nm laser, while the LSPR maximum of the large particles is at 450 nm. In addition, these particles have a wider size distribution.

4.7.2 Plasmonic enhancement of photoluminescence of PbS quantum dots

The photoluminescence spectra of similar structures based on lead sulfide (PbS) quantum dots are shown in Fig. 4.20. The corresponding enhancement factors versus the shell thickness of the plasmonic particles, $K(h_s)$, are shown in Fig. 4.21. As for CdSe/ZnS QDs, the spectral shape and peak position of lead sulfide particles stay the same; only the photoluminescence intensity changes, depending on the type of plasmonic particles.

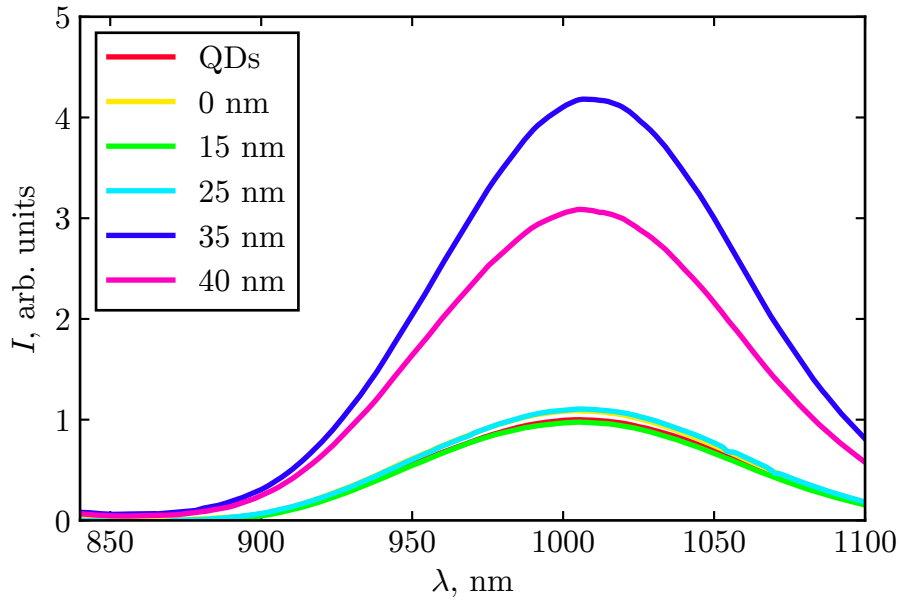


Figure 4.20. Photoluminescence spectra of PbS QDs without and near Ag/SiO₂ plasmonic particles with $d_c \approx 20$ nm and different SiO₂ shell thicknesses [79]

Lead sulfide quantum dots show the same form of $K(h_s)$. With small plasmonic particles (Fig. 4.21), the largest enhancement for PbS QDs, $K = 4.2$, is observed at $h_s = 35$ nm. Without a shell on the plasmonic particles, and at $h_s = 15$ and 25 nm, the photoluminescence intensity is close to the initial level, while at $h_s = 40$ nm an enhancement of $K = 3.1$ is observed. With large nanoparticles (Fig. 4.22), the dependence was similar, but, as in the other cases, the maximum enhancement factor was lower than for similar smaller particles, $K = 2.9$.

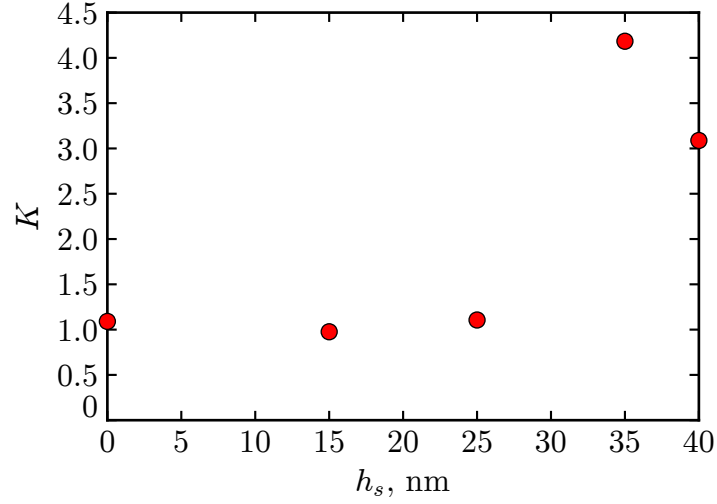


Figure 4.21. Photoluminescence enhancement factor versus SiO_2 shell thickness in hybrid structures of PbS QDs and Ag/ SiO_2 plasmonic particles with $d_c \approx 20$ nm [79]

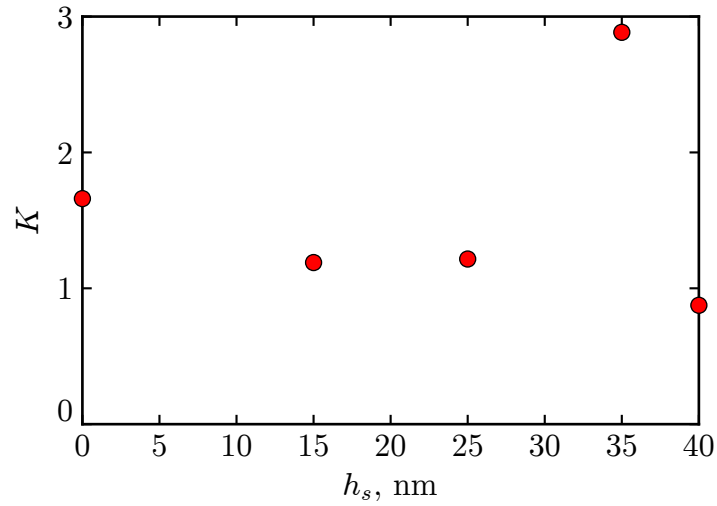


Figure 4.22. Photoluminescence enhancement factor versus SiO_2 shell thickness in hybrid structures of PbS QDs and Ag/ SiO_2 plasmonic particles with $d_c \approx 60$ nm

The change in h_{opt} compared with CdSe/ZnS structures may have at least two causes: a) the shell in CdSe/ZnS increases the total spacer between the particles, b) the ZnS barrier layer in CdSe/ZnS reduces the probability of carrier tunneling compared with a structure without a shell. Thus, other things being equal, h_{opt} should be smaller in structures based on core/shell QDs than in structures based on core-only QDs, which is confirmed by this experiment.

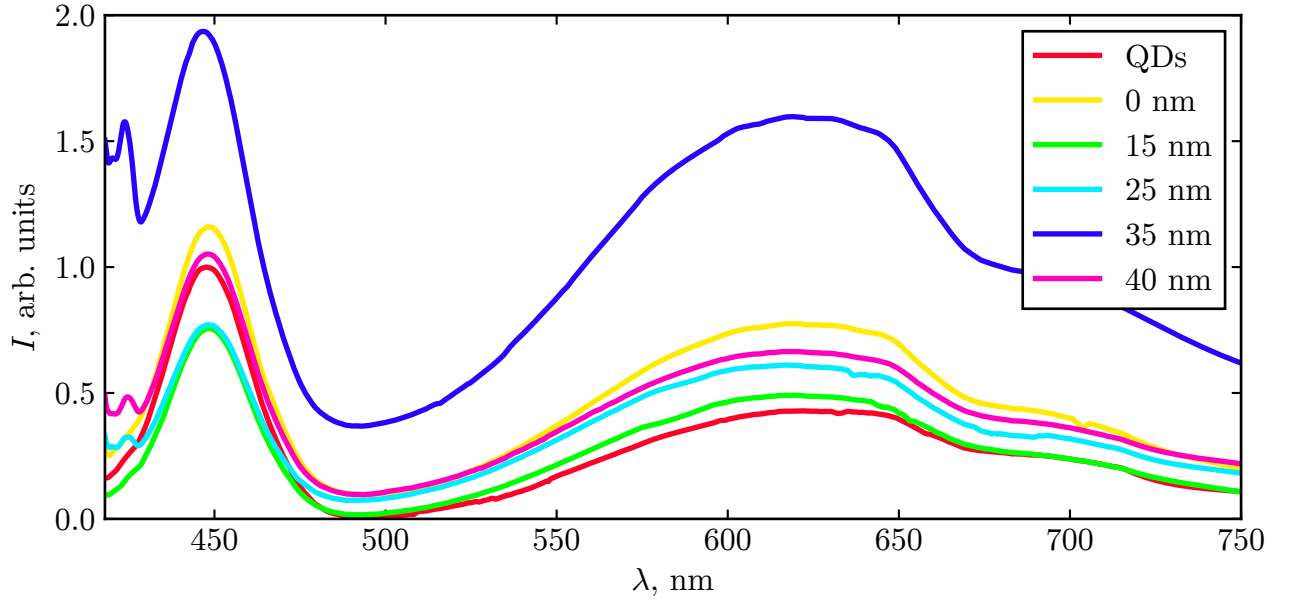


Figure 4.23. Photoluminescence spectra of CdS QDs without and near Ag/SiO₂ plasmonic particles with $d_c \approx 20$ nm and different SiO₂ shell thicknesses [79]

4.7.3 Plasmonic enhancement of photoluminescence of CdS quantum dots

The photoluminescence spectra of structures based on cadmium sulfide quantum dots are shown in Fig. 4.23. The corresponding dependence $K(h_s)$ is shown in Fig. 4.24.

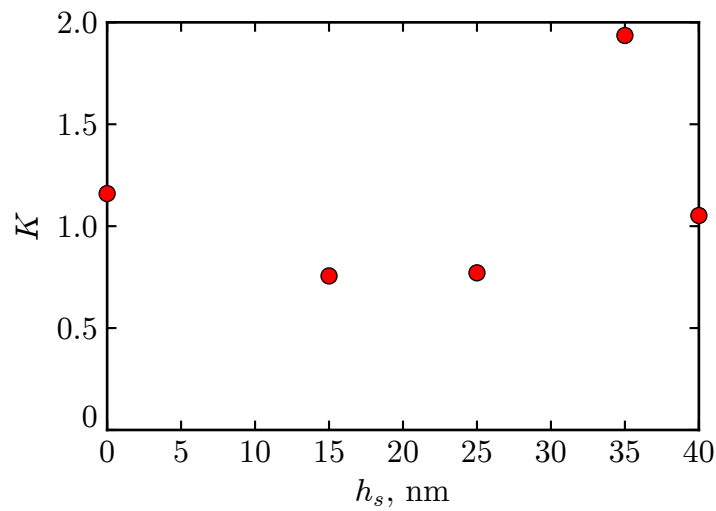


Figure 4.24. Photoluminescence enhancement factor versus SiO₂ shell thickness in hybrid structures of CdS QDs and Ag/SiO₂ plasmonic particles with $d_c \approx 20$ nm [79]

For hybrid structures based on CdS QDs, the largest enhancement, $K = 1.9$, is observed at $h_s = 35$ nm. For this QD type, this enhancement factor is the lowest among the studied structures. Note, however, that all the systems studied share one observation: luminescence is enhanced in the presence of plasmonic particles at an optimal spacer thickness h_{opt} .

At $h_s = 40$ nm, the photoluminescence intensity falls to the initial level. For the peak associated with surface states, the largest enhancement is also observed at $h_s = 35$ nm. For shell thicknesses $h_s = 25\text{--}40$ nm, peaks appear in the luminescence spectra near 425 nm. This may be because the plasmonic particles have their LSPR at this wavelength. One can assume that photons emitted by the quantum dots at these wavelengths additionally excite the LSPR in the plasmonic particles, which in turn enhances the QD PL at these wavelengths.

The results on plasmonic enhancement for lead and cadmium sulfide QDs were also published in [79].

4.8 Possible optical applications of the hybrid structures

The effects described above are of considerable interest for optical applications. Let us consider possible excitation schemes for a structure based on the studied nanoparticles, with external optical pumping (Fig. 4.25) and an optical resonator made of photonic crystals (PCs). Let a layer of colloidal QDs with emission frequency ω_{qd} be placed between two layers of plasmonic particles with plasmon resonance frequency ω_{pl} , which in turn lie between two photonic crystals with reflection spectra $R_1(\omega)$ and $R_2(\omega)$ (different in the general case). Let monochromatic excitation ω_{in} fall on the first photonic crystal and generate radiation that leaves either in the direction of the incident beam (frequency $\omega_{\parallel}$) or perpendicular to it (frequency $\omega_{\perp}$).

For efficient excitation of the plasmon resonance, the frequencies must satisfy:

$$\begin{aligned}\omega_{\text{in}} &= \omega_{\text{pl}}, \\ \omega_{\text{in}} &> \omega_{\text{qd}}.\end{aligned}$$

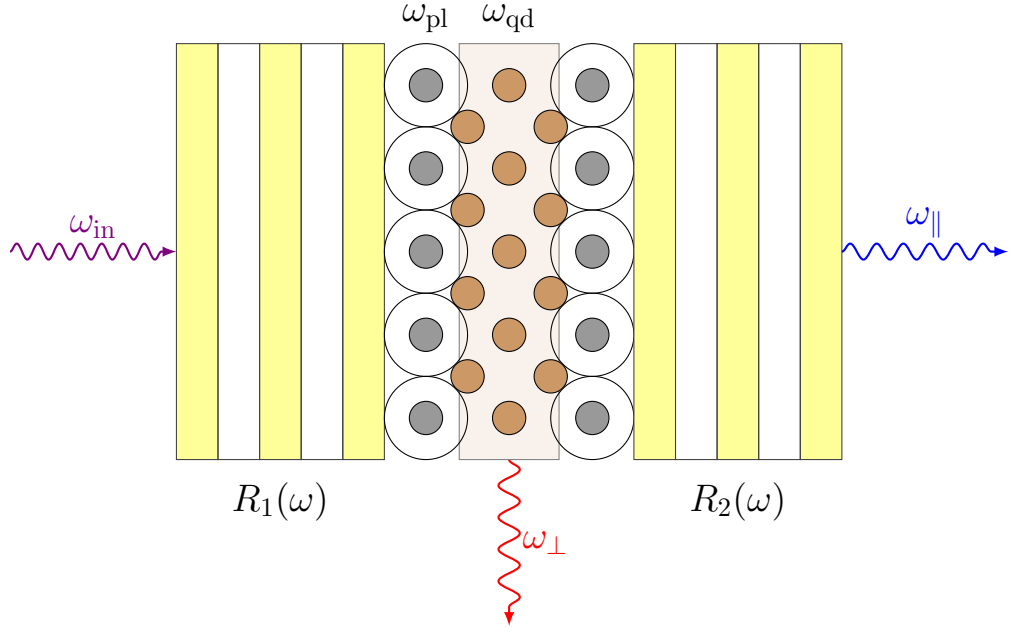


Figure 4.25. Model of a structure with external optical pumping

Table 4.2. Emission of the structure for different reflection spectra of the second PC in Fig. 4.25

	$R_2(\omega_{pl})$	$R_2(\omega_{qd})$	$\omega_{\perp}$	$\omega_{\parallel}$
1	0	0	-	ω_{in}, ω_{qd}
2	0	1	ω_{qd}	ω_{in}
3	1	0	-	ω_{qd}
4	1	1	ω_{qd}	-

Since PC_1 must transmit the excitation, the only structures of interest for PC_1 are those with $R_1(\omega_{in}) = R_1(\omega_{pl}) = 0$ and $R_1(\omega_{qd}) = 1$, i.e., the photonic crystal acts as a selective mirror for the radiation excited in the QD layer.

In this case, several interesting variants of PC_2 are possible, with the reflection spectra listed in Table 4.2. In case 1, PC_2 may be absent: the excitation and the quantum dot emission are mixed. In case 2, PC_1 and PC_2 are identical: the quantum dot emission is reflected many times by the photonic crystals, while the unabsorbed excitation passes through the sample. In case 3, the excitation is reflected back, and the excited emission can pass further. In case 4, PC_2 acts as a mirror. Thus, structures 2–4 may show the properties of optical amplifiers.

The corresponding methods for fabricating photonic crystals were described in our papers [132–135] and summarized in [136]. However, fabricating the structure shown in Fig. 4.25, namely precisely joining its two halves, is still

particularly difficult, so these results are not included in the thesis. This task can be regarded as a direction for further work on this topic.

4.9 Conclusions to Chapter 4

A method was proposed for the synthesis of monodisperse quasi-spherical silver particles with a diameter of $d_c \approx 20$ nm. For larger particles ($d_c \approx 60$ nm), a two-step method was proposed and implemented, in which small particles serve as seeds for the growth of larger particles in the second step. To determine the optimal distance between a plasmonic particle and a QD, Ag/SiO₂ core/shell particles with different shell thicknesses were synthesized by the modified Stöber method. The plasmonic properties of the particles were studied by optical spectroscopy.

To study plasmonic enhancement, structures of plasmonic particle and quantum dot layers on quartz substrates were fabricated. Plasmonic enhancement of photoluminescence was studied for core-only CdS and PbS quantum dots and for core/shell CdSe/ZnS quantum dots in the presence of Ag/SiO₂ plasmonic particles with different mean metal core diameters (≈ 20 and 60 nm) and different dielectric shell thicknesses (15–40 nm).

For all the QDs studied, photoluminescence enhancement is observed at an optimal distance h_{opt} , in agreement with the literature review (Section 1.3). For CdSe/ZnS and PbS QDs, which were studied with both particle sizes, the strongest photoluminescence enhancement is observed near plasmonic particles of small diameter. For CdSe/ZnS, the largest enhancement, $K = 8.7$, is observed at $h_{\text{opt}} = 25$ nm; for PbS, $K = 4.2$ at $h_{\text{opt}} = 35$ nm; for CdS, $K = 1.9$ at $h_{\text{opt}} = 35$ nm. For CdSe/ZnS QDs, this enhancement factor is the highest among the QDs studied.

Our result that the optimal thickness is of the order of 25–35 nm agrees with experimental studies [4, 75] on the range of surface-enhanced Raman scattering and luminescence on nanostructured metal surfaces, and with numerical simulations showing that noticeable fluorescence in such systems can be expected for spacer thicknesses above 15 nm [78].

As a result of this part of the work, the following main statement was formulated: the photoluminescence enhancement in a hybrid system of CdSe/ZnS core/shell quantum dots and Ag/SiO₂ plasmonic nanoparticles arises from a manifold increase in the extinction of the combined system. The effect

is controlled by varying the thickness of the dielectric shell.

Conclusion

The work achieved the following results:

1. Within a single technological approach, a technology was developed and implemented for the synthesis of monodisperse nanocrystals: colloidal quantum dots of metal chalcogenides and cesium lead halides, and silver-based plasmonic nanoparticles.
2. Automated experimental setups were proposed and built for 1) nanoparticle synthesis in a flow of immiscible liquids, 2) deposition of nanoparticle layers from colloidal solutions, and 3) optical monitoring of nanoparticle synthesis and nanoparticle layer deposition by measuring light-scattering indicatrices. These engineering solutions are protected by three Russian utility model patents.
3. Scientifically grounded synthesis conditions were developed for nanocrystals capable of oriented attachment, of CsPbX_3 ($X = \text{Cl}, \text{Br}, \text{I}$) perovskites and solid solutions based on them. They rely on the concept of the decisive role of the $[\text{PbX}_6]^{4-}$ anionic octahedral complex.
4. Ag nanoparticles 20 nm in diameter with a 25 nm SiO_2 shell increased the photoluminescence intensity of a CdSe/ZnS layer more than eightfold compared with the same layer without plasmonic nanoparticles.
5. The synthesized luminescent CsPbBr_3 nanoparticles were shown to be more radiation-hard than commercial CdSe/ZnS nanoparticles.
6. XRD demonstrated that layers of halide perovskite nanocrystals up to several hundred nanometers thick can be deposited with a strong (100) orientation.

The methods proposed by the author and the materials synthesized were used in joint work with other research groups on biomarkers based on colloidal quantum dots [137], on charge transport in nanoparticles dispersed in polymer matrices [138], and on optical orientation and alignment of excitons in layers of perovskite nanocrystals [139].

Main Publications on the Topic of the Thesis

1. Ryzhov O.A., **Matyushkin L.B.**, Moshnikov V.A., Aleksandrova O.A. Reactor for the synthesis of nanostructures. RU Patent 166323; filed 24.12.2015; published 20.11.2016.
2. Moshnikov V.A., Maksimov A.I., Aleksandrova O.A., **Matyushkin L.B.**, Ryabko A.A. Device for studying materials by the light-scattering indicatrix method. RU Patent 167044; filed 14.06.2016; published 20.12.2016.
3. **Matyushkin L.B.**, Khondryukov D.V., Aleksandrova O.A. Automated setup for depositing thin-film coatings of nanoscale thickness by molecular layering. RU Patent 156478; filed 21.04.2015; published 10.11.2015.
4. Aleshin A.N., Shcherbakov I.P., Gushchina E.V., **Matyushkin L.B.**, Moshnikov V.A. Solution-processed field-effect transistors based on polyfluorene–cesium lead halide nanocrystals composite films with small hysteresis of output and transfer characteristics // Organic Electronics. 2017. V.50. P.213–219.
5. Nestoklon M.O., Goupalov S.V., Dzhioev R.I., Ken O.S., Korenev V.L., Kusrayev Yu.G., Sapega V.F., de Weerd C., Gomez L., Gregorkiewicz T., Lin J., Suenaga K., Fujiwara Y., **Matyushkin L.B.**, Yassievich I.N. Optical orientation and alignment of excitons in ensembles of inorganic perovskite nanocrystals // Physical Review B. 2018. V.97, No. 23. P.235304.
6. **Matyushkin L.B.**, Pertsova A., Moshnikov V.A. Enhanced luminescence of quantum dots near a layer of Ag/SiO₂ nanoparticles // Technical Physics Letters. 2018. V.44, No. 4. P.331–333 (Russian original: Pis'ma v Zhurnal Tekhnicheskoi Fiziki. 2018. V.44, No. 8. P.35–41).
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10. **Matyushkin L.B.**, Reshetnikova A.A., Andronov A.O., Afonicheva P.K., Myakin S.V., Permiakov N.V., Moshnikov V.A. Morphology, optical, and adsorption properties of copper-oxide layers deposited from complex compound solutions // Semiconductors. 2017. V.51, No. 5. P.586–590 (Russian original: Fizika i Tekhnika Poluprovodnikov. 2017. V.51, No. 5. P.615–619).
11. Mazing D.S., Shul'ga A.I., **Matyushkin L.B.**, Aleksandrova O.A., Moshnikov V.A. Synthesis and characterization of colloidal nanocrystals of ternary chalcogenide compounds // Optics and Spectroscopy. 2017. V.122, No. 1. P.110–113 (Russian original: Optika i Spektroskopiya. 2017. V.122, No. 1. P.122–125).
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15. Tarasov S.A., Aleksandrova O.A., Maksimov A.I., Maraeva E.V., **Matyushkin L.B.**, Men'kovich E.A., Moshnikov V.A., Musikhin S.F. Study of the self-organization processes in lead sulfide quantum dots // Semiconductors. 2014. V.48, No. 13. P.1729–1731 (Russian original: Izvestiya Vysshikh Uchebnykh Zavedenii. Elektronika. 2013. No. 3. P.28–32).
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Papers [6–15] were published in journals listed by the Higher Attestation Commission (VAK) of Russia; publications [4–23] are indexed in Web of Science and Scopus.

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